DAVID WILLIAMS

Probability with Martingales

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Probability with Martingales

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Contents

Preface – please read! xi

A Question of Terminology xiii

A Guide to Notation xiv

Chapter 0: A Branching-Process Example 1

0.0. Introductory remarks. 0.1. Typical number of children, X. 0.2. Size of n^{th} generation, Z_n. 0.3. Use of conditional expectations. 0.4. Extinction probability, π. 0.5. Pause for thought: measure. 0.6. Our first martingale. 0.7. Convergence (or not) of expectations. 0.8. Finding the distribution of M_∞. 0.9. Concrete example.

PART A: FOUNDATIONS

Chapter 1: Measure Spaces 14

1.0. Introductory remarks. 1.1. Definitions of algebra, σ-algebra. 1.2. Examples. Borel σ-algebras, $\mathcal{B}(S)$, $\mathcal{B} = \mathcal{B}(\mathbf{R})$. 1.3. Definitions concerning set functions. 1.4. Definition of measure space. 1.5. Definitions concerning measures. 1.6. Lemma. Uniqueness of extension, π-systems. 1.7. Theorem. Carathéodory's extension theorem. 1.8. Lebesgue measure Leb on $((0,1], \mathcal{B}(0,1])$. 1.9. Lemma. Elementary inequalities. 1.10. Lemma. Monotone-convergence properties of measures. 1.11. Example/Warning.

Chapter 2: Events 23

2.1. Model for experiment: $(\Omega, \mathcal{F}, \mathbf{P})$. 2.2. The intuitive meaning. 2.3. Examples of $(\Omega, \mathcal{F})$ pairs. 2.4. Almost surely (a.s.) 2.5. Reminder: lim sup, lim inf, $\downarrow$ lim, etc. 2.6. Definitions. $\limsup E_n$, $(E_n, \text{i.o.})$. 2.7.

First Borel-Cantelli Lemma (BC1). 2.8. Definitions. $\liminf E_n, (E_n, \text{ev})$. 2.9. Exercise.

Chapter 3: Random Variables 29

3.1. Definitions. Σ-measurable function, $\text{m}\Sigma$, $(\text{m}\Sigma)^+$,$\text{b}\Sigma$. 3.2. Elementary Propositions on measurability. 3.3. Lemma. Sums and products of measurable functions are measurable. 3.4. Composition Lemma. 3.5. Lemma on measurability of infs, liminfs of functions. 3.6. Definition. Random variable. 3.7. Example. Coin tossing. 3.8. Definition. σ-algebra generated by a collection of functions on Ω. 3.9. Definitions. Law, Distribution Function. 3.10. Properties of distribution functions. 3.11. Existence of random variable with given distribution function. 3.12. Skorokod representation of a random variable with prescribed distribution function. 3.13. Generated σ-algebras – a discussion. 3.14. The Monotone-Class Theorem.

Chapter 4: Independence 38

4.1. Definitions of independence. 4.2. The π-system Lemma; and the more familiar definitions. 4.3. Second Borel-Cantelli Lemma (BC2). 4.4. Example. 4.5. A fundamental question for modelling. 4.6. A coin-tossing model with applications. 4.7. Notation: IID RVs. 4.8. Stochastic processes; Markov chains. 4.9. Monkey typing Shakespeare. 4.10. Definition. Tail σ-algebras. 4.11. Theorem. Kolmogorov's 0-1 law. 4.12. Exercise/Warning.

Chapter 5: Integration 49

5.0. Notation, etc. $\mu(f) =: \int f d\mu$, $\mu(f; A)$. 5.1. Integrals of non-negative simple functions, SF^+. 5.2. Definition of $\mu(f)$, $f \in (\text{m}\Sigma)^+$. 5.3. Monotone-Convergence Theorem (MON). 5.4. The Fatou Lemmas for functions (FATOU). 5.5. 'Linearity'. 5.6. Positive and negative parts of f. 5.7. Integrable function, $\mathcal{L}^1(S, \Sigma, \mu)$. 5.8. Linearity. 5.9. Dominated Convergence Theorem (DOM). 5.10. Scheffé's Lemma (SCHEFFÉ). 5.11. Remark on uniform integrability. 5.12. The standard machine. 5.13. Integrals over subsets. 5.14. The measure $f\mu$, $f \in (m\Sigma)^+$.

Chapter 6: Expectation 58

Introductory remarks. 6.1. Definition of expectation. 6.2. Convergence theorems. 6.3. The notation $\text{E}(X; F)$. 6.4. Markov's inequality. 6.5. Sums of non-negative RVs. 6.6. Jensen's inequality for convex functions. 6.7. Monotonicity of $\mathcal{L}^p$ norms. 6.8. The Schwarz inequality. 6.9. $\mathcal{L}^2$: Pythagoras, covariance, etc. 6.10. Completeness of $\mathcal{L}^p$ $(1 \leq p < \infty)$. 6.11. Orthogonal projection. 6.12. The 'elementary formula' for expectation. 6.13. Hölder from Jensen.

Chapter 7: An Easy Strong Law 71

7.1. 'Independence means multiply' – again! 7.2. Strong Law – first version. 7.3. Chebyshev's inequality. 7.4. Weierstrass approximation theorem.

Chapter 8: Product Measure 75

8.0. Introduction and advice. 8.1. Product measurable structure, $\Sigma_1 \times \Sigma_2$. 8.2. Product measure, Fubini's Theorem. 8.3. Joint laws, joint pdfs. 8.4. Independence and product measure. 8.5. $\mathcal{B}(\mathbf{R})^n = \mathcal{B}(\mathbf{R}^n)$. 8.6. The n-fold extension. 8.7. Infinite products of probability triples. 8.8. Technical note on the existence of joint laws.

PART B: MARTINGALE THEORY

Chapter 9: Conditional Expectation 83

9.1. A motivating example. 9.2. Fundamental Theorem and Definition (Kolmogorov, 1933). 9.3. The intuitive meaning. 9.4. Conditional expectation as least-squares-best predictor. 9.5. Proof of Theorem 9.2. 9.6. Agreement with traditional expression. 9.7. Properties of conditional expectation: a list. 9.8. Proofs of the properties in Section 9.7. 9.9. Regular conditional probabilities and pdfs. 9.10. Conditioning under independence assumptions. 9.11. Use of symmetry: an example.

Chapter 10: Martingales 93

10.1. Filtered spaces. 10.2. Adapted processes. 10.3. Martingale, supermartingale, submartingale. 10.4. Some examples of martingales. 10.5. Fair and unfair games. 10.6. Previsible process, gambling strategy. 10.7. A fundamental principle: you can't beat the system! 10.8. Stopping time. 10.9. Stopped supermartingales are supermartingales. 10.10. Doob's Optional-Stopping Theorem. 10.11. Awaiting the almost inevitable. 10.12. Hitting times for simple random walk. 10.13. Non-negative superharmonic functions for Markov chains.

Chapter 11: The Convergence Theorem 106

11.1. The picture that says it all. 11.2. Upcrossings. 11.3. Doob's Upcrossing Lemma. 11.4. Corollary. 11.5. Doob's 'Forward' Convergence Theorem. 11.6. Warning. 11.7. Corollary.

Chapter 12: Martingales bounded in $\mathcal{L}^2$ 110

12.0. Introduction. 12.1. Martingales in $\mathcal{L}^2$: orthogonality of increments. 12.2. Sums of zero-mean independent random variables in $\mathcal{L}^2$. 12.3. Random signs. 12.4. A symmetrization technique: expanding the sample space. 12.5. Kolmogorov's Three-Series Theorem. 12.6. Cesàro's Lemma. 12.7. Kronecker's Lemma. 12.8. A Strong Law under variance constraints. 12.9. Kolmogorov's Truncation Lemma. 12.10. Kolmogorov's Strong Law of Large Numbers (SLLN). 12.11. Doob decomposition. 12.12. The angle-brackets process $\langle M \rangle$. 12.13. Relating convergence of M to finiteness of $\langle M \rangle_\infty$. 12.14. A trivial 'Strong Law' for martingales in $\mathcal{L}^2$. 12.15. Lévy's extension of the Borel-Cantelli Lemmas. 12.16. Comments.

Chapter 13: Uniform Integrability 126

13.1. An 'absolute continuity' property. 13.2. Definition. UI family. 13.3. Two simple sufficient conditions for the UI property. 13.4. UI property of conditional expectations. 13.5. Convergence in probability. 13.6. Elementary proof of (BDD). 13.7. A necessary and sufficient condition for $\mathcal{L}^1$ convergence.

Chapter 14: UI Martingales 133

14.0. Introduction. 14.1. UI martingales. 14.2. Lévy's 'Upward' Theorem. 14.3. Martingale proof of Kolmogorov's 0-1 law. 14.4. Lévy's 'Downward' Theorem. 14.5. Martingale proof of the Strong Law. 14.6. Doob's Submartingale Inequality. 14.7. Law of the Iterated Logarithm: special case. 14.8. A standard estimate on the normal distribution. 14.9. Remarks on exponential bounds; large deviation theory. 14.10. A consequence of Hölder's inequality. 14.11. Doob's $\mathcal{L}^p$ inequality. 14.12. Kakutani's Theorem on 'product' martingales. 14.13.The Radon-Nikodým theorem. 14.14. The Radon-Nikodým theorem and conditional expectation. 14.15. Likelihood ratio; equivalent measures. 14.16. Likelihood ratio and conditional expectation. 14.17. Kakutani's Theorem revisited; consistency of LR test. 14.18. Note on Hardy spaces, etc.

Chapter 15: Applications 153

15.0. Introduction – please read! 15.1. A trivial martingale-representation result. 15.2. Option pricing; discrete Black-Scholes formula. 15.3. The Mabinogion sheep problem. 15.4. Proof of Lemma 15.3(c). 15.5. Proof of result 15.3(d). 15.6. Recursive nature of conditional probabilities. 15.7. Bayes' formula for bivariate normal distributions. 15.8. Noisy observation of a single random variable. 15.9. The Kalman-Bucy filter. 15.10. Harnesses entangled. 15.11. Harnesses unravelled, 1. 15.12. Harnesses unravelled, 2.

PART C: CHARACTERISTIC FUNCTIONS

Chapter 16: Basic Properties of CFs 172

16.1. Definition. 16.2. Elementary properties. 16.3. Some uses of characteristic functions. 16.4. Three key results. 16.5. Atoms. 16.6. Lévy's Inversion Formula. 16.7. A table.

Chapter 17: Weak Convergence 179

17.1. The 'elegant' definition. 17.2. A 'practical' formulation. 17.3. Skorokhod representation. 17.4. Sequential compactness for Prob($\overline{\mathbf{R}}$). 17.5. Tightness.

Chapter 18: The Central Limit Theorem 185

18.1. Lévy's Convergence Theorem. 18.2. o and O notation. 18.3. Some important estimates. 18.4. The Central Limit Theorem. 18.5. Example. 18.6. CF proof of Lemma 12.4.

APPENDICES

Chapter A1: Appendix to Chapter 1 192

A1.1. A non-measurable subset A of S^1. A1.2. d-systems. A1.3. Dynkin's Lemma. A1.4. Proof of Uniqueness Lemma 1.6. A1.5. λ-sets: 'algebra' case. A1.6. Outer measures. A1.7. Carathéodory's Lemma. A1.8. Proof of Carathéodory's Theorem. A1.9. Proof of the existence of Lebesgue measure on $((0,1], \mathcal{B}(0,1])$. A1.10. Example of non-uniqueness of extension. A1.11. Completion of a measure space. A1.12. The Baire category theorem.

Chapter A3: Appendix to Chapter 3 205

A3.1. Proof of the Monotone-Class Theorem 3.14. A3.2. Discussion of generated σ-algebras.

Chapter A4: Appendix to Chapter 4 208

A4.1. Kolmogorov's Law of the Iterated Logarithm. A4.2. Strassen's Law of the Iterated Logarithm. A4.3. A model for a Markov chain.

Chapter A5: Appendix to Chapter 5 211

A5.1. Doubly monotone arrays. A5.2. The key use of Lemma 1.10(a). A5.3. 'Uniqueness of integral'. A5.4. Proof of the Monotone-Convergence Theorem.

Chapter A9: Appendix to Chapter 9 214

A9.1. Infinite products: setting things up. A9.2. Proof of A9.1(e).

Chapter A13: Appendix to Chapter 13 217

A13.1. Modes of convergence: definitions. A13.2. Modes of convergence: relationships.

Chapter A14: Appendix to Chapter 14 219

A14.1. The σ-algebra $\mathcal{F}_T$, T a stopping time. A14.2. A special case of OST. A14.3. Doob's Optional-Sampling Theorem for UI martingales. A14.4. The result for UI submartingales.

Chapter A16: Appendix to Chapter 16 222

A16.1. Differentiation under the integral sign.

Chapter E: Exercises 224

References 243

Index 246

Preface – please read!

The most important chapter in this book is *Chapter E: Exercises.* I have left the interesting things for *you* to do. You can start *now* on the 'EG' exercises, but see 'More about exercises' later in this Preface.

The book, which is essentially the set of lecture notes for a third-year undergraduate course at Cambridge, is as lively an introduction as I can manage to the rigorous theory of probability. Since much of the book is devoted to martingales, it is bound to become very lively: look at those Exercises on Chapter 10! But, of course, there is that initial plod through the measure-theoretic foundations. It must be said however that measure theory, that most arid of subjects when done for its own sake, becomes amazingly more alive when used in probability, not only because it is then applied, but also because it is immensely enriched.

You cannot avoid measure theory: an *event* in probability is a measurable set, a *random variable* is a measurable function on the sample space, the *expectation* of a random variable is its integral with respect to the probability measure; and so on. To be sure, one can take some central results from measure theory as axiomatic in the main text, giving careful proofs in appendices; and indeed that is exactly what I have done.

Measure theory for its own sake is based on the fundamental addition rule for measures. Probability theory supplements that with the multiplication rule which describes independence; and things are already looking up. But what really enriches and enlivens things is that we deal with lots of σ-algebras, not just the one σ-algebra which is the concern of measure theory.

In planning this book, I decided for every topic what things I considered just a bit too advanced, and, often with sadness, I have ruthlessly omitted them.

For a more thorough training in many of the topics covered here, see Billingsley (1979), Chow and Teicher (1978), Chung (1968), Kingman and

Taylor (1966), Laha and Rohatgi (1979), and Neveu (1965). As regards measure theory, I learnt it from Dunford and Schwartz (1958) and Halmos (1959). After reading this book, you must read the still-magnificent Breiman (1968), and, for an excellent indication of what can be done with discrete martingales, Hall and Heyde (1980).

Of course, intuition is much more important than knowledge of measure theory, and you should take every opportunity to sharpen your intuition. There is no better whetstone for this than Aldous (1989), though it is a very demanding book. For appreciating the scope of probability and for learning how to think about it, Karlin and Taylor (1981), Grimmett and Stirzaker (1982), Hall (1988), and Grimmett's recent superb book, Grimmett (1989), on percolation are strongly recommended.

More about exercises. In compiling Chapter E, which consists exactly of the homework sheet I give to the Cambridge students, I have taken into account the fact that this book, like any other mathematics book, implicitly contains a vast number of other exercises, many of which are easier than those in Chapter E. I refer of course to the exercises *you* create by reading the statement of a result, and then trying to prove it for yourself, before you read the given proof. One other point about exercises: you will, for example, surely forgive my using expectation E in Exercises on Chapter 4 before E is treated with full rigour in Chapter 6.

Acknowledgements. My first thanks must go to the students who have endured the course on which the book is based and whose quality has made me try hard to make it worthy of them; and to those, especially David Kendall, who had developed the course before it became my privilege to teach it. My thanks to David Tranah and other staff of CUP for their help in converting the course into this book. Next, I must thank Ben Garling, James Norris and Chris Rogers without whom the book would have contained more errors and obscurities. (The many faults which surely remain in it are my responsibility.) Helen Rutherford and I typed part of the book, but the vast majority of it was typed by Sarah Shea-Simonds in a virtuoso performance worthy of Horowitz. My thanks to Helen and, most especially, to Sarah. Special thanks to my wife, Sheila, too, for all her help.

But my best thanks – and yours if you derive any benefit from the book – must go to three people whose names appear in capitals in the Index: J.L. Doob, A.N. Kolmogorov and P. Lévy: without them, there wouldn't have been much to write about, as Doob (1953) splendidly confirms.

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David Williams
October 1990

A Question of Terminology

Random variables: functions or equivalence classes?

At the level of this book, the theory would be more 'elegant' if we regarded a random variable as an *equivalence class* of measurable functions on the sample space, two functions belonging to the same equivalence class if and only if they are equal almost everywhere. Then the conditional-expectation map

$$X \mapsto \mathsf{E}(X|\mathcal{G})$$

would be a truly well-defined contraction map from $L^p(\Omega, \mathcal{F}, \mathbf{P})$ to $L^p(\Omega, \mathcal{G}, \mathbf{P})$ for $p \geq 1$; and we would not have to keep mentioning versions (representatives of equivalence classes) and would be able to avoid the endless 'almost surely' qualifications.

I have however chosen the 'inelegant' route: firstly, I prefer to work with *functions*, and confess to preferring

$$4 + 5 = 2 \bmod 7 \qquad \text{to} \qquad [4]_7 + [5]_7 = [2]_7.$$

But there is a substantive reason. I hope that this book will tempt you to progress to the much more interesting, and more important, theory where the parameter set of our process is uncountable (e.g. it may be the time-parameter set $[0, \infty)$). There, the equivalence-class formulation just will not work: the 'cleverness' of introducing quotient spaces loses the subtlety which is essential even for formulating the fundamental results on existence of continuous modifications, etc., unless one performs contortions which are hardly elegant. Even if these contortions allow one to *formulate* results, one would still have to use genuine functions to *prove* them; so where does the reality lie?!

A Guide to Notation

► signifies something important, ►► something very important, and ►►► the Martingale Convergence Theorem.

I use ':=' to signify 'is defined to equal'. This Pascal notation is particularly convenient because it can also be used in the reversed sense.

I use analysts' (as opposed to category theorists') conventions:

► $$\mathbf{N} := \{1, 2, 3, \ldots\} \subseteq \{0, 1, 2, \ldots\} =: \mathbf{Z}^+.$$

Everyone is agreed that $\mathbf{R}^+ := [0, \infty)$.

For a set B contained in some universal set S, I_B denotes the indicator function of B: that is $I_B : S \to \{0, 1\}$ and

$$I_B(s) := \begin{cases} 1 & \text{if } s \in B, \\ 0 & \text{otherwise.} \end{cases}$$

For $a, b \in \mathbf{R}$,

$$a \wedge b := \min(a, b), \qquad a \vee b := \max(a, b).$$

CF:characteristic function; DF: distribution function; pdf: probability density function.

σ-algebra, $\sigma(\mathcal{C})$ (1.1); $\sigma(Y_\gamma : \gamma \in \mathcal{C})$ (3.8, 3.13). π-system (1.6); d-system (A1.2).

---

a.e.:	almost everywhere (1.5)
a.s.:	almost surely (2.4)
bΣ:	the space of bounded Σ-measurable functions (3.1)

$\mathcal{B}(S)$:	the Borel σ-algebra on S, $\mathcal{B} := \mathcal{B}(\mathbf{R})$ (1.2)
$C \bullet X$:	discrete stochastic integral (10.6)
$d\lambda/d\mu$:	Radon-Nikodým derivative (5.14)
$d\mathbf{Q}/d\mathbf{P}$:	Likelihood Ratio (14.13)
$\mathsf{E}(X)$:	expectation $\mathsf{E}(X) := \int_\Omega X(\omega)\mathbf{P}(d\omega)$ of X (6.3)
$\mathsf{E}(X;F)$:	$\int_F X d\mathbf{P}$ (6.3)
$\mathsf{E}(X\|\mathcal{G})$:	conditional expectation (9.3)
(E_n, ev):	$\liminf E_n$ (2.8)
$(E_n, \text{i.o.})$:	$\limsup E_n$ (2.6)
f_X:	probability density function (pdf) of X (6.12).
$f_{X,Y}$:	joint pdf (8.3)
$f_{X\|Y}$:	conditional pdf (9.6)
F_X:	distribution function of X (3.9)
lim inf:	for sets, (2.8)
lim sup:	for sets, (2.6)
$x = \uparrow \lim x_n$:	$x_n \uparrow x$ in that $x_n \le x_{n+1}$ $(\forall n)$ and $x_n \to x$.
log:	natural (base e) logarithm
$\mathcal{L}_X$, Λ_X:	law of X (3.9)
$\mathcal{L}^p$, L^p:	Lebesgue spaces (6.7, 6.13)
Leb:	Lebesgue measure (1.8)
$\mathrm{m}\Sigma$:	space of Σ-measurable functions (3.1)
M^T:	process M stopped at time T (10.9)
$\langle M \rangle$:	angle-brackets process (12.12)
$\mu(f)$:	integral of f with respect to μ (5.0, 5.2)
$\mu(f;A)$:	$\int_A f d\mu$ (5.0, 5.2)
φ_X:	CF of X (Chapter 16)
φ:	pdf of standard normal N(0,1) distribution
Φ:	DF of N(0,1) distribution
X^T:	X stopped at time T (10.9)

Chapter 0

A Branching-Process Example

(This Chapter is not essential for the remainder of the book. You can start with Chapter 1 if you wish.)

0.0. Introductory remarks

The purpose of this chapter is threefold: to take something which is probably well known to you from books such as the immortal Feller (1957) or Ross (1976), so that you start on familiar ground; to make you start to think about some of the problems involved in making the elementary treatment into rigorous mathematics; and to indicate what new results appear if one applies the somewhat more advanced theory developed in this book. We stick to one example: a branching process. This is rich enough to show that the theory has some substance.

0.1. Typical number of children, X

In our model, the number of children of a typical animal (see Notes below for some interpretations of 'child' and 'animal') is a random variable X with values in $\mathbf{Z}^+$. *We assume that*

$$\mathbf{P}(X = 0) > 0.$$

We define the *generating function* f *of* X as the map $f : [0,1] \to [0,1]$, where

$$f(\theta) := \mathbf{E}(\theta^X) = \sum_{k \in \mathbf{Z}^+} \theta^k \mathbf{P}(X = k).$$

Standard theorems on power series imply that, for $\theta \in [0,1]$,

$$f'(\theta) = \mathbf{E}(X\theta^{X-1}) = \sum k\theta^{k-1}\mathbf{P}(X = k)$$

and

$$\mu := \mathbf{E}(X) = f'(1) = \sum k\mathbf{P}(X = k) \leq \infty.$$

Of course, $f'(1)$ is here interpreted as

$$\lim_{\theta \uparrow 1} \frac{f(\theta) - f(1)}{\theta - 1} = \lim_{\theta \uparrow 1} \frac{1 - f(\theta)}{1 - \theta},$$

since $f(1) = 1$. *We assume that*

$$\mu < \infty.$$

Notes. The first application of branching-process theory was to the question of survival of family names; and in that context, animal = man, and child = son.

In another context, 'animal' can be 'neutron', and 'child' of that neutron will signify a neutron released if and when the parent neutron crashes into a nucleus. Whether or not the associated branching process is supercritical can be a matter of real importance.

We can often find branching processes embedded in richer structures and can then use the results of this chapter to start the study of more interesting things.

For superb accounts of branching processes, see Athreya and Ney (1972), Harris (1963), Kendall (1966, 1975).

0.2. Size of n^{th} generation, Z_n

To be a bit formal: suppose that we are given a doubly infinite sequence

(a) $$\left\{X_r^{(m)} : m, r \in \mathbf{N}\right\}$$

of independent identically distributed random variables (IID RVs), each with the same distribution as X:

$$\mathbf{P}(X_r^{(m)} = k) = \mathbf{P}(X = k).$$

The idea is that for $n \in \mathbf{Z}^+$ and $r \in \mathbf{N}$, the variable $X_r^{(n+1)}$ represents the number of children (who will be in the $(n+1)^{\text{th}}$ generation) of the r^{th} animal (if there is one) in the n^{th} generation. The fundamental rule therefore is that if Z_m signifies the size of the n^{th} generation, then

(b) $$Z_{n+1} = X_1^{(n+1)} + \cdots + X_{Z_n}^{(n+1)}.$$

We assume that $Z_0 = 1$, so that (b) gives a full recursive definition of the sequence $(Z_m : m \in \mathbf{Z}^+)$ from the sequence (a). Our first task is

to calculate the distribution function of Z_n, or equivalently to find the generating function

(c) $$f_n(\theta) := \mathsf{E}(\theta^{Z_n}) = \sum \theta^k \mathsf{P}(Z_n = k).$$

0.3. Use of conditional expectations

The first main result is that for $n \in \mathbf{Z}^+$ (and $\theta \in [0,1]$)

(a) $$f_{n+1}(\theta) = f_n(f(\theta)),$$

so that for each $n \in \mathbf{Z}^+$, f_n is the n-fold composition

(b) $$f_n = f \circ f \circ \ldots \circ f.$$

Note that the 0-fold composition is by convention the identity map $f_0(\theta) = \theta$, in agreement with – indeed, forced by – the fact that $Z_0 = 1$.

To prove (a), we use – at the moment in intuitive fashion – the following very special case of the very useful *Tower Property of Conditional Expectation*:

(c) $$\mathsf{E}(U) = \mathsf{E}\mathsf{E}(U|V);$$

to find the expectation of a random variable U, first find the conditional expectation $\mathsf{E}(U|V)$ of U given V, and then find the expectation of *that*. We prove the ultimate form of (c) at a later stage.

We apply (c) with $U = \theta^{Z_{n+1}}$ and $V = Z_n$:

$$\mathsf{E}(\theta^{Z_{n+1}}) = \mathsf{E}\mathsf{E}(\theta^{Z_{n+1}}|Z_n).$$

Now, for $k \in \mathbf{Z}^+$, the conditional expectation of $\theta^{Z_{n+1}}$ given that $Z_n = k$ satisfies

(d) $$\mathsf{E}(\theta^{Z_{n+1}}|Z_n = k) = \mathsf{E}(\theta^{X_1^{(n+1)}+\cdots+X_k^{(n+1)}}|Z_n = k).$$

But Z_n is constructed from variables $X_s^{(r)}$ with $r \leq n$, and so Z_n is independent of $X_1^{(n+1)}, \ldots, X_k^{(n+1)}$. The conditional expectation given $Z_n = k$ in the right-hand term in (d) must therefore agree with the absolute expectation

(e) $$\mathsf{E}(\theta^{X_1^{(n+1)}} \ldots \theta^{X_k^{(n+1)}}).$$

But the expression at (e) is a *expectation of the product of independent random variables* and as part of the family of '*Independence means multiply*' results, we know that this expectation of a product may be rewritten as the product of expectations. Since (for every n and r)

$$\mathsf{E}(\theta^{X_r^{(n+1)}}) = f(\theta),$$

we have proved that

$$\mathsf{E}(\theta^{Z_{n+1}} | Z_n = k) = f(\theta)^k,$$

and this is what it means to say that

$$\mathsf{E}(\theta^{Z_{n+1}} | Z_n) = f(\theta)^{Z_n}.$$

[If V takes only integer values, then when $V = k$, the conditional expectation $\mathsf{E}(U|V)$ of U given V is equal to the conditional expectation $\mathsf{E}(U|V = k)$ of U given that $V = k$. (Sounds reasonable!)] Property (c) now yields

$$\mathsf{E}\theta^{Z_{n+1}} = \mathsf{E}f(\theta)^{Z_n},$$

and, since

$$\mathsf{E}(\alpha^{Z_n}) = f_n(\alpha),$$ □

result (a) is proved.

Independence and *conditional expectations* are two of the main topics in this course.

0.4. Extinction probability, π

Let $\pi_n := \mathsf{P}(Z_n = 0)$. Then $\pi_n = f_n(0)$, so that, by (0.3,b),

(a) $$\pi_{n+1} = f(\pi_n).$$

Measure theory confirms our intuition about the extinction probability:

(b) $$\pi := \mathsf{P}(Z_m = 0 \text{ for some } m) = \uparrow \lim \pi_n.$$

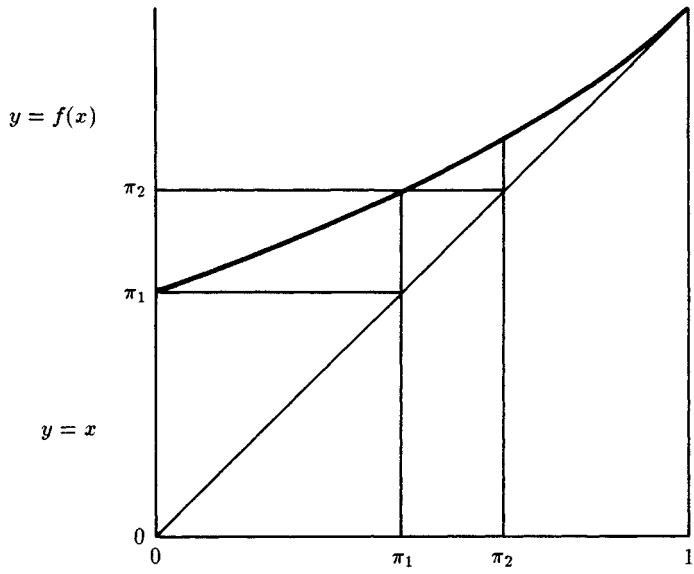
Because f is continuous, it follows from (a) that

(c) $$\pi = f(\pi).$$

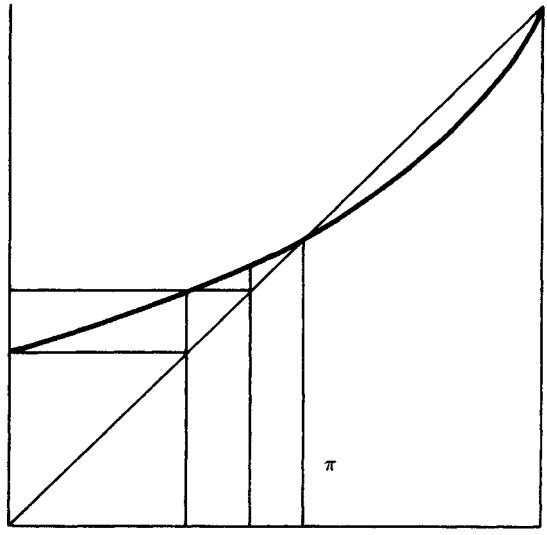
The function f is analytic on (0,1), and is non-decreasing and convex (of non-decreasing slope). Also, $f(1) = 1$ and $f(0) = \mathsf{P}(X = 0) > 0$. The slope $f'(1)$ of f at 1 is $\mu = \mathsf{E}(X)$. The celebrated pictures opposite now make the following Theorem obvious.

THEOREM

If $\mathsf{E}(X) > 1$, *then the extinction probability* π *is the unique root of the equation* $\pi = f(\pi)$ *which lies strictly between 0 and 1. If* $\mathsf{E}(X) \leq 1$, *then* $\pi = 1$.



Case 1: subcritical, $\mu = f'(1) < 1$. Clearly, $\pi = 1$.
The *critical case* $\mu = 1$ has a similar picture.



Case 2: supercritical, $\mu = f'(1) > 1$. Now, $\pi < 1$.

0.5. Pause for thought: measure

Now that we have finished revising what introductory courses on probability theory say about branching-process theory, let us think about why we must find a more precise language. To be sure, the claim at (0.4,b) that

(a) $$\pi = \uparrow \lim \pi_n$$

is intuitively plausible, but how could one *prove* it? We certainly cannot prove it at present because we have no means of stating with pure-mathematical precision what it is supposed to mean. Let us discuss this further.

Back in Section 0.2, we said 'Suppose that we are given a doubly infinite sequence $\{X_r^{(m)} : m, r \in \mathbf{N}\}$ of independent identically distributed random variables each with the same distribution as X'. What does this mean? A random variable is a (certain kind of) function on a sample space Ω. We could follow elementary theory in taking Ω to be the set of all outcomes, in other words, taking Ω to be the Cartesian product

$$\Omega = \prod_{r,s} \mathbf{Z}^+,$$

the typical element ω of Ω being

$$\omega = (\omega_s^{(r)} : r \in \mathbf{N}, s \in \mathbf{N}),$$

and then setting $X_s^{(r)}(\omega) = \omega_s^{(r)}$. Now Ω is an uncountable set, so that we are outside the 'combinatorial' context which makes sense of π_n in the elementary theory. Moreover, if one assumes the Axiom of Choice, one can *prove* that it is impossible to assign to *all* subsets of Ω a probability satisfying the 'intuitively obvious' axioms and making the X's IID RVs with the correct common distribution. So, we have to know that the set of ω corresponding to the event 'extinction occurs' *is* one to which one can uniquely assign a probability (which will then provide a definition of π). Even then, we have to prove (a).

Example. Consider for a moment what is in some ways a bad attempt to construct a 'probability theory'. Let $\mathcal{C}$ be the class of subsets C of $\mathbf{N}$ for which the 'density'

$$\rho(C) := \lim_{n \uparrow \infty} \sharp\{k : 1 \leq k \leq n; k \in C\}$$

exists. Let $C_n := \{1, 2, \ldots, n\}$. Then $C_n \in \mathcal{C}$ and $C_n \uparrow \mathbf{N}$ in the sense that $C_n \subseteq C_{n+1}, \forall n$ and also $\bigcup C_n = \mathbf{N}$. However, $\rho(C_n) = 0, \forall n$, but $\rho(\mathbf{N}) = 1$.

Hence the logic which will allow us correctly to deduce (a) from the fact that

$$\{Z_n = 0\} \uparrow \{\text{extinction occurs}\}$$

fails for the $(\mathbf{N}, \mathcal{C}, \rho)$ set-up: $(\mathbf{N}, \mathcal{C}, \rho)$ is not 'a probability triple'. □

There *are* problems. Measure theory resolves them, but provides a huge bonus in the form of much deeper results such as the Martingale Convergence Theorem which we now take a first look at – at an intuitive level, I hasten to add.

0.6. Our first martingale

Recall from (0.2,b) that

$$Z_{n+1} = X_1^{(n+1)} + \cdots + X_{Z_n}^{(n+1)},$$

where the $X^{(n+1)}$ variables are independent of the values $Z_1, Z_2, \ldots, Z_n$. It is clear from this that

$$\mathsf{P}(Z_{n+1} = j | Z_0 = i_0, Z_1 = i_1, \ldots, Z_n = i_n) = \mathsf{P}(Z_{n+1} = j | Z_n = i_n),$$

a result which you will probably recognize as stating that the process $Z = (Z_n : n \geq 0)$ *is a Markov chain.* We therefore have

$$\begin{aligned}\mathsf{E}(Z_{n+1} | Z_0 = i_0, Z_1 = i_1, \ldots, Z_n = i_n) &= \sum_j j \mathsf{P}(Z_{n+1} = j | Z_n = i_n) \\ &= \mathsf{E}(Z_{n+1} | Z_n = i_n),\end{aligned}$$

or, in a condensed and better notation,

$$\text{(a)} \qquad \mathsf{E}(Z_{n+1} | Z_0, Z_1, \ldots, Z_n) = \mathsf{E}(Z_{n+1} | Z_n).$$

Of course, it is intuitively obvious that

$$\text{(b)} \qquad \mathsf{E}(Z_{n+1} | Z_n) = \mu Z_n,$$

because each of the Z_n animals in the n^{th} generation has on average μ children. We can confirm result (b) by differentiating the result

$$\mathsf{E}(\theta^{Z_{n+1}} | Z_n) = f(\theta)^{Z_n}$$

with respect to θ and setting $\theta = 1$.

Now define

(c) $$M_n := Z_n/\mu^n, \qquad n \geq 0.$$

Then

$$\mathsf{E}(M_{n+1}|Z_0, Z_1, \ldots, Z_n) = M_n,$$

which exactly says that

(d) *M is a martingale relative to the Z process.*

Given the history of Z up to stage n, the next value M_{n+1} of M is on average what it is now: M is 'constant on average' in this very sophisticated sense of conditional expectation given 'past' and 'present'. The true statement

(e) $$\mathsf{E}(M_n) = 1, \quad \forall n$$

is of course infinitely cruder.

A statement $\mathcal{S}$ is said to be true **almost surely** (a.s.) or **with probability** 1 if (surprise, surprise!)

$$\mathsf{P}(\mathcal{S} \text{ is true}) = 1.$$

Because our martingale M is *non-negative* ($M_n \geq 0, \forall n$), the **Martingale Convergence Theorem** implies that *it is almost surely true that*

(f) $$M_\infty := \lim M_n \quad \textit{exists.}$$

Note that if $M_\infty > 0$ for some outcome (which can happen with positive probability only when $\mu > 1$), then the statement

$$Z_n/\mu^n \to M_\infty \qquad \text{(a.s.)}$$

is a precise formulation of 'exponential growth'. A particularly fascinating question is: *suppose that $\mu > 1$; what is the behaviour of Z conditional on the value of M_∞?*

0.7. Convergence (or not) of expectations

We know that $M_\infty := \lim M_n$ exists with probability 1, and that $\mathsf{E}(M_n) = 1$, $\forall n$. We might be tempted to believe that $\mathsf{E}(M_\infty) = 1$. However, we already know that if $\mu \leq 1$, then, almost surely, the process dies out and M_n is eventually 0. Hence

(a) *if $\mu \leq 1$, then $M_\infty = 0$* (a.s.) *and*

$$0 = \mathsf{E}(M_\infty) \neq \lim \mathsf{E}(M_n) = 1.$$

This is an excellent example to keep in mind when we come to study *Fatou's Lemma*, valid for any sequence (Y_n) of non-negative random variables:

$$\mathsf{E}(\liminf Y_n) \leq \liminf \mathsf{E}(Y_n).$$

What is 'going wrong' at (a) is that (when $\mu \leq 1$) for large n, the chances are that M_n will be large if M_n is not 0 and, very roughly speaking, this large value times its small probability will keep $\mathsf{E}(M_n)$ at 1. See the concrete examples in Section 0.9.

Of course, it is very important to know when

(b) $$\lim \mathsf{E}(\cdot) = \mathsf{E}(\lim \cdot),$$

and we do spend quite a considerable time studying this. The best general theorems are rarely good enough to get the best results for concrete problems, as is evidenced by the fact that

(c) $$\mathsf{E}(M_\infty) = 1 \textit{ if and only if } \text{both } \mu > 1 \textit{ and } \mathsf{E}(X \log X) < \infty,$$

where X is the typical number of children. Of course $0 \log 0 = 0$. If $\mu > 1$ and $\mathsf{E}(X \log X) = \infty$, then, even though the process may not die out, $M_\infty = 0$, a.s.

0.8. Finding the distribution of M_∞

Since $M_n \to M_\infty$ (a.s.), it is obvious that for $\lambda > 0$,

$$\exp(-\lambda M_n) \to \exp(-\lambda M_\infty) \qquad \text{(a.s.)}$$

Now since each $M_n \geq 0$, the whole sequence $(\exp(-\lambda M_n))$ is bounded in absolute value by the constant 1, independently of the outcome of our experiment. The *Bounded Convergence Theorem* says that we *can* now assert what we would wish:

(a) $$\mathsf{E} \exp(-\lambda M_\infty) = \lim \mathsf{E} \exp(-\lambda M_n).$$

Since $M_n = Z_n/\mu^n$ and $\mathsf{E}(\theta^{Z_n}) = f_n(\theta)$, we have

(b) $$\mathsf{E} \exp(-\lambda M_n) = f_n(\exp(-\lambda/\mu^n)),$$

so that, in principle (if very rarely in practice), we can calculate the left-hand side of (a). However, for a non-negative random variable Y, *the distribution function* $y \mapsto \mathsf{P}(Y \leq y)$ *is completely determined by the map*

$$\lambda \mapsto \mathsf{E} \exp(-\lambda Y) \quad \textit{on} \quad (0, \infty).$$

Hence, in principle, we can find the distribution of M_∞.

We have seen that the real problem is to calculate the function

$$L(\lambda) := \mathsf{E} \exp(-\lambda M_\infty).$$

Using (b), the fact that $f_{n+1} = f \circ f_n$, and the continuity of L (another consequence of the Bounded Convergence Theorem), you can immediately establish the functional equation:

(c) $$L(\lambda\mu) = f(L(\lambda)).$$

0.9. Concrete example

This concrete example is just about the only one in which one can calculate everything explicitly, but, in the way of mathematics, it is useful in many contexts.

We take the 'typical number of children' X to have a *geometric distribution*:

(a) $$\mathsf{P}(X = k) = pq^k \quad (k \in \mathbf{Z}^+),$$

where

$$0 < p < 1, \quad q := 1 - p.$$

Then, as you can easily check,

(b) $$f(\theta) = \frac{p}{1 - q\theta}, \quad \mu = \frac{q}{p},$$

and

$$\pi = \begin{cases} p/q & \text{if } q > p, \\ 1 & \text{if } q \leq p. \end{cases}$$

To calculate $f \circ f \circ \ldots \circ f$, we use a device familiar from the geometry of the upper half-plane. If

$$G = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}$$

is a non-singular 2×2 matrix, define the fractional linear transformation:

(c) $$G(\theta) = \frac{g_{11}\theta + g_{12}}{g_{21}\theta + g_{22}}.$$

Then you can check that if H is another such matrix, then

$$G(H(\theta)) = (GH)(\theta),$$

so that composition of fractional linear transformations corresponds to matrix multiplication.

Suppose that $p \neq q$. Then, by the $S^{-1}AS = \Lambda$ method, for example, we find that the n^{th} power of the matrix corresponding to f is

$$\begin{pmatrix} 0 & p \\ -q & 1 \end{pmatrix}^n = (q-p)^{-1} \begin{pmatrix} 1 & p \\ 1 & q \end{pmatrix} \begin{pmatrix} p^n & 0 \\ 0 & q^n \end{pmatrix} \begin{pmatrix} q & -p \\ -1 & 1 \end{pmatrix},$$

so that

(d) $$f_n(\theta) = \frac{p\mu^n(1-\theta) + q\theta - p}{q\mu^n(1-\theta) + q\theta - p}.$$

If $\mu = q/p \leq 1$, then $\lim_n f_n(\theta) = 1$, corresponding to the fact that the process dies out.

Suppose now that $\mu > 1$. Then you can easily check that, for $\lambda > 0$,

$$\begin{aligned} L(\lambda) :&= \mathsf{E}\exp(-\lambda M_\infty) = \lim f_n(\exp(-\lambda/\mu^n)) \\ &= \frac{p\lambda + q - p}{q\lambda + q - p} \\ &= \pi e^{-\lambda \cdot 0} + \int_0^\infty (1-\pi)^2 e^{-\lambda x} e^{-(1-\pi)x} dx, \end{aligned}$$

from which we deduce that

and
$$\mathsf{P}(M_\infty = 0) = \pi,$$
$$\mathsf{P}(x < M_\infty < x + dx) = (1-\pi)^2 e^{-(1-\pi)x} dx \quad (x > 0),$$

or, better,

$$\mathsf{P}(M_\infty > x) = (1-\pi)e^{-(1-\pi)x} \qquad (x > 0).$$

Suppose that $\mu < 1$. In this case, it is interesting to ask: what is the distribution of Z_n conditioned by $Z_n \neq 0$? We find that

$$\mathsf{E}(\theta^{Z_n} | Z_n \neq 0) = \frac{f_n(\theta) - f_n(0)}{1 - f_n(0)} = \frac{\alpha_n \theta}{1 - \beta_n \theta},$$

where

$$\alpha_n = \frac{p-q}{p-q\mu^n}, \quad \beta_n = \frac{q - q\mu^n}{p - q\mu^n},$$

so $0 < \alpha_n < 1$ and $\alpha_n + \beta_n = 1$. As $n \to \infty$, we see that

$$\alpha_n \to 1 - \mu, \quad \beta_n \to \mu,$$

so (this is justified)

(e) $$\lim_{n\to\infty} \mathbf{P}(Z_n = k | Z_n \neq 0) = (1-\mu)\mu^{k-1} \qquad (k \in \mathbf{N}).$$

Suppose that $\mu = 1$. You can show by induction that

$$f_n(\theta) = \frac{n - (n-1)\theta}{(n+1) - n\theta},$$

and that

$$\mathbf{E}(e^{-\lambda Z_n/n} | Z_n \neq 0) \to 1/(1+\lambda),$$

corresponding to

(f) $$\mathbf{P}(Z_n/n > x | Z_n \neq 0) \to e^{-x}, \qquad x > 0.$$

'The Fatou factor'

We know that when $\mu \leq 1$, we have $\mathbf{E}(M_n) = 1$, $\forall n$, but $\mathbf{E}(M_\infty) = 0$. Can we get some insight into this?

First consider the case when $\mu < 1$. Result (e) makes it plausible that for large n,

$$\mathbf{E}(Z_n | Z_n \neq 0) \text{ is roughly } (1-\mu)\sum k\mu^{k-1} = 1/(1-\mu).$$

We know that

$$\mathbf{P}(Z_n \neq 0) = 1 - f_n(0) \text{ is roughly } (1-\mu)\mu^n,$$

so we should have (roughly)

$$\begin{aligned} \mathbf{E}(M_n) &= \mathbf{E}\left(\frac{Z_n}{\mu^n} \middle| Z_n \neq 0\right) \mathbf{P}(Z_n \neq 0) \\ &= \frac{1}{(1-\mu)\mu^n}(1-\mu)\mu^n = 1, \end{aligned}$$

which might help explain how the 'balance' $\mathbf{E}(M_n) = 1$ is achieved by big values times small probabilities.

Now consider the case when $\mu = 1$. Then

$$\mathbf{P}(Z_n \neq 0) = 1/(n+1),$$

and, from (f), Z_n/n conditioned by $Z_n \neq 0$ is roughly exponential with mean 1, so that $M_n = Z_n$ conditioned by $Z_n \neq 0$ is on average of size about n, the correct order of magnitude for balance.

Warning. We have just been using for 'correct intuitive explanations' exactly the type of argument which might have misled us into thinking that $\mathbf{E}(M_\infty) = 1$ in the first place. But, of course, the result

$$\mathbf{E}(M_n) = \mathbf{E}(M_n | Z_n \neq 0)\mathbf{P}(Z_n \neq 0) = 1$$

is a matter of obvious *fact.*

PART A: FOUNDATIONS

Chapter 1
Measure Spaces

1.0. Introductory remarks

Topology is about *open* sets. The characterizing property of a *continuous* function f is that the inverse image $f^{-1}(G)$ of an open set G is open.

Measure theory is about *measurable* sets. The characterizing property of a *measurable* function f is that the inverse image $f^{-1}(A)$ of any measurable set is measurable.

In topology, one axiomatizes the notion of 'open set', insisting in particular that the union of *any* collection of open sets is open, and that the intersection of a *finite* collection of open sets is open.

In measure theory, one axiomatizes the notion of 'measurable set', insisting that the union of a *countable* collection of measurable sets is measurable, and that the intersection of a *countable* collection of measurable sets is also measurable. Also, the complement of a measurable set must be measurable, and the whole space must be measurable. Thus the measurable sets form a σ-algebra, a structure stable (or 'closed') under countably many set operations. Without the insistence that 'only countably many operations are allowed', measure theory would be self-contradictory – a point lost on certain philosophers of probability.

The probability that a point chosen at random on the surface of the unit sphere S^2 in $\mathbf{R}^3$ falls into the subset F of S^2 is just the area of F divided by the total area 4π. What could be easier?

However, Banach and Tarski showed (see Wagon (1985)) that if the Axiom of Choice is assumed, as it is throughout conventional mathematics, then there exists a subset F of the unit sphere S^2 in $\mathbf{R}^3$ such that for

$3 \leq k < \infty$ (and even for $k = \infty$), S^2 is the disjoint union of k exact copies of F:

$$S^2 = \bigcup_{i=1}^{k} \tau_i^{(k)} F,$$

where each $\tau_i^{(k)}$ is a rotation. If F has an 'area', then that area must simultaneously be $4\pi/3, 4\pi/4, \ldots, 0$. The only conclusion is that the set F is *non-measurable* (not *Lebesgue* measurable): it is so complicated that one cannot assign an area to it. Banach and Tarski have not broken the Law of Conservation of Area: they have simply operated outside its jurisdiction.

Remarks. (i) Because every rotation τ has a fixed point x on S^2 such that $\tau(x) = x$, it is not possible to find a subset A of S^2 and a rotation τ such that $A \cup \tau(A) = S^2$ and $A \cap \tau(A) = \emptyset$. So, we could not have taken $k = 2$.

(ii) Banach and Tarski even proved that given any two bounded subsets A and B of $\mathbf{R}^3$ each with non-empty interior, it is possible to decompose A into a certain finite number n of disjoint pieces $A = \bigcup_{i=1}^{n} A_i$ and B into the same number n of disjoint pieces $B = \bigcup_{i=1}^{n} B_i$, in such a way that, for each i, A_i is Euclid-congruent to B_i!!! So, we can disassemble A and rebuild it as B.

(iii) Section A1.1 (optional!) in the appendix to this chapter gives an Axiom-of-Choice construction of a non-measurable subset of S^1.

This chapter introduces

σ-algebras, π-systems, and *measures*

and emphasizes *monotone-convergence properties of measures*. We shall see in later chapters that, although not all sets are measurable, it is always the case for probability theory that enough sets are measurable.

1.1. Definitions of algebra, σ-algebra

Let S be a set.

Algebra on S

A collection Σ_0 of subsets of S is called an *algebra on S* (or algebra of subsets of S) if

(i) $S \in \Sigma_0$,

(ii) $F \in \Sigma_0 \quad \Rightarrow \quad F^c := S \backslash F \in \Sigma_0$,

(iii) $F, G \in \Sigma_0 \quad \Rightarrow \quad F \cup G \in \Sigma_0$.

[Note that $\emptyset = S^c \in \Sigma_0$ and

$$F, G \in \Sigma_0 \quad \Rightarrow \quad F \cap G = (F^c \cup G^c)^c \in \Sigma_0.]$$

Thus, an algebra on S is a family of subsets of S stable under finitely many set operations.

Exercise (optional). Let $\mathcal{C}$ be the class of subsets C of $\mathbb{N}$ for which the 'density'

$$\lim_{m\uparrow\infty} m^{-1}\sharp\{k : 1 \leq k \leq m; k \in C\}$$

exists. We might like to think of this density (if it exists) as 'the probability that a number chosen at random belongs to C'. But there are many reasons why this does not conform to a proper probability theory. (We saw one in Section 0.5.) For example, you should find elements F and G in $\mathcal{C}$ for which $F \cap G \notin \mathcal{C}$.

Note on terminology ('algebra versus field'). An algebra in our sense is a true algebra in the algebraists' sense with $\cap$ as product, and symmetric difference

$$A\triangle B := (A \cup B)\backslash(A \cap B)$$

as 'sum', the underlying field of the algebra being the field with 2 elements. (This is why we prefer 'algebra of subsets' to 'field of subsets': there is no way that an algebra of subsets is a field in the algebraists' sense - unless Σ_0 is *trivial*, that is, $\Sigma_0 = \{S, \emptyset\}$.)

σ-algebra on S

A collection Σ of subsets of S is called a *σ-algebra on S* (or σ-algebra of subsets of S) if Σ is an algebra on S such that *whenever* $F_n \in \Sigma$ $(n \in \mathbb{N})$, *then*

$$\bigcup_n F_n \in \Sigma.$$

[Note that if Σ is a σ-algebra on S and $F_n \in \Sigma$ for $n \in \mathbb{N}$, then

$$\bigcap_n F_n = (\bigcup_n F_n^c)^c \in \Sigma.]$$

Thus, a σ-algebra on S is a family of subsets of S 'stable under any countable collection of set operations'.

Note. Whereas it is usually possible to write in 'closed form' the typical element of many of the algebras of sets which we shall meet (see Section 1.8 below for a first example), it is usually impossible to write down the typical element of a σ-algebra. This is the reason for our concentrating where possible on the much simpler 'π-systems'.

Measurable space

A pair (S, Σ), where S is a set and Σ is a σ-algebra on S, is called a *measurable space*. An element of Σ is called a Σ-measurable subset of S.

$\sigma(\mathcal{C})$, σ-algebra generated by a class $\mathcal{C}$ of subsets

Let $\mathcal{C}$ be a class of subsets of S. Then $\sigma(\mathcal{C})$, the *σ-algebra generated by $\mathcal{C}$*, is the smallest σ-algebra Σ on S such that $\mathcal{C} \subseteq \Sigma$. It is the intersection of all σ-algebras on S which have $\mathcal{C}$ as a subclass. (Obviously, the class of *all* subsets of S is a σ-algebra which extends $\mathcal{C}$.)

1.2. Examples. Borel σ-algebras, $\mathcal{B}(S)$, $\mathcal{B} = \mathcal{B}(\mathbf{R})$

Let S be a topological space.

$\mathcal{B}(S)$

$\mathcal{B}(S)$, the Borel σ-algebra on S, is the σ-algebra generated by the family of open subsets of S. With slight abuse of notation,

$$\mathcal{B}(S) := \sigma(\text{open sets}).$$

$\mathcal{B} := \mathcal{B}(\mathbf{R})$

It is standard shorthand that $\mathcal{B} := \mathcal{B}(\mathbf{R})$.

The σ-algebra $\mathcal{B}$ is the most important of all σ-algebras. Every subset of $\mathbf{R}$ which you meet in everyday use is an element of $\mathcal{B}$; and indeed it is difficult (but possible!) to find a subset of $\mathbf{R}$ constructed explicitly (without the Axiom of Choice) which is not in $\mathcal{B}$.

Elements of $\mathcal{B}$ can be quite complicated. However, the collection

$$\pi(\mathbf{R}) := \{(-\infty, x] : x \in \mathbf{R}\}$$

(not a standard notation) is very easy to understand, and it is often the case that all we need to know about $\mathcal{B}$ is that

(a) $$\mathcal{B} = \sigma(\pi(\mathbf{R})).$$

Proof of (a). For each x in $\mathbf{R}$, $(-\infty, x] = \bigcap_{n \in \mathbf{N}} (-\infty, x + n^{-1})$, so that as a *countable* intersection of open sets, the set $(-\infty, x]$ is in $\mathcal{B}$.

All that remains to be proved is that every open subset G of $\mathbf{R}$ is in $\sigma(\pi(\mathbf{R}))$. But every such G is a *countable* union of open intervals, so we need only show that, for $a, b \in \mathbf{R}$ with $a < b$,

$$(a, b) \in \sigma(\pi(\mathbf{R})).$$

But, for any u with $u > a$,

$$(a, u] = (-\infty, u] \cap (-\infty, a]^c \in \sigma(\pi(\mathbf{R})),$$

and since, for $\varepsilon = \frac{1}{2}(b - a)$,

$$(a, b) = \bigcup_n (a, b - \varepsilon n^{-1}],$$

we see that $(a, b) \in \sigma(\pi(\mathbf{R}))$, and the proof is complete. □

1.3. Definitions concerning set functions

Let S be a set, let Σ_0 be an algebra on S, and let μ_0 be a non-negative *set function*

$$\mu_0 : \Sigma_0 \to [0, \infty].$$

Additive

Then μ_0 is called *additive* if $\mu_0(\emptyset) = 0$ and, for $F, G \in \Sigma_0$,

$$F \cap G = \emptyset \quad \Rightarrow \quad \mu_0(F \cup G) = \mu_0(F) + \mu_0(G).$$

Countably additive

The map μ_0 is called *countably additive* (or σ-additive) if $\mu(\emptyset) = 0$ and whenever $(F_n : n \in \mathbb{N})$ is a sequence of disjoint sets in Σ_0 with union $F = \bigcup F_n$ in Σ_0 (note that this is an assumption since Σ_0 need not be a σ-algebra), then

$$\mu_0(F) = \sum_n \mu_0(F_n).$$

Of course (why?), a countably additive set function is additive.

1.4. Definition of measure space

Let (S, Σ) be a measurable space, so that Σ is a σ-algebra on S.

A map

$$\mu : \Sigma \to [0, \infty].$$

is called a *measure* on (S, Σ) if μ is countably additive. The triple (S, Σ, μ) is then called a *measure space.*

1.5. Definitions concerning measures

Let (S, Σ, μ) be a measure space. Then μ (or indeed the measure space (S, Σ, μ)) is called

finite

if $\mu(S) < \infty$,

σ-finite

if there is a sequence $(S_n : n \in \mathbb{N})$ of elements of Σ such that

$$\mu(S_n) < \infty \ (\forall n \in \mathbb{N}) \text{ and } \bigcup S_n = S.$$

Warning. Intuition is usually OK for finite measures, and adapts well for σ-finite measures. However, measures which are not σ-finite can be crazy; fortunately, there are no such measures in this book.

Probability measure, probability triple

Our measure μ is called a *probability measure* if

$$\mu(S) = 1,$$

and (S, Σ, μ) is then called a *probability triple.*

μ-null element of Σ, almost everywhere (a.e.)

An element F of Σ is called *μ-null* if $\mu(F) = 0$. A statement $\mathcal{S}$ about points s of S is said to hold *almost everywhere* (a.e.) if

$$F := \{s : \mathcal{S}(s) \text{ is false}\} \in \Sigma \text{ and } \mu(F) = 0.$$

1.6. LEMMA. Uniqueness of extension, π-systems

Moral: σ-algebras are 'difficult', but π-systems are 'easy'; so we aim to work with the latter.

►(a) *Let S be a set. Let $\mathcal{I}$ be a* **π-system on** *S, that is, a family of subsets of S* **stable under finite intersection**:

$$I_1, I_2 \in \mathcal{I} \quad \Rightarrow \quad I_1 \cap I_2 \in \mathcal{I}.$$

Let $\Sigma := \sigma(\mathcal{I})$. Suppose that μ_1 and μ_2 are measures on (S, Σ) such that $\mu_1(S) = \mu_2(S) < \infty$ and $\mu_1 = \mu_2$ on $\mathcal{I}$. Then

$$\mu_1 = \mu_2 \text{ on } \Sigma.$$

►►(b) **Corollary. If two probability measures agree on a π-system, then they agree on the σ-algebra generated by that π-system.**

The example $\mathcal{B} = \sigma(\pi(\mathbf{R}))$ is of course the most important example of the $\Sigma = \sigma(\mathcal{I})$ in the theorem.

This result will play an important rôle. Indeed, it will be applied more frequently than will the celebrated existence result in Section 1.7. Because of this, the proof of Lemma 1.6 given in Sections A1.2-1.4 of the appendix to this chapter should perhaps be consulted – but read the remainder of this chapter first.

1.7. THEOREM. Carathéodory's Extension Theorem

► *Let S be a set, let Σ_0 be an algebra on S, and let*

$$\Sigma := \sigma(\Sigma_0).$$

If μ_0 is a countably additive map $\mu_0 : \Sigma_0 \to [0, \infty]$, then there exists a measure μ on (S, Σ) such that

$$\mu = \mu_0 \text{ on } \Sigma_0.$$

If $\mu_0(S) < \infty$, then, by Lemma 1.6, this extension is unique – an algebra is a π-system!

In a sense, this result should have more ► signs than any other, for without it we could not construct any interesting models. However, once we have our model, we make no further use of the theorem.

The proof of this result given in Sections A1.5-1.8 of the appendix is there for completeness. It will do no harm to assume the result for this course. Let us now see how the theorem is *used.*

1.8. Lebesgue measure Leb **on** $((0,1], \mathcal{B}(0,1])$

Let $S = (0,1]$. For $F \subseteq S$, say that $F \in \Sigma_0$ if F may be written as a finite union

$$(*) \qquad F = (a_1, b_1] \cup \ldots \cup (a_r, b_r]$$

where $r \in \mathbb{N}$, $0 \leq a_1 \leq b_1 \leq \cdots \leq a_r \leq b_r \leq 1$. Then Σ_0 is an algebra on $(0,1]$ and

$$\Sigma := \sigma(\Sigma_0) = \mathcal{B}(0,1].$$

(We write $\mathcal{B}(0,1]$ instead of $\mathcal{B}((0,1])$.) For F as at $(*)$, let

$$\mu_0(F) = \sum_{k \leq r} (b_k - a_k).$$

Then μ_0 is well-defined and additive on Σ_0 (this is easy). Moreover, μ_0 is *countably* additive on Σ_0. (This is not trivial. See Section A1.9.) Hence, by Theorem 1.7, there exists a unique measure μ on $((0,1], \mathcal{B}(0,1])$ extending μ_0 on Σ_0. This measure μ is called *Lebesgue measure* on $((0,1], \mathcal{B}(0,1])$ or (loosely) Lebesgue measure on (0,1]. We shall often denote μ by Leb. Lebesgue measure (still denoted by Leb) on $([0,1], \mathcal{B}[0,1])$ is of course obtained by a trivial modification, the set $\{0\}$ having Lebesgue measure 0. Of course, Leb makes precise the concept of length.

In a similar way, we can construct (σ-finite) Lebesgue measure (which we also denote by Leb) on $\mathbf{R}$ (more strictly, on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$.

1.9. LEMMA. Elementary inequalities

Let (S, Σ, μ) be a measure space. Then

(a) $\mu(A \cup B) \leq \mu(A) + \mu(B) \qquad (A, B \in \Sigma)$,

►(b) $\mu(\bigcup_{i \leq n} F_i) \leq \sum_{i \leq n} \mu(F_i) \qquad (F_1, F_2, \ldots, F_n \in \Sigma)$.

Furthermore, if $\mu(S) < \infty$, then

(c) $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B) \qquad (A, B \in \Sigma)$,

(d) **(inclusion-exclusion formula)**: *for $F_1, F_2, \ldots, F_n \in \Sigma$,*

$$\mu(\bigcup_{i \leq n} F_i) = \sum_{i \leq n} \mu(F_i) - \sum \sum_{i<j \leq n} \mu(F_i \cap F_j)$$

$$+ \sum \sum \sum_{i<j<k \leq n} \mu(F_i \cap F_j \cap F_k) - \cdots + (-1)^{n-1} \mu(F_1 \cap F_2 \cap \ldots \cap F_n),$$

successive partial sums alternating between over- and under-estimates.

You will surely have seen some version of these results previously. Result (c) is obvious because $A \cup B$ is the disjoint union $A \cup (B \backslash (A \cap B))$. But (c)⇒(a)⇒(b) – check that 'infinities do not matter'. You can deduce (d) from (c) by induction, but, as we shall see later, the neat way to prove (d) is by integration.

1.10. LEMMA. Monotone-convergence properties of measures

These results are often needed for making things rigorous. (Peep ahead to the 'Monkey typing Shakespeare' Section 4.9.) Again, let (S, Σ, μ) be a measure space.

►(a) **If $F_n \in \Sigma \quad (n \in \mathbb{N})$ and $F_n \uparrow F$, then $\mu(F_n) \uparrow \mu(F)$.**

Notes. $F_n \uparrow F$ means: $F_n \subseteq F_{n+1} \quad (\forall n \in \mathbb{N}), \quad \bigcup F_n = F$. Result (a) is *the* fundamental property of measure.

Proof of (a). Write $G_1 := F_1, \qquad G_n := F_n \backslash F_{n-1} \quad (n \geq 2)$. Then the sets $G_n \quad (n \in \mathbb{N})$ are *disjoint*, and

$$\mu(F_n) = \mu(G_1 \cup G_2 \cup \ldots \cup G_n) = \sum_{k \leq n} \mu(G_k) \uparrow \sum_{k < \infty} \mu(G_k) = \mu(F). \quad \square$$

Application. In a proper formulation of the branching-process example of Chapter 0,

$$\{Z_n = 0\} \uparrow \{\text{extinction occurs}\}, \text{ so that } \pi_n \uparrow \pi.$$

(A proper formulation of the branching-process example *will* be given later.)

►(b) **If $G_n \in \Sigma$, $G_n \downarrow G$ and $\mu(G_k) < \infty$ for some k, then $\mu(G_n) \downarrow \mu(G)$.**
Proof of (b). For $n \in \mathbf{N}$, let $F_n := G_k \backslash G_{k+n}$, and now apply part (a). □

Example - *to indicate what can 'go wrong'.* For $n \in \mathbf{N}$, let

$$H_n := (n, \infty).$$

Then $\text{Leb}(H_n) = \infty, \forall n$, but $H_n \downarrow \emptyset$.

►(c) *The union of a countable number of μ-null sets is μ-null.*
This is a trivial corollary of results (1.9,b) and (1.10,a).

1.11. Example/Warning

Let (S, Σ, μ) be $([0,1], \mathcal{B}[0,1], \text{Leb})$. Let $\varepsilon(k)$ be a sequence of strictly positive numbers such that $\varepsilon(k) \downarrow 0$. For a single point x of S, we have

$$\text{(a)} \qquad \{x\} \subseteq (x - \varepsilon(k), x + \varepsilon(k)) \cap S,$$

so that for every k, $\mu(\{x\}) < 2\varepsilon(k)$, and so $\mu(\{x\}) = 0$. That $\{x\}$ is $\mathcal{B}(S)$-measurable follows because $\{x\}$ is the intersection of the countable number of open subsets of S on the right-hand side of (a).

Let $V = \mathbf{Q} \cap [0,1]$, the set of rationals in [0,1]. Since V is a countable union of singletons: $V = \{v_n : n \in \mathbf{N}\}$, it is clear that V is $\mathcal{B}[0,1]$-measurable and that $\text{Leb}(V) = 0$. We can include V in an open subset of S of measure at most $4\varepsilon(k)$ as follows:

$$V \subseteq G_k = \bigcup_{n \in \mathbf{N}} [(v_n - \varepsilon(k)2^{-n}, v_n + \varepsilon(k)2^{-n}) \cap S] =: \bigcup_n I_{n,k}.$$

Clearly, $H := \bigcap_k G_k$ satisfies $\text{Leb}(H) = 0$ and $V \subseteq H$. Now, it is a consequence of the *Baire category theorem* (see the appendix to this chapter) that *H is uncountable*, so

(b) *the set H is an uncountable set of measure 0; moreover,*

$$H = \bigcap_k \bigcup_n I_{n,k} \neq \bigcup_n \bigcap_k I_{n,k} = V.$$

Throughout the subject, we have to be careful about interchanging orders of operations.

Chapter 2

Events

2.1. Model for experiment: $(\Omega, \mathcal{F}, \mathbf{P})$

A model for an experiment involving randomness takes the form of a *probability triple* $(\Omega, \mathcal{F}, \mathbf{P})$ in the sense of Section 1.5.

Sample space

Ω is a set called the *sample space.*

Sample point

A point ω of Ω is called a *sample point.*

Event

The σ-algebra $\mathcal{F}$ on Ω is called the family of events, so that an *event* is an element of $\mathcal{F}$, that is, an $\mathcal{F}$-measurable subset of Ω.

By definition of probability triple, $\mathbf{P}$ is a probability measure on $(\Omega, \mathcal{F})$.

2.2. The intuitive meaning

Tyche, Goddess of Chance, chooses a point ω of Ω 'at random' according to the law $\mathbf{P}$ in that, for F in $\mathcal{F}$, $\mathbf{P}(F)$ represents the 'probability' (in the sense understood by our intuition) that the point ω chosen by Tyche belongs to F.

The chosen point ω determines the outcome of the experiment. Thus there is a map

$$\Omega \to \text{set of outcomes},$$
$$\omega \mapsto \text{outcome}.$$

There is no reason why this 'map' (the co-domain lies in our intuition!) should be one-one. Often it is the case that although there is some obvious 'minimal' or 'canonical' model for an experiment, it is better to use some richer model. (For example, we can read off many properties of coin tossing by imbedding the associated random walk in a Brownian motion.)

2.3. Examples of $(\Omega, \mathcal{F})$ pairs

We leave the question of assigning probabilities until later.

(a) *Experiment: Toss coin twice.* We can take

$$\Omega = \{HH, HT, TH, TT\}, \quad \mathcal{F} = \mathcal{P}(\Omega) := \text{set of all subsets of } \Omega.$$

In this model, the intuitive event 'At least one head is obtained' is described by the mathematical event (element of $\mathcal{F}$) $\{HH, HT, TH\}$.

(b) *Experiment: Toss coin infinitely often.* We can take

$$\Omega = \{H, T\}^{\mathbf{N}},$$

so that a typical point ω of Ω is a sequence

$$\omega = (\omega_1, \omega_2, \ldots), \quad \omega_n \in \{H, T\}.$$

We certainly wish to speak of the intuitive event '$\omega_n = W$', where $W \in \{H, T\}$, and it is natural to choose

$$\mathcal{F} = \sigma(\{\omega \in \Omega : \omega_n = W\} : n \in \mathbf{N}, W \in \{H, T\}).$$

Although $\mathcal{F} \neq \mathcal{P}(\Omega)$ (accept this!), it turns out that $\mathcal{F}$ is big enough; for example, we shall see in Section 3.7 that the truth set

$$F = \left\{\omega : \frac{\sharp(k \leq n : \omega_k = H)}{n} \to \frac{1}{2}\right\}$$

of the statement

$$\frac{\text{number of heads in } n \text{ tosses}}{n} \to \frac{1}{2}$$

is an element of $\mathcal{F}$.

Note that we can use the current model as a more informative model for the experiment in (a), using the map $\omega \mapsto (\omega_1, \omega_2)$ of sample points to outcomes.

(c) *Experiment: Choose a point between* 0 *and* 1 *uniformly at random.* Take $\Omega = [0, 1], \mathcal{F} = \mathcal{B}[0, 1], \omega$ signifying the point chosen. In this case, we obviously take $\mathbf{P}$ =Leb. The sense in which this model contains model (b) for the case of a fair coin will be explained later.

2.4. Almost surely (a.s.)

►A statement S about outcomes is said to be true *almost surely (a.s.)*, or *with probability* 1 *(w.p.1)*, if

$$F := \{\omega : S(\omega) \text{ is true}\} \in \mathcal{F} \text{ and } \mathsf{P}(F) = 1.$$

(a) **Proposition.** *If* $F_n \in \mathcal{F} \quad (n \in \mathbb{N})$ *and* $\mathsf{P}(F_n) = 1, \forall n$, *then*

$$\mathsf{P}(\bigcap_n F_n) = 1.$$

Proof. $\mathsf{P}(F_n^c) = 0, \forall n$, so, by Lemma 1.10(c), $\mathsf{P}(\bigcup_n F_n^c) = 0$. But $\bigcap F_n = (\bigcup F_n^c)^c$. □

(b) *Something to think about.* Some distinguished philosophers have tried to develop probability without measure theory. One of the reasons for difficulty is the following.

When the discussion (2.3,b) is extended to define the appropriate probability measure for fair coin tossing, the Strong Law of Large Numbers (SLLN) states that $F \in \mathcal{F}$ and $\mathsf{P}(F) = 1$, where F, the truth set of the statement 'proportion of heads in n tosses $\to \frac{1}{2}$', is defined formally in (2.3,b).

Let $\mathcal{A}$ be the set of all maps $\alpha : \mathbb{N} \to \mathbb{N}$ such that $\alpha(1) < \alpha(2) < \ldots$. For $\alpha \in \mathcal{A}$, let

$$F_\alpha = \left\{\omega : \frac{\sharp(k \le n : \omega_{\alpha(k)} = H)}{n} \to \frac{1}{2}\right\}$$

the 'truth set of the Strong Law for the subsequence α'. Then, of course, we have $\mathsf{P}(F_\alpha) = 1, \forall \alpha \in \mathcal{A}$.

Exercise. Prove that

$$\bigcap_{\alpha \in \mathcal{A}} F_\alpha = \emptyset.$$

(*Hint.* For any given ω, find an α ... .)

The moral is that the concept of 'almost surely' gives us (i) absolute precision, but also (ii) enough flexibility to avoid the self-contradictions into which those innocent of measure theory too easily fall. (Of course, since philosophers are pompous where we are precise, they are thought to think deeply ... .)

2.5. Reminder: lim sup, lim inf, ↓ lim, etc.

(a) Let $(x_n : n \in \mathbb{N})$ be a sequence of real numbers. We define

$$\limsup x_n := \inf_m \left\{\sup_{n \ge m} x_n\right\} = \downarrow \lim_m \left\{\sup_{n \ge m} x_n\right\} \in [-\infty, \infty].$$

Obviously, $y_m := \sup_{n \geq m} x_n$ is monotone non-increasing in m, so that the limit of the sequence y_m exists in $[-\infty, \infty]$. The use of $\uparrow\lim$ or $\downarrow\lim$ to signify monotone limits will be handy, as will $y_n \downarrow y_\infty$ to signify $y_\infty = \downarrow \lim y_n$.

(b) Analogously,

$$\liminf x_n := \sup_m \left\{ \inf_{n \geq m} x_n \right\} = \uparrow \lim_m \left\{ \inf_{n \geq m} x_n \right\} \in [-\infty, \infty].$$

(c) We have

$$x_n \text{ converges in } [-\infty, \infty] \iff \limsup x_n = \liminf x_n,$$

and then $\lim x_n = \limsup x_n = \liminf x_n$.

►(d) Note that

(i) if $z > \limsup x_n$, then

$x_n < z$ eventually (that is, for all sufficiently large n)

(ii) if $z < \limsup x_n$, then

$x_n > z$ infinitely often (that is, for infinitely many n).

2.6. Definitions. $\limsup E_n, (E_n, \text{ i.o.})$

The event (in the rigorous formulation: the truth set of the statement)

'number of heads/ number of tosses $\to \frac{1}{2}$'

is built out of simple events such as 'the n^{th} toss results in heads' in a rather complicated way. We need a systematic method of being able to handle complicated combinations of events. The idea of taking lim infs and lim sups of sets provides what is required.

It might be helpful to note the tautology that, if E is an event, then

$$E = \{\omega : \omega \in E\}.$$

Suppose now that $(E_n : n \in \mathbb{N})$ *is a sequence of events.*

►(a) We define

$$\begin{aligned}(E_n, \text{ i.o.}) :&= (E_n \text{ infinitely often}) \\ &:= \limsup E_n := \bigcap_m \bigcup_{n \geq m} E_n \\ &= \{\omega : \text{for every } m, \quad \exists n(\omega) \geq m \text{ such that } \omega \in E_{n(\omega)}\} \\ &= \{\omega : \omega \in E_n \text{ for infinitely many } n\}.\end{aligned}$$

►(b) (**Reverse Fatou Lemma** - *needs FINITENESS of* P)

$$\mathsf{P}(\limsup E_n) \geq \limsup \mathsf{P}(E_n).$$

Proof. Let $G_m := \bigcup_{n \geq m} E_n$. Then (look at the definition in (a)) $G_m \downarrow G$, where $G := \limsup E_n$. By result (1.10,b), $\mathsf{P}(G_m) \downarrow \mathsf{P}(G)$. But, clearly,

$$\mathsf{P}(G_m) \geq \sup_{n \geq m} \mathsf{P}(E_n).$$

Hence,

$$\mathsf{P}(G) \geq \downarrow \lim_m \left\{ \sup_{n \geq m} \mathsf{P}(E_n) \right\} =: \limsup \mathsf{P}(E_n). \qquad \square$$

2.7. First Borel-Cantelli Lemma (BC1)

►► **Let $(E_n : n \in \mathbb{N})$ be a sequence of events such that $\sum_n \mathsf{P}(E_n) < \infty$. Then**

$$\mathsf{P}(\limsup E_n) = \mathsf{P}(E_n, \text{ i.o.}) = 0.$$

Proof. With the notation of (2.6,b), we have, for each m,

$$\mathsf{P}(G) \leq \mathsf{P}(G_m) \leq \sum_{n \geq m} \mathsf{P}(E_n),$$

using (1.9,b) and (1.10,a). Now let $m \uparrow \infty$. $\square$

Notes. (i) An instructive proof by integration will be given later.

(ii) Many applications of the First Borel-Cantelli Lemma will be given within this course. Interesting applications require concepts of independence, random variables, etc..

2.8. Definitions. $\liminf E_n$, $(E_n, \text{ ev})$

Again suppose that $(E_n : n \in \mathbb{N})$ is a sequence of events.

►(a) We define

$$\begin{aligned}(E_n, \text{ ev}) &:= (E_n \text{ eventually}) \\ &:= \liminf E_n := \bigcup_m \bigcap_{n \geq m} E_n \\ &= \{\omega : \text{ for some } m(\omega), \quad \omega \in E_n, \forall n \geq m(\omega)\} \\ &= \{\omega : \omega \in E_n \text{ for all large } n\}.\end{aligned}$$

(b) Note that $(E_n, \text{ ev})^c = (E_n^c, \text{ i.o.})$.

►►(c) (**Fatou's Lemma for sets** - *true for ALL measure spaces*)

$$\mathsf{P}(\liminf E_n) \leq \liminf \mathsf{P}(E_n).$$

Exercise. Prove this in analogy with the proof of result (2.6,b), using (1.10,a) rather than (1.10,b).

2.9. Exercise

For an event E, define the indicator function I_E on Ω via

$$\mathrm{I}_E(\omega) := \begin{cases} 1, & \text{if } \omega \in E, \\ 0, & \text{if } \omega \notin E. \end{cases}$$

Let $(E_n : n \in \mathbb{N})$ be a sequence of events. Prove that, for each ω,

$$\mathrm{I}_{\limsup E_n}(\omega) = \limsup \mathrm{I}_{E_n}(\omega),$$

and establish the corresponding result for lim infs.

Chapter 3

Random Variables

Let (S, Σ) be a measurable space, so that Σ is a σ-algebra on S.

3.1. Definitions. Σ-measurable function, $m\Sigma, (m\Sigma)^+, b\Sigma$

Suppose that $h : S \to \mathbf{R}$. For $A \subseteq \mathbf{R}$, define

$$h^{-1}(A) := \{s \in S : h(s) \in A\}.$$

Then h is called Σ-*measurable* if $h^{-1} : \mathcal{B} \to \Sigma$, that is, $h^{-1}(A) \in \Sigma$, $\forall A \in \mathcal{B}$.

So, here is a picture of a Σ-measurable function h:

$$S \xrightarrow{h} \mathbf{R}$$

$$\Sigma \xleftarrow{h^{-1}} \mathcal{B}$$

We write $m\Sigma$ for the class of Σ-measurable functions on S, and $(m\Sigma)^+$ for the class of non-negative elements in $m\Sigma$. We denote by $b\Sigma$ the class of bounded Σ-measurable functions on S.

Note. Because lim sups of sequences even of finite-valued functions may be infinite, and for other reasons, it is convenient to extend these definitions to functions h taking values in $[-\infty, \infty]$ in the obvious way: h is called Σ-*measurable* if $h^{-1} : \mathcal{B}[-\infty, \infty] \to \Sigma$.

Which of the various results stated for real-valued functions extend to functions with values in $[-\infty, \infty]$, and what these extensions are, should be obvious.

Borel function

A function h from a topological space S to $\mathbf{R}$ is called **Borel** if h is $\mathcal{B}(S)$-measurable. The most important case is when S itself is $\mathbf{R}$.

3.2. Elementary Propositions on measurability

(a) *The map* h^{-1} *preserves all set operations:*

$$h^{-1}(\bigcup_\alpha A_\alpha) = \bigcup_\alpha h^{-1}(A_\alpha), \quad h^{-1}(A^c) = (h^{-1}(A))^c, \quad \text{etc.}$$

Proof. This is just definition chasing. □

►(b) *If* $\mathcal{C} \subseteq \mathcal{B}$ *and* $\sigma(\mathcal{C}) = \mathcal{B}$, *then* $h^{-1} : \mathcal{C} \to \Sigma \quad \Rightarrow \quad h \in \mathrm{m}\Sigma$.

Proof. Let $\mathcal{E}$ be the class of elements B in $\mathcal{B}$ such that $h^{-1}(B) \in \Sigma$. By result (a), $\mathcal{E}$ is a σ-algebra, and, by hypothesis, $\mathcal{E} \supseteq \mathcal{C}$. □

(c) *If* S *is topological and* $h : S \to \mathbf{R}$ *is continuous, then* h *is Borel.*

Proof. Take $\mathcal{C}$ to be the class of open subsets of $\mathbf{R}$, and apply result (b). □

►(d) *For any measurable space* (S, Σ), *a function* $h : S \to \mathbf{R}$ *is* Σ*-measurable if*

$$\{h \le c\} := \{s \in S : h(s) \le c\} \in \Sigma \quad (\forall c \in \mathbf{R}).$$

Proof. Take $\mathcal{C}$ to be the class $\pi(\mathbf{R})$ of intervals of the form $(-\infty, c]$, $c \in \mathbf{R}$, and apply result (b). □

Note. Obviously, similar results apply in which $\{h \le c\}$ is replaced by $\{h > c\}$, $\{h \ge c\}$, etc.

3.3. LEMMA. Sums and products of measurable functions are measurable

► $\mathrm{m}\Sigma$ *is an algebra over* $\mathbf{R}$, *that is,*

if $\lambda \in \mathbf{R}$ *and* $h, h_1, h_2 \in \mathrm{m}\Sigma$, *then*

$$h_1 + h_2 \in \mathrm{m}\Sigma, \quad h_1 h_2 \in \mathrm{m}\Sigma, \quad \lambda h \in \mathrm{m}\Sigma.$$

Example of proof. Let $c \in \mathbf{R}$. Then for $s \in S$, it is clear that $h_1(s) + h_2(s) > c$ if and only if for some rational q, we have

$$h_1(s) > q > c - h_2(s).$$

In other words,

$$\{h_1 + h_2 > c\} = \bigcup_{q \in \mathbf{Q}} (\{h_1 > q\} \cap \{h_2 > c - q\}),$$

a countable union of elements of Σ. □

3.4. Composition Lemma.

If $h \in \mathrm{m}\Sigma$ *and* $f \in \mathrm{m}\mathcal{B}$, *then* $f \circ h \in \mathrm{m}\Sigma$.

Proof. Draw the picture:

$$S \xrightarrow{h} \mathbf{R} \xrightarrow{f} \mathbf{R}$$

$$\Sigma \xleftarrow{h^{-1}} \mathcal{B} \xleftarrow{f^{-1}} \mathcal{B}$$

Note. There are obvious generalizations based on the definition (important in more advanced theory): if (S_1, Σ_1) and (S_2, Σ_2) are measurable spaces and $h : S_1 \to S_2$, then h is called Σ_1/Σ_2-measurable if $h^{-1} : \Sigma_2 \to \Sigma_1$. From this point of view, what we have called Σ-*measurable* should read $\Sigma/\mathcal{B}$-*measurable* (or perhaps $\Sigma/\mathcal{B}[-\infty, \infty]$-measurable).

3.5. LEMMA on measurability of infs, lim infs of functions

►► *Let* $(h_n : n \in \mathbf{N})$ *be a sequence of elements of* $\mathrm{m}\Sigma$. *Then*

(i) $\inf h_n$, (ii) $\liminf h_n$, (iii) $\limsup h_n$

are Σ-*measurable (into* $([-\infty, \infty], \mathcal{B}[-\infty, \infty])$, *but we shall still write* $\inf h_n \in \mathrm{m}\Sigma$ *(for example)). Further,*

(iv) $\{s : \lim h_n(s) \text{ exists in } \mathbf{R}\} \in \Sigma$.

Proof. (i) $\{\inf h_n \geq c\} = \bigcap_n \{h_n \geq c\}$.

(ii) Let $L_n(s) := \inf\{h_r(s) : r \geq n\}$. Then $L_n \in \mathrm{m}\Sigma$, by part (i). But

$$L(s) := \liminf h_n(s) = \uparrow \lim L_n(s) = \sup L_n(s),$$

and $\{L \leq c\} = \bigcap_n \{L_n \leq c\} \in \Sigma$.

(iii) This part is now obvious.

(iv) This is also clear because the set on which $\lim h_n$ exists in $\mathbf{R}$ is

$$\{\limsup h_n < \infty\} \cap \{\liminf h_n > -\infty\} \cap g^{-1}(\{0\}),$$

where

$$g := \limsup h_n - \liminf h_n. \qquad \square$$

3.6. Definition. Random variable

►Let $(\Omega, \mathcal{F})$ be our (sample space, family of events). A *random variable* is an element of $\mathrm{m}\mathcal{F}$. Thus,

$$X : \Omega \to \mathbf{R}, \quad X^{-1} : \mathcal{B} \to \mathcal{F}.$$

3.7. Example. Coin tossing

Let $\Omega = \{H,T\}^{\mathbf{N}}$, $\omega = (\omega_1, \omega_2, \ldots)$, $\omega_n \in \{H,T\}$. As in (2.3,b), we define

$$\mathcal{F} = \sigma(\{\omega : \omega_n = W\} : n \in \mathbf{N}, W \in \{H,T\}).$$

Let

$$X_n(\omega) := \begin{cases} 1 & \text{if } \omega_n = H, \\ 0 & \text{if } \omega_n = T. \end{cases}$$

The definition of $\mathcal{F}$ guarantees that each X_n is a random variable. By Lemma 3.3,

$$S_n := X_1 + X_2 + \cdots + X_n = \text{number of heads in } n \text{ tosses}$$

is a random variable.

Next, for $p \in [0,1]$, we have

$$\Lambda := \left\{\omega : \frac{\text{number of heads}}{\text{number of tosses}} \to p\right\} = \{\omega : L^+(\omega) = p\} \cap \{\omega : L^-(\omega) = p\},$$

where $L^+ := \limsup n^{-1} S_n$ and L^- is the corresponding lim inf. By Lemma 3.5, $\Lambda \in \mathcal{F}$.

►► Thus, we have taken an important step towards the Strong Law: the result is *meaningful!* It only remains to prove that it is true!

3.8. Definition. σ-algebra generated by a collection of functions on Ω

This is an important idea, discussed further in Section 3.14. (Compare the weakest topology which makes every function in a given family continuous, etc.)

In Example 3.7, we have

a given set Ω,

a family $(X_n : n \in \mathbf{N})$ of maps $X_n : \Omega \to \mathbf{R}$.

The best way to think of the σ-algebra $\mathcal{F}$ in that example is as

$$\mathcal{F} = \sigma(X_n : n \in \mathbf{N})$$

in the sense now to be described.

►►Generally, if we have a collection $(Y_\gamma : \gamma \in C)$ of maps $Y_\gamma : \Omega \to \mathbf{R}$, then

$$\mathcal{Y} := \sigma(Y_\gamma : \gamma \in C)$$

is defined to be the smallest σ-algebra $\mathcal{Y}$ on Ω such that each map Y_γ ($\gamma \in C$) is $\mathcal{Y}$-measurable. Clearly,

$$\sigma(Y_\gamma : \gamma \in C) = \sigma(\{\omega \in \Omega : Y_\gamma(\omega) \in B\} : \gamma \in C, B \in \mathcal{B}).$$

If X is a random variable for some $(\Omega, \mathcal{F})$, then, of course, $\sigma(X) \subseteq \mathcal{F}$.

Remarks. (i) The idea introduced in this section is something which you will pick up gradually as you work through the course. Don't *worry* about it now; *think* about it, yes!

(ii) Normally, π-systems come to our aid. For example, if $(X_n : n \in \mathbf{N})$ is a collection of functions on Ω, and $\mathcal{X}_n$ denotes $\sigma(X_k : k \leq n)$, then the union $\bigcup \mathcal{X}_n$ is a π-system (indeed, an algebra) which generates $\sigma(X_n : n \in \mathbf{N})$.

3.9. Definitions. Law, distribution function

Suppose that X is a random variable carried by some probability triple $(\Omega, \mathcal{F}, \mathbf{P})$. We have

$$\Omega \xrightarrow{X} \mathbf{R}$$

$$[0,1] \xleftarrow{\mathbf{P}} \mathcal{F} \xleftarrow{X^{-1}} \mathcal{B},$$

$$\text{or indeed } [0,1] \xleftarrow{\mathbf{P}} \sigma(X) \xleftarrow{X^{-1}} \mathcal{B}.$$

Define the *law* $\mathcal{L}_X$ of X by

$$\mathcal{L}_X := \mathbf{P} \circ X^{-1}, \qquad \mathcal{L}_X : \mathcal{B} \to [0,1].$$

Then (**Exercise!**) $\mathcal{L}_X$ is a probability measure on $(\mathbf{R}, \mathcal{B})$. Since $\pi(\mathbf{R}) = \{(-\infty, c] : c \in \mathbf{R}\}$ is a π-system which generates $\mathcal{B}$, Uniqueness Lemma 1.6 shows that $\mathcal{L}_X$ is determined by the function $F_X : \mathbf{R} \to [0,1]$ defined as follows:

$$F_X(c) := \mathcal{L}_X(-\infty, c] = \mathbf{P}(X \leq c) = \mathbf{P}\{\omega : X(\omega) \leq c\}.$$

The function F_X is called the *distribution function* of X.

3.10. Properties of distribution functions

Suppose that F is the distribution function $F = F_X$ of some random variable X. Then

(a) $F : \mathbf{R} \to [0,1]$, $F \uparrow$ (that is, $x \leq y \Rightarrow F(x) \leq F(y)$),

(b) $\lim_{x \to \infty} F(x) = 1$, $\lim_{x \to -\infty} F(x) = 0$,

(c) *F is right-continuous.*

Proof of (c). By using Lemma (1.10,b), we see that

$$\mathbf{P}(X \leq x + n^{-1}) \downarrow \mathbf{P}(X \leq x),$$

and this fact together with the monotonicity of F_X shows that F_X is right-continuous.

Exercise! Clear up any loose ends.

3.11. Existence of random variable with given distribution function

►If F has the properties (a,b,c) in Section 3.10, then, by analogy with Section 1.8 on the existence of Lebesgue measure, we can construct a unique probability measure $\mathcal{L}$ on $(\mathbf{R}, \mathcal{B})$ such that

$$\mathcal{L}(-\infty, x] = F(x), \forall x.$$

Take $(\Omega, \mathcal{F}, \mathbf{P}) = (\mathbf{R}, \mathcal{B}, \mathcal{L})$, $\quad X(\omega) = \omega$. Then it is tautological that

$$F_X(x) = F(x), \forall x.$$

Note. The measure $\mathcal{L}$ just described is called the *Lebesgue-Stieltjes measure associated with* F. Its existence is proved in the next section.

3.12. Skorokhod representation of a random variable with prescribed distribution function

Again let $F : \mathbf{R} \to [0,1]$ have properties (3.10,a,b,c). We can construct a random variable with distribution function F carried by

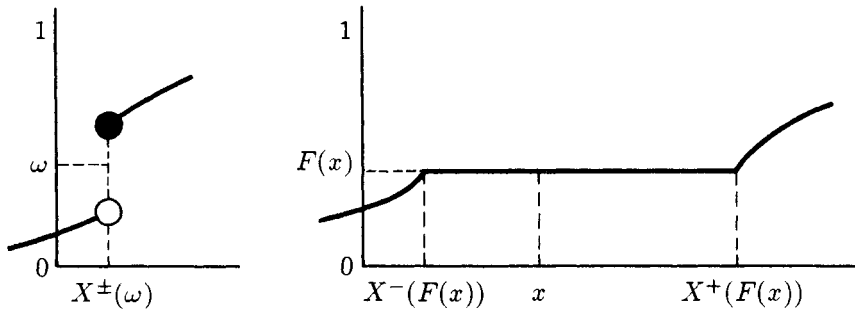
$$(\Omega, \mathcal{F}, \mathbf{P}) = ([0,1], \mathcal{B}[0,1], \text{Leb})$$

as follows. Define (the right-hand equalities, which *you* can prove, are there for clarification only)

(a1) $$X^+(\omega) := \inf\{z : F(z) > \omega\} = \sup\{y : F(y) \leq \omega\},$$

(a1) $$X^-(\omega) := \inf\{z : F(z) \geq \omega\} = \sup\{y : F(y) < \omega\}.$$

The following picture shows cases to watch out for.



By definition of X^-,

$$(\omega \leq F(c)) \quad \Rightarrow \quad (X^-(\omega) \leq c).$$

Now,

$$(z > X^-(\omega)) \quad \Rightarrow \quad (F(z) \geq \omega),$$

so, by the right-continuity of F, $F(X^-(\omega)) \geq \omega$, and

$$(X^-(\omega) \leq c) \quad \Rightarrow \quad \Big(\omega \leq F(X^-(\omega)) \leq F(c)\Big).$$

Thus, $(\omega \leq F(c)) \iff (X^-(\omega) \leq c)$, so that

$$\mathbf{P}(X^- \leq c) = F(c).$$

(b) *The variable X^- therefore has distribution function F, and the measure $\mathcal{L}$ in Section 3.11 is just the law of X^-.*

It will be important later to know that

(c) *X^+ also has distribution function F, and that, indeed,*

$$\mathbf{P}(X^+ = X^-) = 1.$$

Proof of (c). By definition of X^+,

$$(\omega < F(c)) \quad \Rightarrow \quad (X^+(\omega) \leq c),$$

so that $F(c) \leq \mathbf{P}(X^+ \leq c)$. Since $X^- \leq X^+$, it is clear that

$$\{X^- \neq X^+\} = \bigcup_{c \in \mathbf{Q}} \{X^- \leq c < X^+\}.$$

But, for every $c \in \mathbf{R}$,

$$\mathbf{P}(X^- \leq c < X^+) = P(\{X^- \leq c\} \backslash \{X^+ \leq c\}) \leq F(c) - F(c) = 0.$$

Since $\mathbf{Q}$ is countable, the result follows. □

Remark. It is in fact true that every experiment you will meet in this (or any other) course can be modelled via the triple $([0,1], \mathcal{B}[0,1], \text{Leb})$. (You will start to be convinced of this by the end of the next chapter.) However, this observation normally has only curiosity value.

3.13. Generated σ-algebras – a discussion

Suppose that $(\Omega, \mathcal{F}, \mathbf{P})$ is a model for some experiment, and that the experiment has been performed, so that (see Section 2.2) Tyche has made her choice of ω.

Let $(Y_\gamma : \gamma \in C)$ be a collection of random variables associated with our experiment, and suppose that someone reports to you the following information about the chosen point ω:

(*) *the values* $Y_\gamma(\omega)$*, that is, the observed values of the random variables* Y_γ $(\gamma \in C)$.

Then the intuitive significance of the σ-algebra $\mathcal{Y} := \sigma(Y_\gamma : \gamma \in C)$ is that it consists precisely of those events F for which, for each and every ω, you can decide whether or not F has occurred (that is, whether or not $\omega \in F$) on the basis of the information (*); the information (*) is precisely equivalent to the following information:

(**) *the values* $\mathrm{I}_F(\omega)$ $(F \in \mathcal{Y})$.

(a) **Exercise.** Prove that the σ-algebra $\sigma(Y)$ generated by a single random variable Y is given by

$$\sigma(Y) = Y^{-1}(\mathcal{B}) := (\{\omega : Y(\omega) \in B\} : B \in \mathcal{B}),$$

and that $\sigma(Y)$ is generated by the π-system

$$\pi(Y) := (\{\omega : Y(\omega) \leq x\} : x \in \mathbf{R}) = Y^{-1}(\pi(\mathbf{R})). \qquad \square$$

The following results might help clarify things. Good advice: stop reading this section after (c)! Results (b) and (c) are proved in the appendix to this chapter.

(b) If $Y : \Omega \to \mathbf{R}$, then $Z : \Omega \to \mathbf{R}$ is an $\sigma(Y)$-measurable function if and only if there exists a Borel function $f : \mathbf{R} \to \mathbf{R}$ such that $Z = f(Y)$.

(c) If $Y_1, Y_2, \ldots, Y_n$ are functions from Ω to $\mathbf{R}$, then a function $Z : \Omega \to \mathbf{R}$ is $\sigma(Y_1, Y_2, \ldots, Y_n)$-measurable if and only if there exists a Borel function f on $\mathbf{R}^n$ such that $Z = f(Y_1, Y_2, \ldots, Y_n)$. We shall see in the appendix that the more correct measurability condition on f is that f be '$\mathcal{B}^n$-measurable'.

(d) If $(Y_\gamma : \gamma \in C)$ is a collection (parametrized by the infinite set C) of functions from Ω to $\mathbf{R}$, then $Z : \Omega \to \mathbf{R}$ is $\sigma(Y_\gamma : \gamma \in C)$-measurable if and only if there exists a countable sequence $(\gamma_i : i \in \mathbf{N})$ of elements of C and a Borel function f on $\mathbf{R}^{\mathbf{N}}$ such that

$$Z = f(Y_{\gamma_1}, Y_{\gamma_2}, \ldots).$$

Warning – for the over-enthusiastic only. For uncountable C, $\mathcal{B}(\mathbf{R}^C)$ is much larger than the C-fold product measure space $\prod_{\gamma \in C} \mathcal{B}(\mathbf{R})$. It is the latter rather than the former which gives the appropriate type of f in (d).

3.14. The Monotone-Class Theorem

In the same way that Uniqueness Lemma 1.6 allows us to deduce results about σ-algebras from results about π-systems, the following 'elementary' version of the Monotone-Class Theorem allows us to deduce results about general measurable functions from results about indicators of elements of π-systems. Generally, we shall not use the theorem in the main text, preferring 'just to use bare hands'. However, for product measure in Chapter 8, it becomes indispensable.

THEOREM.

►► *Let $\mathcal{H}$ be a class of bounded functions from a set S into $\mathbf{R}$ satisfying the following conditions:*

(i) *$\mathcal{H}$ is a vector space over $\mathbf{R}$;*

(ii) *the constant function 1 is an element of $\mathcal{H}$;*

(iii) *if (f_n) is a sequence of non-negative functions in $\mathcal{H}$ such that $f_n \uparrow f$ where f is a bounded function on S, then $f \in \mathcal{H}$.*

Then if $\mathcal{H}$ contains the indicator function of every set in some π-system $\mathcal{I}$, then $\mathcal{H}$ contains every bounded $\sigma(\mathcal{I})$-measurable function on S.

For proof, see the appendix to this chapter.

Chapter 4

Independence

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability triple.

4.1. Definitions of independence

Note. We focus attention on the σ-algebra formulation (and describe the more familiar forms of independence in terms of it) to acclimatize ourselves to thinking of σ-algebras as the natural means of summarizing information. Section 4.2 shows that the fancy σ-algebra definitions agree with the ones from elementary courses.

Independent σ-algebras

►Sub-σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \ldots$ of $\mathcal{F}$ are called *independent* if, whenever $G_i \in \mathcal{G}_i \quad (i \in \mathbf{N})$ and $i_1, \ldots, i_n$ are distinct, then

$$\mathbf{P}(G_{i_1} \cap \ldots \cap G_{i_n}) = \prod_{k=1}^{n} \mathbf{P}(G_{i_k}).$$

Independent random variables

►Random variables $X_1, X_2, \ldots$ are called *independent* if the σ-algebras

$$\sigma(X_1), \sigma(X_2), \ldots$$

are independent.

Independent events

►Events $E_1, E_2, \ldots$ are called *independent* if the σ-algebras $\mathcal{E}_1, \mathcal{E}_2, \ldots$ are independent, where

$$\mathcal{E}_n \text{ is the } \sigma\text{-algebra } \{\emptyset, E_n, \Omega \backslash E_n, \Omega\}.$$

Since $\mathcal{E}_n = \sigma(I_{E_n})$, it follows that events $E_1, E_2, \ldots$ are independent if and only if the random variables $I_{E_1}, I_{E_2}, \ldots$ are independent.

4.2. The π-system Lemma; and the more familiar definitions

We know from elementary theory that events $E_1, E_2, \ldots$ are independent if and only if whenever $n \in \mathbf{N}$ and $i_1, \ldots, i_n$ are distinct, then

$$\mathbf{P}(E_{i_1} \cap \cdots \cap E_{i_n}) = \prod_{k=1}^{n} \mathbf{P}(E_{i_k}),$$

corresponding results involving complements of the E_i, etc., being consequences of this.

We now use the Uniqueness Lemma 1.6 to obtain a significant generalization of this idea, **allowing us to study independence via (manageable) π-systems rather than (awkward) σ-algebras.**

Let us concentrate on the case of two σ-algebras.

►►(a) **LEMMA.** *Suppose that $\mathcal{G}$ and $\mathcal{H}$ are sub-σ-algebras of $\mathcal{F}$, and that $\mathcal{I}$ and $\mathcal{J}$ are π-systems with*

$$\sigma(\mathcal{I}) = \mathcal{G}, \quad \sigma(\mathcal{J}) = \mathcal{H}.$$

Then $\mathcal{G}$ and $\mathcal{H}$ are independent if and only if $\mathcal{I}$ and $\mathcal{J}$ are independent in that

$$\mathbf{P}(I \cap J) = \mathbf{P}(I)\mathbf{P}(J), \qquad I \in \mathcal{I},\ J \in \mathcal{J}.$$

Proof. Suppose that $\mathcal{I}$ and $\mathcal{J}$ are independent. For fixed I in $\mathcal{I}$, the *measures* (check that they *are* measures!)

$$H \mapsto \mathbf{P}(I \cap H) \text{ and } H \mapsto \mathbf{P}(I)\mathbf{P}(H)$$

on $(\Omega, \mathcal{H})$ have the same total mass $\mathbf{P}(I)$, and agree on $\mathcal{J}$. By Lemma 1.6, they therefore agree on $\sigma(\mathcal{J}) = \mathcal{H}$. Hence,

$$\mathbf{P}(I \cap H) = \mathbf{P}(I)\mathbf{P}(H), \qquad I \in \mathcal{I},\ H \in \mathcal{H}.$$

Thus, for fixed H in $\mathcal{H}$, the measures

$$G \mapsto \mathbf{P}(G \cap H) \text{ and } G \mapsto \mathbf{P}(G)\mathbf{P}(H)$$

on $(\Omega, \mathcal{G})$ have the same total mass $\mathbf{P}(H)$, and agree on $\mathcal{I}$. They therefore agree on $\sigma(\mathcal{I}) = \mathcal{G}$; and this is what we set out to prove. □

Suppose now that X and Y are two random variables on $(\Omega, \mathcal{F}, \mathbf{P})$ such that, whenever $x, y \in \mathbf{R}$,

(b) $$\mathbf{P}(X \le x; Y \le y) = \mathbf{P}(X \le x)\mathbf{P}(Y \le y).$$

Now, (b) says that the π-systems $\pi(X)$ and $\pi(Y)$ (see Section 3.13) are independent. Hence $\sigma(X)$ and $\sigma(Y)$ are independent: that is, X and Y are independent in the sense of Definition 4.1.

In the same way, we can prove that random variables $X_1, X_2, \ldots, X_n$ are independent if and only if

$$\mathbf{P}(X_k \le x_k : 1 \le k \le n) = \prod_{k=1}^{n} \mathbf{P}(X_k \le x_k),$$

and all the familiar things from elementary theory.

Command: Do Exercise E4.1 now.

4.3. Second Borel-Cantelli Lemma (BC2)

►► *If $(E_n : n \in \mathbf{N})$ is a sequence of* **independent** *events, then*

$$\sum \mathbf{P}(E_n) = \infty \quad \Rightarrow \quad \mathbf{P}(E_n, \text{ i.o.}) = \mathbf{P}(\limsup E_n) = 1.$$

Proof. First, we have

$$(\limsup E_n)^c = \liminf E_n^c = \bigcup_m \bigcap_{n \ge m} E_n^c.$$

With p_n denoting $\mathbf{P}(E_n)$, we have

$$\mathbf{P}\left(\bigcap_{n \ge m} E_n^c \right) = \prod_{n \ge m} (1 - p_n),$$

this equation being true if the condition $\{n \ge m\}$ is replaced by condition $\{r \ge n \ge m\}$, because of independence, and the limit as $r \uparrow \infty$ being justified by the monotonicity of the two sides.

For $x \ge 0$, $1 - x \le \exp(-x)$, so that, since $\sum p_n = \infty$,

$$\prod_{n \ge m} (1 - p_n) \le \exp\left(- \sum_{n \ge m} p_n \right) = 0.$$

So, $\mathbf{P}\left[(\limsup E_n)^c\right] = 0$. □

Exercise. Prove that if $0 \le p_n < 1$ and $S := \sum p_n < \infty$, then $\prod(1 - p_n) > 0$. *Hint.* First show that if $S < 1$, then $\prod(1 - p_n) \ge 1 - S$.

4.4. Example

Let $(X_n : n \in \mathbf{N})$ be a sequence of *independent* random variables, each exponentially distributed with rate 1:

$$\mathbf{P}(X_n > x) = e^{-x}, \quad x \geq 0.$$

Then, for $\alpha > 0$,

$$\mathbf{P}(X_n > \alpha \log n) = n^{-\alpha},$$

so that, using (BC1) and (BC2),

$$\text{(a0)} \qquad \mathbf{P}(X_n > \alpha \log n \text{ for infinitely many } n) = \begin{cases} 0 & \text{if } \alpha > 1, \\ 1 & \text{if } \alpha \leq 1. \end{cases}$$

Now let $L := \limsup(X_n / \log n)$. Then

$$\mathbf{P}(L \geq 1) \geq \mathbf{P}(X_n > \log n, \text{ i.o.}) = 1,$$

and, for $k \in \mathbf{N}$,

$$\mathbf{P}(L > 1 + 2k^{-1}) \leq \mathbf{P}\left(X_n > (1 + k^{-1}) \log n, \text{ i.o.}\right) = 0.$$

Thus, $\{L > 1\} = \bigcup_k \{L > 1 + 2k^{-1}\}$ is $\mathbf{P}$-null, and hence

$$L = 1 \text{ almost surely.}$$

Something to think about

In the same way, we can prove the finer result

$$\text{(a1)} \qquad \mathbf{P}(X_n > \log n + \alpha \log\log n, \text{ i.o. }) = \begin{cases} 0 & \text{if } \alpha > 1, \\ 1 & \text{if } \alpha \leq 1, \end{cases}$$

or, even finer,

$$\text{(a2)} \quad \mathbf{P}(X_n > \log n + \log\log n + \alpha \log\log\log n, \text{ i.o. }) = \begin{cases} 0 & \text{if } \alpha > 1, \\ 1 & \text{if } \alpha \leq 1; \end{cases}$$

or etc. By combining in an appropriate way (think about this!) the sequence of statements (a0),(a1),(a2),... with the statement that the union of a countable number of null sets is null while the intersection of a sequence of probability-1 sets has probability 1, we can obviously make remarkably precise statements about the size of the big elements in the sequence (X_n).

I have included in the appendix to this chapter the statement of a truly fantastic theorem about precise description of long-term behaviour: *Strassen's Law.*

A number of exercises in Chapter E are now accessible to you.

4.5. A fundamental question for modelling

Can we construct a sequence $(X_n : n \in \mathbb{N})$ *of independent random variables,* X_n *having prescribed distribution function* F_n*?* We *have* to be able to answer Yes to this question – for example, to be able to construct a rigorous model for the branching-process model of Chapter 0, or indeed for Example 4.4 to make sense. Equation (0.2,b) makes it clear that a Yes answer to our question is *all* that is needed for a rigorous branching-process model.

The trick answer based on the existence of Lebesgue measure given in the next section does settle the question. A more satisfying answer is provided by the theory of product measure, a topic deferred to Chapter 8.

4.6. A coin-tossing model with applications

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be $([0,1], \mathcal{B}[0,1], \text{Leb})$. For $\omega \in \Omega$, expand ω in binary:

$$\omega = 0.\omega_1\omega_2\ldots$$

(The existence of two different expansions of a dyadic rational is not going to cause any problems because the set $\mathbf{D}$ (say) of dyadic rationals in $[0,1]$ has Lebesgue measure 0 – it is a countable set!) An an **Exercise**, you can prove that the sequence $(\xi_n : n \in \mathbb{N})$, where

$$\xi_n(\omega) := \omega_n,$$

is a sequence of independent variables each taking the values 0 or 1 with probability $\frac{1}{2}$ for either possibility. Clearly, $(\xi_n : n \in \mathbb{N})$ provides a model for coin tossing.

Now define

$$Y_1(\omega) := 0.\omega_1\omega_3\omega_6\ldots,$$
$$Y_2(\omega) := 0.\omega_2\omega_5\omega_9\ldots,$$
$$Y_3(\omega) := 0.\omega_4\omega_8\omega_{13}\ldots,$$

and so on. We now need a bit of common sense. Since the sequence

$$\omega_1, \omega_3, \omega_6, \ldots$$

has the same 'coin-tossing' properties as the full sequence $(\omega_n : n \in \mathbb{N})$, it is clear that

$$Y_1 \text{ has the uniform distribution on } [0,1];$$

and similarly for the other Y's.

of SF^+ such that

$$L_n \uparrow L \le f, \quad U_n \downarrow U \ge f$$

and $\mu(L_n) \uparrow 1$, $\mu(U_n) \downarrow 1$. If we define

$$\tilde{f} = \begin{cases} L & \text{if } L = U, \\ \end{cases}$$

then it is clear that $\tilde{f}$ is Borel measurable, while (since $\mu(L) = \mu(U) = 1$) $\{f \neq \tilde{f}\}$ is a subset of the Borel set $\{L \neq U\}$, which Lemma 5.2(b) shows

Step 3: Proof that $\lambda = \mu_0$ *on* Σ_0.

Let $F \in \Sigma_0$. Then, clearly, $\lambda(F) \leq \mu_0(F)$. Now suppose that $F \subseteq \bigcup_n F_n$, where $F_n \in \Sigma_0$. As usual, we can define a sequence (E_n) of disjoint sets:

$$E_1 := F_1, \quad E_n = F_n \cap \left(\bigcup_{k<n} F_k \right)^c$$

such that $E_n \subseteq F_n$ and $\bigcup E_n = \bigcup F_n \supseteq F$. Then

$$\mu_0(F) = \mu_0 \left(\bigcup (F \cap E_n) \right) = \sum \mu_0(F \cap E_n),$$

by using the countable additivity of μ_0 *on* Σ_0. Hence

$$\mu_0(F) \leq \sum \mu_0(E_n) \leq \sum \mu_0(F_n),$$

so that $\lambda(F) \geq \mu_0(F)$. Step 3 is complete.

Step 4: Proof that $\Sigma_0 \subseteq \mathcal{L}$. Let $E \in \Sigma_0$ and $G \in \mathcal{G}$. Then there exists a sequence (F_n) in Σ_0 such that $G \subseteq \bigcup_n F_n$, and

$$\sum_n \mu_0(F_n) \leq \lambda(G) + \varepsilon.$$

Now, by definition of λ,

$$\begin{aligned} \sum_n \mu_0(F_n) &= \sum_n \mu_0(E \cap F_n) + \sum_n \mu_0(E^c \cap F_n) \\ &\geq \lambda(E \cap G) + \lambda(E^c \cap G), \end{aligned}$$

since $E \cap G \subseteq \bigcup (E \cap F_n)$ and $E^c \cap G \subseteq \bigcup (E^c \cap F_n)$. Thus, since ε is arbitrary,

$$\lambda(G) \geq \lambda(E \cap G) + \lambda(E^c \cap G).$$

However, since λ is subadditive,

$$\lambda(G) \leq \lambda(E \cap G) + \lambda(E^c \cap G).$$

We see that E is indeed a λ-set. □

A1.9. Proof of the existence of Lebesgue measure on $((0,1], \mathcal{B}(0,1])$.

Recall the set-up in Section 1.8. Let $S = (0,1]$. For $F \subseteq S$, say that $F \in \Sigma_0$ if F may be written as a finite union

$$(*) \qquad F = (a_1, b_1] \cup \ldots \cup (a_r, b_r]$$

where $r \in \mathbb{N}$, $0 \leq a_1 \leq b_1 \leq \ldots \leq a_r \leq b_r \leq 1$. Then (as you should convince yourself) Σ_0 is an algebra on (0,1] and

$$\Sigma := \sigma(\Sigma_0) = \mathcal{B}(0,1].$$

(We write $\mathcal{B}(0,1]$ rather than $\mathcal{B}((0,1])$.) For F as at (*), let

$$\mu_0(F) = \sum_{k \leq r} (b_k - a_k).$$

Of course, a set F may have different expressions as a finite disjoint union of the form (*): for example,

$$(0,1] = (0, \tfrac{1}{2}] \cup (\tfrac{1}{2}, 1].$$

However, it is easily seen that μ_0 is well defined on Σ_0 and that μ_0 is finitely additive on Σ_0. While this is obvious from a picture, you might (or might not) wish to consider how to make the intuitive argument into a formal proof.

The key thing is to prove that μ_0 *is countably additive on* Σ_0. So, suppose that (F_n) is a sequence of disjoint elements of Σ_0 with union F in Σ_0. We know that if $G_n = \bigcup_{k=1}^{n} F_k$, then

$$\mu_0(G_n) = \sum_{k=1}^{n} \mu_0(F_k) \quad \text{and} \quad G_n \uparrow F.$$

To prove that μ_0 is countably additive it is enough to show that $\mu_0(G_n) \uparrow \mu_0(F)$, for then

$$\mu_0(F) = \uparrow \lim \mu_0(G_n) = \uparrow \lim_n \sum_{k=1}^{n} \mu_0(F_k) = \sum \mu_0(F_k).$$

Let $H_n = F \backslash G_n$. Then $H_n \in \Sigma_0$ and $H_n \downarrow \emptyset$. We need only prove that

$$\mu_0(H_n) \downarrow 0;$$

for then

$$\mu_0(G_n) = \mu_0(F) - \mu_0(H_n) \uparrow \mu_0(F).$$

It is clear that an alternative (and final!) rewording of what we need to show is the following:

(a) *if* (H_n) *is a decreasing sequence of elements of* Σ_0 *such that for some* $\varepsilon > 0$,

$$\mu_0(H_n) \geq 2\varepsilon, \quad \forall n,$$

then $\bigcap_k H_k \neq \emptyset$.

Proof of (a). It is obvious from the definition of Σ_0 that, for each $k \in \mathbb{N}$, we can choose $J_k \in \Sigma_0$ such that, with $\bar{J}_k$ denoting the closure of J_k,

$$\bar{J}_k \subseteq H_k \quad \text{and} \quad \mu(H_k \backslash J_k) \leq \varepsilon 2^{-k}.$$

But then (recall that $H_n \downarrow$)

$$\mu_0\left(H_n \backslash \bigcap_{k \leq n} J_k\right) \leq \mu_0\left(\bigcup_{k \leq n} (H_k \backslash J_k)\right) \leq \sum_{k \leq n} \varepsilon 2^{-k} < \varepsilon.$$

Hence, since $\mu_0(H_n) > 2\varepsilon$, $\forall n$, we see that for every n,

$$\mu_0\left(\bigcap_{k \leq n} J_k\right) > \varepsilon,$$

and hence $\bigcap_{k \leq n} J_k$ is non-empty. *A fortiori* then, for every n,

$$K_n := \bigcap_{k \leq n} \bar{J}_k \quad \text{is non-empty.}$$

That

(b) $$\bigcap \bar{J}_k \neq \emptyset \qquad \text{(whence } \bigcap H_k \neq \emptyset\text{)}$$

now follows from the Heine-Borel theorem: for if (b) is false, then $((\bar{J}_k)^c : k \in \mathbb{N})$ gives a covering of $[0,1]$ by open sets with no finite subcovering. Alternatively, we can argue directly as follows. For each n, choose a point x_n in the non-empty set K_n. Since each x_n belongs to the compact set $\bar{J}_1$, we can find a subsequence (n_q) and a point x of $\bar{J}_1$ such that $x_{n_q} \to x$. However, for each k, $x_{n_q} \in \bar{J}_k$ for all but finitely many q, and since $\bar{J}_k$ is

compact, it follows that $x \in \bar{J}_k$. Hence $x \in \bigcap_k \bar{J}_k$, and property (b) holds. □

Since μ_0 is countably additive on Σ_0 and $\mu_0(0,1] < \infty$, it follows that μ_0 has a unique extension to a measure μ on $((0,1], \mathcal{B}(0,1])$. This is Lebesgue measure Leb on $((0,1], \mathcal{B}(0,1])$.

The μ_0-sets form a σ-algebra strictly larger than $\mathcal{B}(0,1]$, namely the σ-algebra of Lebesgue measurable subsets of (0,1]. See Section A1.11.

A1.10. Example of non-uniqueness of extension

With (S, Σ_0) as in Section A1.9, suppose that for $F \in \Sigma_0$,

(a) $$\nu_0(F) := \begin{cases} 0 & \text{if } F = \emptyset, \\ \infty & \text{if } F \neq \emptyset. \end{cases}$$

The Carathéodory extension of ν_0 will be obtained as the obvious extension of (a) to Σ. However, another extension $\tilde{\nu}$ is given by

$$\tilde{\nu}(F) = \text{number of elements in } F.$$

A1.11. Completion of a measure space

In fact (apart from an 'aside' on the Riemann integral), we do not need completions in this book.

Suppose that (S, Σ, μ) is a measure space. Define a class $\mathcal{N}$ of subsets of S as follows:

$$N \in \mathcal{N} \text{ if and only if } \exists Z \in \Sigma \text{ such that } N \subseteq Z \text{ and } \mu(Z) = 0.$$

It is sometimes philosophically satisfying to be able to make precise the idea that 'N in $\mathcal{N}$ is μ-measurable and $\mu(N) = 0$'. This is done as follows. For any subset F of S, write

$$F \in \Sigma^*$$

if $\exists E, G \in \Sigma$ such that $E \subseteq F \subseteq G$ and $\mu(G \setminus E) = 0$. It is very easy to show that Σ^* is a σ-algebra on S and indeed that $\Sigma^* = \sigma(\Sigma, \mathcal{N})$. With obvious notation we define for $F \in \Sigma^*$,

$$\mu^*(F) = \mu(E) = \mu(G),$$

it being easy to check that μ^* is well defined. Moreover, it is no problem to prove that (S, Σ^*, μ^*) is a measure space, the *completion* of (S, Σ, μ).

For parts of advanced probability, it is essential to complete the basic probability triple $(\Omega, \mathcal{F}, \mathbf{P})$. In other parts of probability, when (for example) S is topological, $\Sigma = \mathcal{B}(S)$, and we wish to consider several different measures on (S, Σ), it is meaningless to insist on completion.

If we begin with $([0,1], \mathcal{B}[0,1], \text{Leb})$, then $\mathcal{B}[0,1]^*$ is the σ-algebra of what are called *Lebesgue-measurable* sets of [0,1]. Then, for example, a function $f : [0,1] \to [0,1]$ is Lebesgue-measurable if the inverse image of every *Borel* set is Lebesgue-measurable: it need not be true that the inverse image of a Lebesgue-measurable set is Lebesgue-measurable.

A1.12. The Baire category theorem

In Section 1.11, we studied a subset H of $S := [0,1]$ such that

(i) $H = \bigcap_k G_k$ for a sequence (G_k) of open subsets of S,

(ii) $H \supseteq V$, where $V = \mathbf{Q} \cap S$.

If H were countable: $H = \{h_r : r \in \mathbf{N}\}$, then we would have

$$S = H \cup H^c = (\bigcup_r \{h_r\}) \cup (\bigcup G_k^c) \tag{a}$$

expressing S as a countable union

$$S = \bigcup_n F_n \tag{b}$$

of closed sets where no F_n contains an open interval. [Since $V \subseteq G_k$ for every k, $G_k^c \subseteq V^c$ so that G_k^c contains only irrational points in S.]

However, the Baire category theorem states that

> *if a complete metric space S may be written as a union of a countable sequence of closed sets:*
>
> $$S = \bigcup F_n$$
>
> *then some F_n contains an open ball.*

Thus the set H must be uncountable.

The Baire category theorem has fundamental applications in functional analysis, and some striking applications to probability too!

Proof of the Baire category theorem. Assume for the purposes of contradiction that no F_n contains an open ball. Since F_1^c is a non-empty open subset of S, we can find x_1 in S and $\varepsilon_1 > 0$ such that

$$B(x_1, \varepsilon_1) \subseteq F_1^c,$$

$B(x_1, \varepsilon_1)$ denoting the open ball of radius ε_1 centred at x_1. Now F_2 contains no open ball, so that the open set

$$U_2 := B(x_1, 2^{-1}\varepsilon_1) \cap F_2^c$$

is non-empty, and we can find x_2 in U_2 and $\varepsilon_2 > 0$ such that

$$B(x_2, \varepsilon_2) \subseteq U_2, \quad \varepsilon_2 < 2^{-1}\varepsilon_1.$$

Inductively, choose a sequence (x_n) in S and (ε_n) in $(0, \infty)$ so that we have $\varepsilon_{n+1} < 2^{-1}\varepsilon_n$ and

$$B(x_{n+1}, \varepsilon_{n+1}) \subseteq U_{n+1} := B(x_n, 2^{-1}\varepsilon_n) \cap F_{n+1}^c.$$

Since $d(x_n, x_{n+1}) < 2^{-1}\varepsilon_n$, it is obvious from the triangle law that (x_n) is Cauchy, so that $x := \lim x_n$ exists, and that

$$x \in \bigcap B(x_n, \varepsilon_n) \subseteq \bigcap F_n^c$$

contradicting the fact that $\bigcup F_n = S$. □

Chapter A3

Appendix to Chapter 3

A3.1. Proof of the Monotone-Class Theorem 3.14

Recall the statement of the theorem.

►► *Let $\mathcal{H}$ be a class of bounded functions from a set S into* $\mathbf{R}$ *satisfying the following conditions:*

(i) *$\mathcal{H}$ is a vector space over* $\mathbf{R}$;

(ii) *the constant function* 1 *is an element of $\mathcal{H}$;*

(iii) *if (f_n) is a sequence of non-negative functions in $\mathcal{H}$ such that $f_n \uparrow f$ where f is a bounded function on S, then $f \in \mathcal{H}$.*

Then if $\mathcal{H}$ contains the indicator function of every set in some π-system $\mathcal{I}$, then $\mathcal{H}$ contains every bounded $\sigma(\mathcal{I})$-measurable function on S.

Proof. Let $\mathcal{D}$ be the class of sets F in S such that $\mathrm{I}_F \in \mathcal{H}$. It is immediate from (i) - (iii) that $\mathcal{D}$ is a d-system. Since $\mathcal{D}$ contains the π-system $\mathcal{I}$, $\mathcal{D}$ contains $\sigma(\mathcal{I})$.

Suppose that f is a $\sigma(\mathcal{I})$-measurable function such that for some K in $\mathbf{N}$,

$$0 \le f(s) \le K, \quad \forall s \in S.$$

For $n \in \mathbf{N}$, define

$$f_n(s) = \sum_{i=0}^{K2^n} i2^{-n}\mathrm{I}_{A(n,i)},$$

where

$$A(n,i) := \{s : i2^{-n} \le f(s) < (i+1)2^{-n}\}.$$

Since f is $\sigma(\mathcal{I})$-measurable, every $A(n,i) \in \sigma(\mathcal{I})$, so that $I_{A(n,i)} \in \mathcal{H}$. Since $\mathcal{H}$ is a vector space, every $f_n \in \mathcal{H}$. But $0 \le f_n \uparrow f$, so that $f \in \mathcal{H}$.

If $f \in b\sigma(\mathcal{I})$, we may write $f = f^+ - f^-$, where $f = \max(f, 0)$ and $f^- = \max(-f, 0)$. Then $f^+, f^- \in b\sigma(\mathcal{I})$ and $f^+, f^- \geq 0$, so that $f^+, f^- \in \mathcal{H}$ by what we established above. □

A3.2. Discussion of generated σ-algebras

This is one of those situations in which it is actually easier to understand things in a more formal abstract setting. So, suppose that

Ω and S are sets, and that $Y : \Omega \to S$;

Σ is a σ-algebra on S:

$X : \Omega \to \mathbf{R}$.

Because Y^{-1} preserves all set operations,

$$Y^{-1}\Sigma := \{Y^{-1}B : B \in \Sigma\}$$

is a σ-algebra on Ω, and because it is tautologically the smallest σ-algebra $\mathcal{Y}$ on Ω such that Y is $\mathcal{Y}/\Sigma$ measurable (in that $Y^{-1} : \Sigma \to \mathcal{Y}$), we call it $\sigma(Y)$:

$$\sigma(Y) = Y^{-1}\Sigma.$$

LEMMA

(a) *X is $\sigma(Y)$-measurable if and only if*

$$X = f(Y)$$

where f is a Σ-measurable function from S to $\mathbf{R}$.

Note. The 'if' part is just the Composition Lemma

Proof of 'only if' part. It is enough to prove that

(b) $X \in b\sigma(Y)$ *if and only if* $\exists f \in b\Sigma$ *such that* $X = f(Y)$.

(Otherwise, consider arc tan X, for example.)

Though we certainly do not need the Monotone-Class Theorem to prove (b), we may as well use it.

So define $\mathcal{H}$ to be the class of all bounded functions X on Ω such that $X = f(Y)$ for some $f \in b\Sigma$. Taking $\mathcal{I} = \sigma(Y)$, note that if $F \in \mathcal{I}$ then $F = Y^{-1}B$ for some B in Σ, so that

$$I_F(\omega) = I_B(Y(\omega)),$$

so that $I_F \in \mathcal{H}$. That $\mathcal{H}$ is a vector space containing constants is obvious.

Finally, suppose that (X_n) is a sequence of elements of $\mathcal{H}$ such that, for some positive real constant K,

$$0 \leq X_n \uparrow X \leq K.$$

For each n, $X_n = f_n(Y)$ for some f_n in $b\Sigma$. Define $f := \limsup f_n$, so that $f \in b\Sigma$. Then $X = f(Y)$. □

One has to be very careful about what Lemma (a) *means in practice.*

To be sure, result (3.13,b) is the special case when $(S, \Sigma) = (\mathbf{R}, \mathcal{B})$.

Discussion of (3.13,c). Suppose that $Y_k : \Omega \to \mathbf{R}$ for $1 \leq k \leq n$. We may define a map $Y : \Omega \to \mathbf{R}^n$ via

$$Y(\omega) := (Y_1(\omega), \ldots, Y_n(\omega)) \in \mathbf{R}^n.$$

The problem mentioned at (3.13,d) and in the Warning following it shows up here because, before we can apply Lemma (a) to prove (3.13,c), we need to prove that

$$\sigma(Y_1, \ldots, Y_n) := \sigma(Y_k^{-1}\mathcal{B}(\mathbf{R}) : 1 \leq k \leq n) = Y^{-1}\mathcal{B}(\mathbf{R}^n) =: \sigma(Y).$$

[This amounts to proving that the product σ-algebra $\prod_{1 \leq k \leq n} \mathcal{B}(\mathbf{R})$ is the same as $\mathcal{B}(\mathbf{R}^n)$. See Section 8.5.] Now $Y_k = \gamma_k \circ Y$, where γ_k is the (continuous, hence Borel) 'k^{th} coordinate' map on $\mathbf{R}^n$, so that Y_k is $\sigma(Y)$-measurable. On the other hand, every open subset of $\mathbf{R}^n$ is a countable union of open rectangles $G_1 \times \cdots \times G_n$ where each G_k is a subinterval of $\mathbf{R}$, and since

$$\{Y \in G_1 \times \cdots \times G_n\} = \bigcap \{Y_k \in G_k\} \in \sigma(Y_1, \ldots, Y_n),$$

things do work out. □

You can already see why we are in an appendix, and why we skip discussion of (3.13,d).

Chapter A4

Appendix to Chapter 4

This appendix gives the statement of Strassen's Law of the Iterated Logarithm. Section A4.3 treats the completely different topic of constructing a rigorous model for a Markov chain.

A4.1. Kolmogorov's Law of the Iterated Logarithm

THEOREM

Let $X_1, X_2, \ldots$ be IID RVs each with mean 0 and variance 1.

Let $S_n := X_1 + X_2 + \cdots + X_n$. Then, almost surely,

$$\limsup \frac{S_n}{\sqrt{2n \log\log n}} = +1, \quad \liminf \frac{S_n}{\sqrt{2n \log\log n}} = -1.$$

This result already gives very precise behaviour on the big values of partial sums. See Section 14.7 for proof in the case when the X's are normally distributed.

A4.2. Strassen's Law of the Iterated Logarithm

Strassen's Law is a staggering extension of Kolmogorov's result.

Let (X_n) and (S_n) be as in the previous section. For each ω, let the map $t \mapsto S_t(\omega)$ on $[0, \infty)$ be the linear interpolation of the map $n \mapsto S_n(\omega)$ on $\mathbf{Z}^+$, so that

$$S_t(\omega) := (t-n)S_{n+1}(\omega) + (n+1-t)S_n(\omega), \qquad t \in [n, n+1).$$

With Kolmogorov's result in mind, define

$$Z_n(t, \omega) := \frac{S_{nt}(\omega)}{\sqrt{2n \log\log n}}, \qquad t \in [0, 1],$$

so that $t \mapsto Z_n(t,\omega)$ on [0,1] is a rescaled version of the random walk S run up to time n. Say that a function $t \mapsto f(t,\omega)$ is in the set $K(\omega)$ of limiting shapes of the path associated with ω if there is a sequence $n_1(\omega), n_2(\omega), \ldots$ in $\mathbf{N}$ such that

$$Z_n(t,\omega) \to f(t,\omega) \text{ uniformly in } t \in [0,1].$$

Now let K consist of those functions f in $C[0,1]$ which can be written in the Lebesgue-integral form

$$f(t) = \int_0^t h(s)ds \quad \text{where} \quad \int_0^1 h(s)^2 ds \leq 1.$$

Strassen's Theorem

$$\mathbf{P}[K(\omega) = K] = 1.$$

Thus, (almost) all paths have the same limiting shapes. Khinchine's law follows from Strassen's precisely because (Exercise!)

$$\sup\{f(1): f \in K\} = 1, \quad \inf\{f(1): f \in K\} = -1.$$

However, the only element of K for which $f(1) = 1$ is the function $f(t) = t$, so the big values of S occur when the whole path (when rescaled) looks like a line of slope 1.

Almost every path will, in its Z rescaling, look infinitely often like the function t and infinitely often like the function $-t$; etc., etc.

References. For a highly-motivated classical proof of Strassen's Law, see Freedman (1971). For a proof for Brownian motion based on the powerful theory of large deviations, see Stroock (1984).

A4.3. A model for a Markov chain

Let E be a countable set; let μ be a probability measure on $(E, \mathcal{E})$, where $\mathcal{E}$ denotes the set of all subsets of E; and let P denote a stochastic $E \times E$ matrix as in Section 4.8.

Complicating the notation somewhat for reasons we shall discover later, we wish to construct a probability triple $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbf{P}}^\mu)$ carrying an E-valued stochastic process $(\tilde{Z}_n : n \in \mathbf{Z}^+)$ such that for $n \in \mathbf{Z}^+$ and $i_0, i_1, \ldots, i_n \in E$, we have

$$\tilde{\mathbf{P}}^\mu(\tilde{Z}_0 = i_0; \ldots; \tilde{Z}_n = i_n) = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.$$

The trick is to make $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbf{P}}^{\mu})$ carry *independent* E-valued variables

$$(\tilde{Z}_0; \tilde{Y}(i,n): i \in E; n \in \mathbf{N})$$

$\tilde{Z}_0$ having law μ and such that

$$\tilde{\mathbf{P}}^{\mu}(\tilde{Y}(i,n) = j) = p(i,j), \quad (i,j \in \mathbf{E}).$$

We can obviously do this via the construction in Section 4.6.

For $\tilde{\omega} \in \tilde{\Omega}$ and $n \in \mathbf{N}$, define

$$\tilde{Z}_n(\tilde{\omega}) := \tilde{Y}(\tilde{Z}_{n-1}(\tilde{\omega}), n);$$

and that's it!

Chapter A5

Appendix to Chapter 5

Our task is to prove the Monotone-Convergence Theorem 5.3. We need an elementary preliminary result.

A5.1. Doubly monotone arrays

Proposition. *Let*

$$(y_n^{(r)} : r \in \mathbf{N}, n \in \mathbf{N})$$

be an array of numbers in $[0, \infty]$ *which is doubly monotone:*

$$\textit{for fixed } r,\ y_n^{(r)} \uparrow \textit{ as } n \uparrow \textit{ so that } y^{(r)} := \uparrow \lim_n y_n^{(r)} \textit{ exists};$$

$$\textit{for fixed } n,\ y_n^{(r)} \uparrow \textit{ as } r \uparrow \textit{ so that } y_n := \uparrow \lim_r y_n^{(r)} \textit{ exists}.$$

Then

$$y^{(\infty)} := \uparrow \lim_r y^{(r)} = \uparrow \lim_n y_n =: y_\infty.$$

Proof. The result is almost trivial. By replacing each $(y_n^{(r)})$ by arc tan $y_n^{(r)}$, we can assume that the $y_\cdot^{(\cdot)}$ are uniformly bounded.

Let $\varepsilon > 0$ be given. Choose n_0 such that $y_{n_0} > y_\infty - \frac{1}{2}\varepsilon$. Then choose r_0 such that $y_{n_0}^{(r_0)} > y_{n_0} - \frac{1}{2}\varepsilon$. Then

$$y^{(\infty)} \geq y^{(r_0)} \geq y_{n_0}^{(r_0)} > y_\infty - \varepsilon,$$

so that $y^{(\infty)} \geq y_\infty$. Similarly, $y_\infty \geq y^{(\infty)}$. □

A5.2. The key use of Lemma 1.10(a)

This is where the fundamental monotonicity property of measures is used. Please re-read Section 5.1 at this stage.

LEMMA

(a) *Suppose that* $A \in \Sigma$ *and that*

$$h_n \in SF^+ \quad \textit{and} \quad h_n \uparrow \mathrm{I}_A.$$

Then $\mu_0(h_n) \uparrow \mu(A)$.

Proof. From (5.1,e), $\mu_0(h_n) \leq \mu(A)$, so we need only prove that

$$\liminf \mu_0(h_n) \geq \mu(A).$$

Let $\varepsilon > 0$, and define $A_n := \{s \in A : h_n(s) > 1 - \varepsilon\}$. Then $A_n \uparrow A$ so that, by Lemma 1.10(a), $\mu(A_n) \uparrow \mu(A)$. But

$$(1 - \varepsilon)\mathrm{I}_{A_n} \leq h_n$$

so that, by (5.1,e), $(1 - \varepsilon)\mu(A_n) \leq \mu_0(h_n)$. Hence

$$\liminf \mu_0(h_n) \geq (1 - \varepsilon)\mu(A).$$

Since this is true for every $\varepsilon > 0$, the result follows. □

LEMMA

(b) *Suppose that* $f \in SF^+$ *and that*

$$g_n \in SF^+ \quad \textit{and} \quad g_n \uparrow f.$$

Then $\mu_0(g_n) \uparrow \mu_0(f)$.

Proof. We can write f as a finite sum $f = \sum a_k \mathrm{I}_{A_k}$ where the sets A_k are disjoint and each $a_k > 0$. Then

$$a_k^{-1}\mathrm{I}_{A_k} g_n \uparrow \mathrm{I}_{A_k} \qquad (n \uparrow \infty),$$

and the result follows from Lemma (a). □

A5.3. 'Uniqueness of integral'

LEMMA

(a) *Suppose that* $f \in (\mathrm{m}\Sigma)^+$ *and that we have two sequences* $(f^{(r)})$ *and* (f_n) *of elements of* SF^+ *such that*

$$f^{(r)} \uparrow f, \quad f_n \uparrow f.$$

Then

$$\uparrow \lim \mu_0(f^{(r)}) = \uparrow \lim \mu_0(f_n).$$

Proof. Let $f_n^{(r)} := f^{(r)} \wedge f_n$. Then as $r \uparrow \infty$, $f_n^{(r)} \uparrow f_n$, and as $n \uparrow \infty$, $f_n^{(r)} \uparrow f^{(r)}$. Hence, by Lemma A5.2(b),

$$\mu_0(f_n^{(r)}) \uparrow \mu_0(f_n) \text{ as } r \uparrow \infty,$$
$$\mu_0(f_n^{(r)}) \uparrow \mu_0(f^{(r)}) \text{ as } n \uparrow \infty.$$

The result now follows from Proposition A5.1. □

Recall from Section 5.2 that for $f \in (\mathrm{m}\Sigma)^+$, we define

$$\mu(f) := \sup\{\mu_0(h) : h \in SF^+; h \leq f\} \leq \infty.$$

By definition of $\mu(f)$, we may choose a sequence h_n in SF^+ such that $h_n \leq f$ and $\mu_0(h_n) \uparrow \mu(f)$. Let us also choose a sequence (g_n) of elements of SF^+ such that $g_n \uparrow f$. (We can do this via the 'staircase function' in Section 5.3.) Now let

$$f_n := \max(g_n, h_1, h_2, \ldots, h_n).$$

Then $f_n \in SF^+$, $f_n \leq f$, and since $f_n \geq g_n$, $f_n \uparrow f$. Since $f_n \leq f$, $\mu_0(f_n) \leq \mu(f)$, and since $f_n \geq h_n$, we see that

$$\mu_0(f_n) \uparrow \mu(f).$$

On combining this fact with Lemma (a), we obtain the next result. (Lemma (a) 'changes our particular sequence to *any* sequence'.)

LEMMA

(b) *Let $f \in (\mathrm{m}\Sigma)^+$ and let (f_n) be any sequence in SF^+ such that $f_n \uparrow f$. Then*

$$\mu(f_n) = \mu_0(f_n) \uparrow \mu(f).$$

A5.4. Proof of the Monotone-Convergence Theorem

Recall the statement:

Let (f_n) be a sequence of elements of $(\mathrm{m}\Sigma)^+$ such that $f_n \uparrow f$. Then

$$\mu(f_n) \uparrow \mu(f).$$

Proof. Let $\alpha^{(r)}$ denote the r^{th} staircase function defined in Section 5.3. Now set $f_n^{(r)} := \alpha^{(r)}(f_n)$, $f^{(r)} := \alpha^{(r)}(f)$. Since $\alpha^{(r)}$ is left-continuous, $f_n^{(r)} \uparrow f^{(r)}$ as $n \uparrow \infty$. Since $\alpha^{(r)}(x) \uparrow x$, $\forall x$, $f_n^{(r)} \uparrow f_n$ as $r \uparrow \infty$. By Lemma A5.2(b), $\mu(f_n^{(r)}) \uparrow \mu(f^{(r)})$ as $n \uparrow \infty$; and by Lemma A5.3(b), $\mu(f_n^{(r)}) \uparrow \mu(f_n)$ as $r \uparrow \infty$. We also know from Lemma A5.3(b) that $\mu(f^{(r)}) \uparrow \mu(f)$. The result now follows from Proposition A5.1. □

Chapter A9

Appendix to Chapter 9

This chapter is solely devoted to the proof of the 'infinite-product' Theorem 8.6. It may be read after Section 9.10. It is probably something which a keen student who has read all previous appendices should study with a tutor.

A9.1. Infinite products: setting things up

Let $(\Lambda_n :\in \mathbf{N})$ be a sequence of probability measures on $(\mathbf{R}, \mathcal{B})$. Let

$$\Omega := \prod_{n\in\mathbf{N}} \mathbf{R},$$

so that a typical element ω of Ω is a sequence $\omega = (\omega_n : n \in \mathbf{N})$ of elements of $\mathbf{R}$. Define $X_n(\omega) := \omega_n$, and set

$$\mathcal{F}_n := \sigma(X_1, X_2, \ldots, X_n).$$

The typical element F_n of $\mathcal{F}_n$ has the form

(a) $$F_n = G_n \times \prod_{k>n} \mathbf{R}, \qquad G_n \in \prod_{1\le k\le n} \mathcal{B}.$$

Fubini's Theorem shows that on the *algebra* (NOT σ-algebra)

$$\mathcal{F}^- = \bigcup \mathcal{F}_n,$$

we may unambiguously use (a) to define a map $\mathbf{P}^- : \mathcal{F}^- \to [0,1]$ via

(b) $$\mathbf{P}^-(F_n) = (\Lambda_1 \times \cdots \times \Lambda_n)(G_n),$$

and that $\mathbf{P}^-$ is *finitely additive on the algebra* $\mathcal{F}^-$.

However, for each fixed n,

(c) $(\Omega, \mathcal{F}_n, \mathbf{P}^-)$ *is a bona fide probability triple which may be identified with* $\prod_{1\le k\le n}(\mathbf{R}, \mathcal{B}, \Lambda_k)$ *via* (a) *and* (b). *Moreover,* $X_1, X_2, \ldots, X_n$ *are independent RVs on* $(\Omega, \mathcal{F}_n, \mathbf{P}^-)$.

We want to prove that

(d) $\mathbf{P}^-$ *is countably additive on* $\mathcal{F}^-$

(obviously with the intention of using Carathéodory's Theorem 1.7). Now we know from our proof of the existence of Lebesgue measure (see (A1.9,a)) that it is enough to show that

(e) *if* (H_r) *is a sequence of sets in* $\mathcal{F}^-$ *such that* $H_r \supseteq H_{r+1}, \forall r$, *and if for some* $\varepsilon > 0$, $\mathbf{P}^-(H_r) \ge \varepsilon$ *for every* r, *then* $\bigcap H_r \ne \emptyset$.

A9.2. Proof of (A9.1,e)

Step 1: For every r, there is some $n(r)$ such that $H_r \in \mathcal{F}_{n(r)}$ and so

$$I_{H_r}(\omega) = h_r(\omega_1, \omega_2, \ldots, \omega_{n(r)}) \quad \text{for some} \quad h_r \in b\mathcal{B}^{n(r)}.$$

Recall that $X_k(\omega) = \omega_k$, and look again at Section A3.2.

Step 2: We have

$$\text{(a0)} \qquad \mathbf{E}^- h_r(X_1, X_2, \ldots, X_{n(r)}) \ge \varepsilon, \forall r,$$

because the left-hand side of (a0) is exactly $\mathbf{P}^-(H_r)$. If we work within the probability triple $(\Omega, \mathcal{F}_{n(r)}, \mathbf{P}^-)$, then we know from Section 9.10 that

$$\gamma_r(\omega) :\dot{=} g_r(\omega_1) := \mathbf{E}^- h_r(\omega_1, X_2, X_3, \ldots, X_{n(r)})$$

is an explicit version of the conditional expectation of I_{H_r} given $\mathcal{F}_1$, and

$$\varepsilon \le \mathbf{P}^-(H_r) = \mathbf{E}^-(\gamma_r) = \Lambda_1(g_r).$$

Now, $0 \le g_r \le 1$, so that

$$\begin{aligned} \varepsilon \le \Lambda_1(g_r) &\le 1\Lambda_1\{g_r \ge \varepsilon 2^{-1}\} + \varepsilon 2^{-1}\Lambda_1\{g_r^{(1)} \le \varepsilon 2^{-1}\} \\ &\le \Lambda_1\{g_r \ge \varepsilon 2^{-1}\} + \varepsilon 2^{-1}. \end{aligned}$$

Thus

$$\Lambda_1\{g_r \ge \varepsilon 2^{-1}\} \ge 2^{-1}\varepsilon.$$

Step 3: However, since $H_r \supseteq H_{r+1}$, we have (working within $(\Omega, \mathcal{F}_m, \mathbf{P}^-)$ where both H_r and H_{r+1} are in $\mathcal{F}_m$)

$$g_r(\omega_1) \ge g_{r+1}(\omega_1), \quad \text{for every } \omega_1 \text{ in } \mathbf{R}.$$

Working on $(\mathbf{R}, \mathcal{B}, \Lambda_1)$ we have

$$\Lambda_1\{g_r \geq \varepsilon 2^{-1}\} \geq \varepsilon 2^{-1}, \forall r,$$

and

$$g_r \downarrow \text{ so that } \{g_r \geq \varepsilon 2^{-1}\} \downarrow;$$

and by Lemma 1.10(b) on the continuity from above of measures, we have

$$\Lambda_1\{\omega_1 : g_r(\omega_1) \geq \varepsilon 2^{-1}, \forall r\} \geq \varepsilon 2^{-1}.$$

Hence, there exists ω_1^* (say) in $\mathbf{R}$ such that

$$\text{(a1)} \qquad \mathsf{E}^- h_r(\omega_1^*, X_2, \ldots, X_{n(r)}) \geq \varepsilon 2^{-1}, \forall r.$$

Step 4: We now repeat Steps 2 and 3 applied to the situation in which

$$(X_1, X_2, \ldots) \quad \text{is replaced by} \quad (X_2, X_3, \ldots),$$

h_r is replaced by $h_r(\omega_1^*)$, where

$$(h_r(\omega_1^*))(\omega_2, \omega_3, \ldots) := h_r(\omega_1^*, \omega_2, \omega_3, \ldots).$$

We find that there exists ω_2^* in $\mathbf{R}$ such that

$$\text{(a2)} \qquad \mathsf{E}^- h_r(\omega_1^*, \omega_2^*, X_3, \ldots, X_{n(r)}) \geq \varepsilon 2^{-2}, \quad \forall r.$$

Proceeding inductively, we obtain a sequence

$$\omega^* = (\omega_n^* : n \in \mathbf{N})$$

with the property that

$$\mathsf{E}^- h_r(\omega_1^*, \omega_2^*, \ldots, \omega_{n(r)}^*) \geq \varepsilon 2^{-n(r)}, \forall r.$$

However,

$$h_r(\omega_1^*, \omega_2^*, \ldots, \omega_{n(r)}^*) = I_{H_r}(\omega^*),$$

and can only be 0 or 1. The only conclusion is that $\omega^* \in H_r, \forall r$; and it was exactly the existence of such an ω^* which we had to prove. □

Chapter A13

Appendix to Chapter 13

This chapter is devoted to comments on **modes of convergence**, regarded by many as good for the souls of students, and certainly easy to set examination questions on.

A13.1. Modes of convergence: definitions

Let $(X_n : n \in \mathbb{N})$ be a sequence of RVs and let X be a RV, all carried by our triple $(\Omega, \mathcal{F}, \mathbf{P})$. Let us collect together definitions known to us.

Almost sure convergence

We say that $X_n \to X$ **almost surely** if

$$\mathbf{P}(X_n \to X) = 1.$$

Convergence in probability

We say that $X_n \to X$ **in probability** if, for every $\varepsilon > 0$,

$$\mathbf{P}(|X_n - X| > \varepsilon) \to 0 \quad \text{as} \qquad n \to \infty.$$

$\mathcal{L}^p$ convergence $(p \geq 1)$

We say that $X_n \to X$ in $\mathcal{L}^p$ if each X_n is in $\mathcal{L}^p$ and $X \in \mathcal{L}^p$ and

$$\|X_n - X\|_p \to 0 \qquad \text{as} \quad n \to \infty,$$

equivalently,

$$\mathbf{E}(|X_n - X|^p) \to 0 \qquad \text{as} \quad n \to \infty.$$

A13.2. Modes of convergence: relationships

Let me state the *facts.*

Convergence in probability is the weakest of the above forms of convergence. Thus

(a) $(X_n \to X, \text{ a.s.}) \Rightarrow (X_n \to X \text{ in prob})$

(b) for $p \geq 1$,

$$(X_n \to X \text{ in } \mathcal{L}^p) \Rightarrow (X_n \to X \text{ in prob}).$$

No other implication between any two of our three forms of convergence is valid. But, of course, for $r \geq p \geq 1$,

(c) $(X_n \to X \text{ in } \mathcal{L}^r) \Rightarrow (X_n \to X \text{ in } \mathcal{L}^p)$.

If we know that 'convergence in probability is happening quickly' in that

(d) $\sum_n \mathsf{P}(|X_n - X| > \varepsilon) < \infty, \quad \forall \varepsilon > 0,$

then (BC1) allows us to conclude that $X_n \to X$, a.s.

The fact that property (d) implies a.s. convergence is used in proving the following result:

(e) *$X_n \to X$ in probability if and only if every subsequence of (X_n) contains a further subsequence along which we have almost sure convergence to X.*

The only other *useful* result is that

(f) *for $p \geq 1$, $X_n \to X$ in $\mathcal{L}^p$ if and only if the following two statements hold:*

(i) *$X_n \to X$ in probability,*

(ii) *the family $(|X_n|^p : n \geq 1)$ is UI.*

There is only one way to gain an understanding of the above facts, and that is to prove them yourself. The exercises under EA13 provide guidance if you need it.

Chapter A14

Appendix to Chapter 14

We work with a filtered space $(\Omega, \mathcal{F}, \{\mathcal{F}_n : n \in \mathbf{Z}^+\}, \mathbf{P})$.

This chapter introduces the σ-algebra $\mathcal{F}_T$, where T is a stopping time. The idea is that $\mathcal{F}_T$ represents the information available to our observer at (or, if you prefer, immediately after) time T. The **Optional-Sampling Theorem** says that if X is a *uniformly integrable* supermartingale and S and T are stopping times with $S \leq T$, then we have the natural extension of the supermartingale property:

$$\mathbf{E}(X_T | \mathcal{F}_S) \leq X_S, \quad \text{a.s.}$$

A14.1. The σ-algebra $\mathcal{F}_T$, T a stopping time

Recall that a map $T : \Omega \to \mathbf{Z}^+ \cup \{\infty\}$ is called a *stopping time* if

$$\{T \leq n\} \in \mathcal{F}_n, \quad n \in \mathbf{Z}^+ \cup \{\infty\},$$

equivalently if

$$\{T = n\} \in \mathcal{F}_n, \quad n \in \mathbf{Z}^+ \cup \{\infty\}.$$

In each of the above statements, the '$n = \infty$' case follows automatically from the validity of the result for every n in $\mathbf{Z}^+$.

Let T be a stopping time. Then, for $F \subseteq \Omega$, we say that $F \in \mathcal{F}_T$ if

►► $$F \cap \{T \leq n\} \in \mathcal{F}_n, \quad n \in \mathbf{Z}^+ \cup \{\infty\},$$

equivalently if

$$F \cap \{T = n\} \in \mathcal{F}_n, \quad n \in \mathbf{Z}^+ \cup \{\infty\}.$$

Then $\mathcal{F}_T = \mathcal{F}_n$ if $T \equiv n$; $\mathcal{F}_T = \mathcal{F}_\infty$ if $T \equiv \infty$; and $\mathcal{F}_T \subseteq \mathcal{F}_\infty$ for every T.

You can easily check that $\mathcal{F}_T$ is a σ-algebra. You can also check that if S is another stopping time, then

$$\mathcal{F}_{S\wedge T} \subseteq \mathcal{F}_T \subseteq \mathcal{F}_{S\vee T}.$$

Hint. If $F \in \mathcal{F}_{S\wedge T}$, then

$$F \cap \{T = n\} = \bigcup_{k\le n} F \cap \{S \wedge T = k\}. \qquad \square$$

Another detail that needs to be checked is that if X is an adapted process and T is a stopping time, then $X_T \in m\mathcal{F}_T$. Here, X_∞ is assumed defined in some way such that X_∞ is $\mathcal{F}_\infty$ measurable.

Proof. For $B \in \mathcal{B}$,

$$\{X_T \in B\} \cap \{T = n\} = \{X_n \in B\} \cap \{T = n\} \in \mathcal{F}_n. \qquad \square$$

A14.2. A special case of OST

LEMMA

Let X be a supermartingale. Let T be a stopping time such that, for some N in $\mathbf{N}$, $T(\omega) \le N$, $\forall\omega$. Then $X_T \in \mathcal{L}^1(\Omega, \mathcal{F}_T, \mathbf{P})$ and

$$\mathsf{E}(X_N|\mathcal{F}_T) \le X_T.$$

Proof. Let $F \in \mathcal{F}_T$. Then

$$\begin{aligned}\mathsf{E}(X_N; F) &= \sum_{n\le N} \mathsf{E}(X_N; F \cap \{T = n\}) \\ &\le \sum_{n\le N} \mathsf{E}(X_n; F \cap \{T = n\}) = \mathsf{E}(X_T; F).\end{aligned}$$

(Of course, the fact that $|X_T| \le |X_1| + \cdots + |X_N|$ guarantees the result that $\mathsf{E}(|X_T|) < \infty$.) $\square$

A14.3. Doob's Optional-Sampling Theorem for UI martingales

►► *Let M be a UI martingale. Then, for any stopping time T,*

$$\mathsf{E}(M_\infty|\mathcal{F}_T) = M_T, \quad \text{a.s.}$$

Corollary 1 (a new Optional-Stopping Theorem!)

If M is a UI martingale, and T is a stopping time, then $\mathsf{E}(|M_T|) < \infty$ *and* $\mathsf{E}(M_T) = \mathsf{E}(M_0)$.

Corollary 2

If M is a UI martingale and S and T are stopping times with $S \leq T$, then

$$\mathsf{E}(M_T|\mathcal{F}_S) = M_S, \quad \text{a.s.}$$

Proof of theorem. By Theorem 14.1 and Lemma A14.2, we have, for $k \in \mathbb{N}$,

$$\mathsf{E}(M_\infty|\mathcal{F}_k) = M_k, \quad \text{a.s.}, \quad \mathsf{E}(M_k|\mathcal{F}_{T\wedge k}) = M_{T\wedge k}, \quad \text{a.s.}$$

Hence, by the Tower Property,

$$(*) \qquad \mathsf{E}(M_\infty|\mathcal{F}_{T\wedge k}) = M_{T\wedge k}, \quad \text{a.s.}$$

If $F \in \mathcal{F}_T$, then (check!) $F \cap \{T \leq k\} \in \mathcal{F}_{T\wedge k}$, so that, by $(*)$,

$$(**) \quad \mathsf{E}(M_\infty; F\cap\{T \leq k\}) = \mathsf{E}(M_{T\wedge k}; F\cap\{T \leq k\}) = \mathsf{E}(M_T; F\cap\{T \leq k\}).$$

We can (and do) restrict attention to the case when $M_\infty \geq 0$, whence $M_n = \mathsf{E}(M_\infty|\mathcal{F}_n) \geq 0$ for all n. Then, on letting $k \uparrow \infty$ in $(**)$ and using (MON), we obtain

$$\mathsf{E}(M_\infty; F \cap \{T < \infty\}) = \mathsf{E}(M_T; F \cap \{T < \infty\}).$$

However, the fact that

$$\mathsf{E}(M_\infty; F \cap \{T = \infty\}) = \mathsf{E}(M_T; F \cap \{T = \infty\}).$$

is tautological. Hence $\mathsf{E}(M_\infty; F) = \mathsf{E}(M_T; F)$. □

Corollary 2 now follows from the Tower Property, and Corollary 1 follows from Corollary 2!

A14.4. The result for UI submartingales

A UI submartingale X has Doob decomposition

$$X = X_0 + M + A,$$

where (**Exercise**: explain why!) $\mathsf{E}(A_\infty) < \infty$ and M is UI. Hence, if T is a stopping time, then, almost surely,

$$\begin{aligned}\mathsf{E}(X_\infty|\mathcal{F}_T) &= X_0 + \mathsf{E}(M_\infty|\mathcal{F}_T) + \mathsf{E}(A_\infty|\mathcal{F}_T)\\ &= X_0 + M_T + \mathsf{E}(A_\infty|\mathcal{F}_T)\\ &\geq X_0 + M_T + \mathsf{E}(A_T|\mathcal{F}_T)\\ &= X_T.\end{aligned}$$

Chapter A16

Appendix to Chapter 16

A16.1. Differentiation under the integral sign

Before stating our theorem on this topic, let us examine the type of application we need in Section 16.3. Suppose that X is a RV such that $\mathbf{E}(|X|) < \infty$ and that $h(t,x) = ixe^{itx}$. (We can treat the real and imaginary parts of h separately.) Note that if $[a,b]$ is a subinterval of $\mathbf{R}$, then the variables $\{h(t,X) : t \in [a,b]\}$ are *dominated* by $|X|$, and so are UI. In the theorem, we shall have

$$\mathbf{E}H(t,X) = \varphi_X(t) - \varphi_X(a), \qquad t \in [a,b],$$

and we can conclude that $\varphi'_X(t)$ exists and equals $\mathbf{E}h(t,X)$.

THEOREM

Let X be a RV carried by $(\Omega, \mathcal{F}, \mathbf{P})$.

Suppose that $a, b \in \mathbf{R}$ with $a < b$, and that

$$h : [a,b] \times \mathbf{R} \to \mathbf{R}$$

has the properties:

(i) *$t \mapsto h(t,x)$ is continuous in t for every x in $\mathbf{R}$,*

(ii) *$x \mapsto h(t,x)$ is $\mathcal{B}$-measurable for every t in $[a,b]$,*

(iii) *the variables $\{h(t,X) : t \in [a,b]\}$ are UI.*

Then

(a) *$t \mapsto \mathbf{E}h(t,X)$ is continuous on $[a,b]$,*

(b) *h is $\mathcal{B}[a,b] \times \mathcal{B}$-measurable,*

(c) *if* $H(t,x) := \int_a^t h(s,x)ds$ *for* $a \le t \le b$, *then for* $t \in (a,b)$,

$$\frac{d}{dt}\mathsf{E}H(t,X) \quad \textit{exists and equals} \quad \mathsf{E}h(t,X).$$

Proof of (a). Since we need only consider the 'sequential case $t_n \to t$', result (a) follows immediately from Theorem 13.7. □

Proof of (b). Define $\delta_n := 2^{-n}(b-a)$, $D_n := (a + \delta\mathbf{Z}^+) \cap [a,b]$,

$$\tau_n(t) := \inf\{\tau \in D_n : \tau \ge t\}, \qquad t \in [a,b],$$
$$h_n(t,x) := h(\tau_n(t),x), \qquad t \in [a,b],\ x \in \mathbf{R}.$$

Then, for $B \in \mathcal{B}$,

$$h_n^{-1}(B) = \bigcup_{\tau \in D_n} (([\tau, \tau+\delta) \cap [a,b]) \times \{x : h(\tau,x) \in B\}),$$

so that h_n is $\mathcal{B}[a,b] \times \mathcal{B}$-measurable. Since $h_n \to h$ on $[a,b] \times \mathbf{R}$, result (b) follows. □

Proof of (c). For $\Gamma \subseteq [a,b] \times \mathbf{R}$, define

$$\alpha(\Gamma) := \{(t,\omega) \in [a,b] \times \Omega : (t, X(\omega)) \in \Gamma\}.$$

If $\Gamma = A \times C$ where $A \in \mathcal{B}[a,b]$ and $C \in \mathcal{B}$, then

$$\alpha(\Gamma) = A \times (X^{-1}C) \in \mathcal{B}[a,b] \times \mathcal{F}.$$

It is now clear that the class of Γ for which $\alpha(\Gamma)$ is an element of $\mathcal{B}[a,b] \times \mathcal{F}$ is a σ-algebra containing $\mathcal{B}[a,b] \times \mathcal{B}$. The point of all this is that

$$(*) \qquad (t,\omega) \mapsto h(t, X(\omega)) \text{ is } \mathcal{B} \times \mathcal{F}\text{- measurable}$$

since for $B \in \mathcal{B}$, $\{(t,\omega) : h(t,X(\omega)) \in B\}$ is $\alpha(h^{-1}B)$. (Yes, I know, we could have obtained $(*)$ more directly using the h_n's, but it is good to have other methods.) Since the family $\{h(t,X) : t \ge 0\}$ is UI, it is bounded in $\mathcal{L}^1$, whence

$$\int_a^b \mathsf{E}|h(t,X)|dt < \infty.$$

Fubini's Theorem now implies that, for $a \le t \le b$,

$$\int_a^t \mathsf{E}h(s,X)ds = \mathsf{E}\int_a^t h(s,X)ds = \mathsf{E}H(t,X),$$

and part (c) now follows. □

Chapter E
Exercises

Starred exercises are more tricky. The first number in an exercise gives a rough indication of which **chapter** it depends on. 'G' stands for 'a bit of gumption is all that's necessary'. A number of exercises may also be found in the main text. Some are repeated here. We begin with an

Antidote to measure-theoretic material – *just for fun, though the point that probability is more than mere measure theory needs hammering home.*

EG.1. Two points are chosen at random on a line AB, each point being chosen according to the uniform distribution on AB, and the choices being made independently of each other. The line AB may now be regarded as divided into three parts. What is the probability that they may be made into a triangle?

EG.2. Planet X is a ball with centre O. Three spaceships A, B and C land at random on its surface, their positions being independent and each uniformly distributed on the surface. Spaceships A and B can communicate directly by radio if $\angle \mathrm{AOB} < 90°$. Show that the probability that they can keep in touch (with, for example, A communicating with B via C if necessary) is $(\pi + 2)/(4\pi)$.

EG.3. Let G be the free group with two generators a and b. Start at time 0 with the unit element 1, the empty word. At each second multiply the current word on the right by one of the four elements a, a^{-1}, b, b^{-1}, choosing each with probability 1/4 (independently of previous choices). The choices

$$a, a, b, a^{-1}, a, b^{-1}, a^{-1}, a, b$$

at times 1 to 9 will produce the reduced word aab of length 3 at time 9. Prove that the probability that the reduced word 1 ever occurs at a positive time is 1/3, and explain why it is intuitively clear that (almost surely)

$$(\text{length of reduced word at time } n)/n \to \tfrac{1}{2}.$$

EG.4.* (Continuation) Suppose now that the elements a, a^{-1}, b, b^{-1} are chosen instead with respective probabilities $\alpha, \alpha, \beta, \beta$, where $\alpha > 0, \beta > 0, \alpha + \beta = \frac{1}{2}$. Prove that the conditional probability that the reduced word 1 ever occurs at a positive time, *given that* the element a is chosen at time 1, is the unique root $x = r(\alpha)$ (say) in (0,1) of the equation

$$3x^3 + (3 - 4\alpha^{-1})x^2 + x + 1 = 0.$$

As time goes on, (it is almost surely true that) more and more of the reduced word becomes fixed, so that a final word is built up. If in the final word, the symbols a and a^{-1} are both replaced by A and the symbols b and b^{-1} are both replaced by B, show that the sequence of A's and B's obtained is a Markov chain on $\{A, B\}$ with (for example)

$$p_{AA} = \frac{\alpha(1-x)}{\alpha(1-x) + 2\beta(1-y)},$$

where $y = r(\beta)$. What is the (almost sure) limiting proportion of occurrence of the symbol a in the final word? (*Note.* This result was used by Professor Lyons of Edinburgh to solve a long-standing problem in potential theory on Riemannian manifolds.)

Algebras, etc.

E1.1. 'Probability' for subsets of $\mathbf{N}$

Let $V \subseteq \mathbf{N}$. Say that V has (Cesàro) density $\gamma(V)$ and write $V \in \mathrm{CES}$ if

$$\gamma(V) := \lim \frac{\sharp(V \cap \{1, 2, 3, \ldots, n\})}{n}$$

exists. Give an example of sets V_1 and V_2 in CES for which $V_1 \cap V_2 \notin \mathrm{CES}$. Thus, CES is not an algebra.

Independence

E4.1. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability triple. Let $\mathcal{I}_1$, $\mathcal{I}_2$ and $\mathcal{I}_3$ be three π-systems on Ω such that, for $k = 1, 2, 3$,

$$\mathcal{I}_k \subseteq \mathcal{F} \text{ and } \Omega \in \mathcal{I}_k.$$

Prove that if

$$\mathbf{P}(I_1 \cap I_2 \cap I_3) = \mathbf{P}(I_1)\mathbf{P}(I_2)\mathbf{P}(I_3)$$

whenever $I_k \in \mathcal{I}_k \quad (k = 1, 2, 3)$, then $\sigma(\mathcal{I}_1), \sigma(\mathcal{I}_2), \sigma(\mathcal{I}_3)$ are independent. Why did we require that $\Omega \in \mathcal{I}_k$?

E4.2. Let $s > 1$, and define $\zeta(s) := \sum_{n \in \mathbb{N}} n^{-s}$, as usual. Let X and Y be independent $\mathbb{N}$-valued random variables with

$$\mathbf{P}(X = n) = \mathbf{P}(Y = n) = n^{-s}/\zeta(s).$$

Prove that the events $(E_p : p \text{ prime})$, where $E_p = \{X \text{ is divisible by } p\}$, are independent. Explain Euler's formula

$$1/\zeta(s) = \prod_p (1 - 1/p^s)$$

probabilistically. Prove that

$$\mathbf{P}(\text{no square other than 1 divides } X) = 1/\zeta(2s).$$

Let H be the highest common factor of X and Y. Prove that

$$\mathbf{P}(H = n) = n^{-2s}/\zeta(2s).$$

E4.3. Let $X_1, X_2, \ldots$ be independent random variables with the same *continuous* distribution function. Let $E_1 := \Omega$, and, for $n \geq 2$, let

$$E_n := \{X_n > X_m, \forall m < n\} = \{\text{a 'Record' occurs at time } n\}.$$

Convince yourself and your tutor that the events $E_1, E_2, \ldots$ are independent, with $\mathbf{P}(E_n) = 1/n$.

Borel-Cantelli Lemmas

E4.4. Suppose that a coin with probability p of heads is tossed repeatedly. Let A_k be the event that a sequence of k (or more) consecutive heads occurs amongst tosses numbered $2^k, 2^k + 1, 2^k + 2, \ldots, 2^{k+1} - 1$. Prove that

$$\mathbf{P}(A_k, \text{ i.o.}) = \begin{cases} 1 & \text{if } p \geq \frac{1}{2}, \\ 0 & \text{if } p < \frac{1}{2}. \end{cases}$$

Hint. Let E_i be the event that there are k consecutive heads beginning at toss numbered $2^k + (i-1)k$. Now make a simple use of the inclusion-exclusion formulae (Lemma 1.9).

E4.5. Prove that if G is a random variable with the normal N(0,1) distribution, then, for $x > 0$,

$$\mathsf{P}(G > x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{1}{2}y^2} dy \leq \frac{1}{x\sqrt{2\pi}} e^{-\frac{1}{2}x^2}.$$

Let $X_1, X_2, \ldots$ be a sequence of independent N(0,1) variables. Prove that, with probability 1, $L \leq 1$, where

$$L := \limsup(X_n / \sqrt{2 \log n}).$$

(*Harder.* Prove that $\mathsf{P}(L = 1) = 1$.) [*Hint.* See Section 14.8.]

Let $S_n := X_1 + X_2 + \cdots + X_n$. Recall that $S_n/\sqrt{n}$ has the N(0,1) distribution. Prove that

$$\mathsf{P}(|S_n| < 2\sqrt{n \log n}, \text{ ev}) = 1.$$

Note that this implies the Strong Law: $\mathsf{P}(S_n/n \to 0) = 1$.

Remark. The *Law of the Iterated Logarithm* states that

$$\mathsf{P}\left(\limsup \frac{S_n}{\sqrt{2n \log\log n}} = 1\right) = 1.$$

Do not attempt to prove this now! See Section 14.7.

E4.6. Converse to SLLN

Let Z be a non-negative RV. Let Y be the integer part of Z. Show that

$$Y = \sum_{n \in \mathbf{N}} \mathrm{I}_{\{Z \geq n\}},$$

and deduce that

$$(*) \qquad \sum_{n \in \mathbf{N}} \mathsf{P}[Z \geq n] \leq \mathsf{E}(Z) \leq 1 + \sum_{n \in \mathbf{N}} \mathsf{P}[Z \geq n].$$

Let (X_n) be a sequence of IID RVs (independent, identically distributed random variables) with $\mathsf{E}(|X_n|) = \infty, \forall n$. Prove that

$$\sum_n \mathsf{P}[X_n| > kn] = \infty \ (k \in \mathbf{N}) \quad \text{and} \quad \limsup \frac{|X_n|}{n} = \infty, \quad \text{a.s.}$$

Deduce that if $S_n = X_1 + X_2 + \cdots + X_n$, then

$$\limsup \frac{|S_n|}{n} = \infty, \quad \text{a.s.}$$

E4.7. What's fair about a fair game?

Let $X_1, X_2, \ldots$ be independent RVs such that

$$X_n = \begin{cases} n^2 - 1 & \text{with probability } n^{-2} \\ -1 & \text{with probability } 1 - n^{-2}. \end{cases}$$

Prove that $\mathbf{E}(X_n) = 0$, $\forall n$, but that if $S_n = X_1 + X_2 + \cdots + X_n$, then

$$\frac{S_n}{n} \to -1, \quad \text{a.s.}$$

E4.8*. Blackwell's test of imagination

This exercise assumes that you are familiar with continuous-parameter Markov chains with two states.

For each $n \in \mathbf{N}$, let $X^{(n)} = \{X^{(n)}(t) : t \geq 0\}$ be a Markov chain with state-space the two-point set $\{0, 1\}$ with Q-matrix

$$Q^{(n)} = \begin{pmatrix} -a_n & a_n \\ b_n & -b_n \end{pmatrix}, \quad a_n, b_n > 0,$$

and transition function $P^{(n)}(t) = \exp(tQ^{(n)})$. Show that, for every t,

$$p_{00}^{(n)}(t) \geq b_n/(a_n + b_n), \quad p_{01}^{(n)}(t) \leq a_n/(a_n + b_n).$$

The processes $(X^{(n)} : n \in \mathbf{N})$ are independent and $X^{(n)}(0) = 0$ for every n. Each $X^{(n)}$ has right-continuous paths.

Suppose that $\sum a_n = \infty$ *and* $\sum a_n/b_n < \infty$.

Prove that if t is a fixed time then

$$(*) \qquad \mathbf{P}\{X^{(n)}(t) = 1 \text{ for infinitely many } n\} = 0.$$

Use Weierstrass's M-test to show that $\sum_n \log p_{00}^{(n)}(t)$ is *uniformly* convergent on $[0, 1]$, and deduce that

$$\mathbf{P}\{X^{(n)}(t) = 0 \text{ for ALL } n\} \to 1 \quad \text{as } t \downarrow 0.$$

Prove that

$$\mathbf{P}\{X^{(n)}(s) = 0, \forall s \leq t, \forall n\} = 0 \text{ for every } t > 0$$

and discuss with your tutor why it is almost surely true that

(**) within every non-empty time interval, infinitely many of the $X^{(n)}$ chains jump.

Now imagine the whole behaviour.

Notes. Almost surely, the process $X = (X^{(n)})$ spends *almost* all its time in the countable subset of $\{0,1\}^{\mathbf{N}}$ consisting of sequences with only finitely many 1's. This follows from (*) and Fubini's Theorem 8.2. However, it is a.s. true that X visits uncountable points of $\{0,1\}^{\mathbf{N}}$ during every non-empty time interval. This follows from (**) and the Baire category theorem A1.12. By using much deeper techniques, one can show that for certain choices of (a_n) and (b_n), X will almost certainly visit *every* point of $\{0,1\}^{\mathbf{N}}$ uncountably often within a finite time.

Tail σ-algebras

E4.9. Let $Y_0, Y_1, Y_2, \ldots$ be independent random variables with

$$\mathbf{P}(Y_n = +1) = \mathbf{P}(Y_n = -1) = \tfrac{1}{2}, \quad \forall n.$$

For $n \in \mathbf{N}$, define

$$X_n := Y_0 Y_1 \ldots Y_n.$$

Prove that the variables $X_1, X_2, \ldots$ are independent. Define

$$\mathcal{Y} := \sigma(Y_1, Y_2, \ldots), \quad \mathcal{T}_n := \sigma(X_r : r > n).$$

Prove that

$$\mathcal{L} := \bigcap_n \sigma(\mathcal{Y}, \mathcal{T}_n) \neq \sigma\left(\mathcal{Y}, \bigcap_n \mathcal{T}_n\right) =: \mathcal{R}.$$

Hint. Prove that $Y_0 \in \mathrm{m}\mathcal{L}$ and that Y_0 is independent of $\mathcal{R}$.

E4.10. *Star Trek*, 2

See E10.11, which you can do now.

Dominated-Convergence Theorem

E5.1. Let $S := [0,1]$, $\Sigma := \mathcal{B}(S)$, $\mu :=$ Leb. Define $f_n := nI_{(0,1/n)}$. Prove that $f_n(s) \to 0$ for every s in S, but that $\mu(f_n) = 1$ for every n. Draw a picture of $g := \sup_n |f_n|$, and show that $g \notin \mathcal{L}^1(S, \Sigma, \mu)$.

Inclusion-Exclusion Formulae

E5.2. Prove the inclusion-exclusion formulae and inequalities of Section 1.9 by considering integrals of indicator functions.

The Strong Law

E7.1. Inverting Laplace transforms

Let f be a bounded continuous function on $[0,\infty)$. The Laplace transform of f is the function L on $(0,\infty)$ defined by

$$L(\lambda) := \int_0^\infty e^{-\lambda x} f(x) dx.$$

Let $X_1, X_2, \ldots$ be independent RVs each with the exponential distribution of rate λ, so $\mathbf{P}[X > x] = e^{-\lambda x}$, $\mathbf{E}(X) = \frac{1}{\lambda}$, $\mathrm{Var}(X) = \frac{1}{\lambda^2}$. Show that

$$(-1)^{n-1} \frac{\lambda^n L^{(n-1)}(\lambda)}{(n-1)!} = \mathbf{E} f(S_n),$$

where $S_n = X_1 + X_2 + \cdots + X_n$, and $L^{(n-1)}$ denotes the $(n-1)^{th}$ derivative of L. Prove that f may be recovered from L as follows: for $y > 0$,

$$f(y) = \lim_{n \uparrow \infty} (-1)^{n-1} \frac{(n/y)^n L^{(n-1)}(n/y)}{(n-1)!}.$$

E7.2. The uniform distribution on the sphere $S^{n-1} \subseteq \mathbf{R}^n$

As usual, write $S^{n-1} = \{x \in \mathbf{R}^n : |x| = 1\}$. You may assume that there is a unique probability measure ν^{n-1} on $\left(S^{n-1}, \mathcal{B}(S^{n-1})\right)$ such that $\nu^{n-1}(A) = \nu^{n-1}(HA)$ for every orthogonal $n \times n$ matrix H and every A in $\mathcal{B}(S^{n-1})$.

Prove that if $\mathbf{X}$ is a vector in $\mathbf{R}_n$, the components of which are independent N(0,1) variables, then for every orthogonal $n \times n$ matrix H, the vector $H\mathbf{X}$ has the same property. Deduce that $\mathbf{X}/|\mathbf{X}|$ has law ν^{n-1}.

Let $Z_1, Z_2, \ldots$ be independent N(0,1) variables and define

$$R_n = (Z_1^2 + Z_2^2 + \cdots + Z_n^2)^{\frac{1}{2}}.$$

Prove that $R_n/\sqrt{n} \to 1$, a.s.

Combine these ideas to prove a rather striking fact which relates the normal distribution to the 'infinite-dimensional' sphere and which is important both for Brownian motion and for Fock-space constructions in quantum mechanics:

If, for each n, $(Y_1^{(n)}, Y_2^{(n)}, \ldots, Y_n^{(n)})$ is a point chosen on S^{n-1} according to the distribution ν^{n-1}, then

$$\lim_{n\to\infty} \mathbf{P}(\sqrt{n}Y_1^{(n)} \le x) = \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-y^2/2} dy,$$

$$\lim_{n\to\infty} \mathbf{P}(\sqrt{n}Y_1^{(n)} \le x_1;\ \sqrt{n}Y_2^{(n)} \le x_2) = \Phi(x_1)\Phi(x_2).$$

Hint. $\mathbf{P}(Y_1^{(n)} \le u) = \mathbf{P}(X_1/R_n \le u)$.

Conditional Expectation

E9.1. Prove that if $\mathcal{G}$ is a sub-σ-algebra of $\mathcal{F}$ and if $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbf{P})$ and if $Y \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbf{P})$ and

$$(*) \qquad \mathbf{E}(X; G) = \mathbf{E}(Y; G)$$

for every G in a π-system which contains Ω and generates $\mathcal{G}$, then (*) holds for every G in $\mathcal{G}$.

E9.2. Suppose that $X, Y \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbf{P})$ and that

$$\mathbf{E}(X|Y) = Y, \quad \text{a.s.}, \qquad \mathbf{E}(Y|X) = X, \quad \text{a.s.}$$

Prove that $\mathbf{P}(X = Y) = 1$.

Hint. Consider $\mathbf{E}(X - Y; X > c, Y \le c) + \mathbf{E}(X - Y; X \le c, Y \le c)$.

Martingales

E10.1. Pólya's urn

At time 0, an urn contains 1 black ball and 1 white ball. At each time 1,2,3,..., a ball is chosen at random from the urn and is replaced together with a new ball of the same colour. Just after time n, there are therefore $n + 2$ balls in the urn, of which $B_n + 1$ are black, where B_n is the number of black balls chosen by time n.

Let $M_n = (B_n + 1)/(n+2)$, the proportion of black balls in the urn just after time n. Prove that (relative to a natural filtration which you should specify) M is a martingale.

Prove that $\mathbf{P}(B_n = k) = (n + 1)^{-1}$ for $0 \le k \le n$. What is the distribution of Θ, where $\Theta := \lim M_n$?

Prove that for $0 < \theta < 1$,

$$N_n^\theta := \frac{(n+1)!}{B_n!(n-B_n)!}\,\theta^{B_n}(1-\theta)^{n-B_n}$$

defines a martingale N^θ. (*Continued at E10.8.*)

E10.2. Martingale formulation of Bellman's Optimality Principle

Your winnings per unit stake on game n are ε_n, where the ε_n are IID RVs with

$$\mathbf{P}(\varepsilon_n = +1) = p, \mathbf{P}(\varepsilon_n = -1) = q, \quad \text{where} \quad \tfrac{1}{2} < p = 1 - q < 1.$$

Your stake C_n on game n must lie between 0 and Z_{n-1}, where Z_{n-1} is your fortune at time $n-1$. Your object is to maximize the expected 'interest rate' $\mathbf{E}\log(Z_N/Z_0)$, where N is a given integer representing the length of the game, and Z_0, your fortune at time 0, is a given constant. Let $\mathcal{F}_n = \sigma(\varepsilon_1, \ldots, \varepsilon_n)$ be your 'history' up to time n. Show that if C is any (previsible) strategy, then $\log Z_n - n\alpha$ is a *super*martingale, where α denotes the '*entropy*'

$$\alpha = p\log p + q\log q + \log 2,$$

so that $\mathbf{E}\log(Z_N/Z_0) \le N\alpha$, but that, for a certain strategy, $\log Z_n - n\alpha$ is a *martingale*. What is the best strategy?

E10.3. Stopping times

Suppose that S and T are stopping times (relative to $(\Omega, \mathcal{F}, \{\mathcal{F}_n\})$). Prove that $S \wedge T$ $(:= \min(S,T))$, $S \vee T (:= \max(S,T))$ and $S + T$ are stopping times.

E10.4. Let S and T be stopping times with $S \le T$. Define the process $1_{(S,T]}$ with parameter set $\mathbf{N}$ via

$$1_{(S,T]}(n,\omega) := \begin{cases} 1 & \text{if } S(\omega) < n \le T(\omega), \\ 0 & \text{otherwise.} \end{cases}$$

Prove that $1_{(S,T]}$ is previsible, and deduce that if X is a supermartingale, then

$$\mathbf{E}(X_{T\wedge n}) \le \mathbf{E}(X_{S\wedge n}), \quad \forall n.$$

E10.5. *'What always stands a reasonable chance of happening will (almost surely) happen – sooner rather than later.'*

Suppose that T is a stopping time such that for some $N \in \mathbf{N}$ and some $\varepsilon > 0$, we have, for every n:

$$\mathbf{P}(T \leq n + N | \mathcal{F}_n) > \varepsilon, \quad \text{a.s.}$$

Prove by induction using $\mathbf{P}(T > kN) = \mathbf{P}(T > kN; T > (k-1)N)$ that for $k = 1, 2, 3, \ldots$

$$\mathbf{P}(T > kN) \leq (1 - \varepsilon)^k.$$

Show that $\mathbf{E}(T) < \infty$.

E10.6. ABRACADABRA

At each of times 1,2,3,..., a monkey types a capital letter at random, the sequence of letters typed forming an IID sequence of RVs each chosen uniformly from amongst the 26 possible capital letters.

Just before **each** time $n = 1, 2, \ldots$, a new gambler arrives on the scene. He bets \$ 1 that

$$\text{the } n^{\text{th}} \text{ letter will be } A.$$

If he loses, he leaves. If he wins, he receives \$ 26 all of which he bets on the event that

$$\text{the } (n+1)^{\text{th}} \text{ letter will be } B.$$

If he loses, he leaves. If he wins, he bets his whole current fortune of \$ 26^2 that

$$\text{the } (n+2)^{\text{th}} \text{ letter will be } R$$

and so on through the ABRACADABRA sequence. Let T be the first time by which the monkey has produced the consecutive sequence ABRACADABRA. Explain why martingale theory makes it intuitively obvious that

$$\mathbf{E}(T) = 26^{11} + 26^4 + 26$$

and use result 10.10(c) to prove this. (See Ross (1983) for other such applications.)

E10.7. Gambler's Ruin

Suppose that $X_1, X_2, \ldots$ are IID RVs with

$$\mathbf{P}[X = +1] = p, \quad \mathbf{P}[X = -1] = q, \quad \text{where} \quad 0 < p = 1 - q < 1,$$

and $p \neq q$. Suppose that a and b are integers with $0 < a < b$. Define

$$S_n := a + X_1 + \cdots + X_n, \quad T := \inf\{n : S_n = 0 \quad \text{or} \quad S_n = b\}.$$

Let $\mathcal{F} = \sigma(X_1, \ldots, X_n)$ ($\mathcal{F}_0 = \{\emptyset, \Omega\}$). Explain why T satisfies the conditions in Question E10.5. Prove that

$$M_n := (\frac{q}{p})^{S_n} \quad \text{and} \quad N_n = S_n - n(p-q)$$

define martingales M and N. Deduce the values of $\mathsf{P}(S_T = 0)$ and $\mathsf{E}(S_T)$.

E10.8. Bayes' urn

A random number Θ is chosen uniformly between 0 and 1, and a coin with probability Θ of heads is minted. The coin is tossed repeatedly. Let B_n be the number of heads in n tosses. Prove that (B_n) has exactly the same probabilistic structure as the (B_n) sequence in (E10.1) on Polya's urn. Prove that N_n^θ is a regular conditional pdf of Θ given $B_1, B_2, \ldots, B_n$.

(*Continued at E18.5.*)

E10.9. Show that if X is a non-negative supermartingale and T is a stopping time, then

$$\mathsf{E}(X_T; T < \infty) \leq \mathsf{E}(X_0).$$

(*Hint.* Recall Fatou's Lemma.) Deduce that $c\,\mathsf{P}(\sup_n X_n \geq c) \leq \mathsf{E}(X_0)$.

E10.10*. The 'Star-ship *Enterprise*' Problem

The control system on the star-ship *Enterprise* has gone wonky. All that one can do is to set a distance to be travelled. The spaceship will then move that distance in a randomly chosen direction, then stop. The object is to get into the Solar System, a ball of radius r. Initially, the *Enterprise* is at a distance $R_0 (> r)$ from the Sun.

Let R_n be the distance from Sun to *Enterprise* after n 'space-hops'. Use Gauss's theorems on potentials due to spherically-symmetric charge distributions to show that whatever strategy is adopted, $1/R_n$ is a *supermartingale*, and that for any strategy which always sets a distance no greater than that from Sun to *Enterprise*, $1/R_n$ is a *martingale*. Use (E10.9) to show that

$$\mathsf{P}[\textit{Enterprise} \text{ gets into Solar System}] \leq r/R_0.$$

For each $\varepsilon > 0$, you can choose a strategy which makes this probability greater than $(r/R_0) - \varepsilon$. What kind of strategy will this be?

E10.11*. *Star Trek*, 2. 'Captain's Log ...

Mr Spock and Chief Engineer Scott have modified the control system so that the *Enterprise* is confined to move for ever in a fixed plane passing through the Sun. However, the next 'hop-length' is now automatically set to be the current distance to the Sun ('next' and 'current' being updated in the obvious way). Spock is muttering something about logarithms and random walks, but I wonder whether it is (almost) certain that we will get into the Solar System sometime ... '

Hint. Let $X_n := \log R_n - \log R_{n-1}$. Prove that $X_1, X_2, \ldots$ is an IID sequence of variables each of mean 0 and finite variance σ^2 (say), where $\sigma > 0$. let

$$S_n := X_1 + X_2 + \cdots + X_n.$$

Prove that if α is a fixed positive number, then

$$\begin{aligned}\mathbf{P}[\inf_n S_n = -\infty] &\geq \mathbf{P}[S_n \leq -\alpha\sigma\sqrt{n}, \text{i.o.}] \\ &\geq \limsup \mathbf{P}[S_n \leq -\alpha\sigma\sqrt{n}] = \Phi(-\alpha) > 0.\end{aligned}$$

(Use the Central Limit Theorem.) Prove that the event $\{\inf_n S_n = -\infty\}$ is in the tail σ-algebra of the (X_n) sequence.

E12.1. Branching Process

A branching process $Z = \{Z_n : n \geq 0\}$ is constructed in the usual way. Thus, a family $\left\{X_k^{(n)} : n, k \geq 1\right\}$ of IID $\mathbf{Z}^+$-valued random variables is supposed given. We define $Z_0 := 1$ and then define recursively:

$$Z_{n+1} := X_1^{(n+1)} + \cdots + X_{Z_n}^{(n+1)} \qquad (n \geq 0).$$

Assume that if X denotes any one of the $X_k^{(n)}$, then

$$\mu := \mathbf{E}(X) < \infty \quad \text{and} \quad 0 < \sigma^2 := \operatorname{Var}(X) < \infty.$$

Prove that $M_n := Z_n/\mu^n$ defines a martingale M relative to the filtration $\mathcal{F}_n = \sigma(Z_0, Z_1, \ldots, Z_n)$. Show that

$$\mathbf{E}\left(Z_{n+1}^2 | \mathcal{F}_n\right) = \mu^2 Z_n^2 + \sigma^2 Z_n$$

and deduce that M is bounded in $\mathcal{L}^2$ if and only if $\mu > 1$. Show that when $\mu > 1$,

$$\operatorname{Var}(M_\infty) = \sigma^2\{\mu(\mu-1)\}^{-1}.$$

E12.2. Use of Kronecker's Lemma

Let $E_1, E_2, \ldots$ be independent events with $\mathbf{P}(E_n) = 1/n$. Let $Y_i = \mathrm{I}_{E_i}$. Prove that $\sum (Y_k - \frac{1}{k}) / \log k$ converges a.s., and use Kronecker's Lemma to deduce that

$$\frac{N_n}{\log n} \longrightarrow 1, \quad \text{a.s.},$$

where $N_n := Y_1 + \cdots + Y_n$. An interesting application is to E4.3, when N_n becomes the *number of records* by time n.

E12.3. *Star Trek*, 3

Prove that if the strategy in E10.11 is (in the obvious sense) employed – and for ever – *in* $\mathbf{R}^3$ *rather than in* $\mathbf{R}^2$, then

$$\sum R_n^{-2} < \infty, \quad \text{a.s.},$$

where R_n is the distance from the *Enterprise* to the Sun at time n.

Note. It should be obvious which result plays the key rôle here, but you should try to make your argument fully rigorous.

Uniform Integrability

E13.1. Prove that a clas $\mathcal{C}$ of RVs is UI if and only if both of the following conditions (i) and (ii) hold:

(i) $\mathcal{C}$ is bounded in $\mathcal{L}^1$, so that $A := \sup\{\mathsf{E}(|X|) : X \in \mathcal{C}\} < \infty$,

(ii) for every $\varepsilon > 0$, $\exists \delta > 0$ such that if $F \in \mathcal{F}$, $\mathbf{P}(F) < \delta$ and $X \in \mathcal{C}$, then $\mathsf{E}(|X|; F) < \varepsilon$.

Hint for 'if'. For $X \in \mathcal{C}$, $\mathbf{P}(|X| > K) \leq K^{-1}A$.

Hint for 'only if'. $\mathsf{E}(|X|; F) \leq \mathsf{E}(|X|; |X| > K) + K\mathbf{P}(F)$.

E13.2. Prove that if $\mathcal{C}$ and $\mathcal{D}$ are UI classes of RVs, and if we define

$$\mathcal{C} + \mathcal{D} := \{X + Y : X \in \mathcal{C},\ Y \in \mathcal{D}\},$$

then $\mathcal{C} + \mathcal{D}$ is UI. *Hint.* One way to prove this is to use E13.1.

E13.3. Let $\mathcal{C}$ be a UI family of RVs. Say that $Y \in \mathcal{D}$ if for some $X \in \mathcal{C}$ and some sub-σ-algebra $\mathcal{G}$ of $\mathcal{F}$, we have $Y = \mathsf{E}(X|\mathcal{G})$, a.s. Prove that $\mathcal{D}$ is UI.

E14.1. Hunt's Lemma

Suppose that (X_n) is a sequence of RVs such that $X := \lim X_n$ exists a.s. and that (X_n) is dominated by Y in $(\mathcal{L}^1)^+$:

$$|X_n(\omega)| \leq Y(\omega), \quad \forall (n, \omega), \quad \text{and} \quad \mathsf{E}(Y) < \infty.$$

Let $\{\mathcal{F}_n\}$ be any filtration. Prove that

$$\mathsf{E}(X_n|\mathcal{F}_n) \to \mathsf{E}(X|\mathcal{F}_\infty) \quad \text{a.s.}$$

Hint. Let $Z_m := \sup_{r \geq m} |X_r - X|$. Prove that $Z_m \to 0$ a.s. and in $\mathcal{L}^1$. Prove that for $n \geq m$, we have, almost surely,

$$|\mathsf{E}(X_n|\mathcal{F}_n) - \mathsf{E}(X|\mathcal{F}_\infty)| \leq |\mathsf{E}(X|\mathcal{F}_n) - \mathsf{E}(X|\mathcal{F}_\infty)| + \mathsf{E}(Z_m|\mathcal{F}_n).$$

E14.2. Azuma-Hoeffding Inequality

(a) Show that if Y is a RV with values in $[-c, c]$ and with $\mathsf{E}(Y) = 0$, then, for $\theta \in \mathbf{R}$,

$$\mathsf{E}e^{\theta Y} \leq \cosh \theta c \leq \exp\left(\frac{1}{2}\theta^2 c^2\right).$$

(b) Prove that if M is a martingale null at 0 such that for some sequence $(c_n : n \in \mathbf{N})$ of positive constants,

$$|M_n - M_{n-1}| \leq c_n, \quad \forall n,$$

then, for $x > 0$,

$$\mathsf{P}\left(\sup_{k \leq n} M_k \geq x\right) \leq \exp\left(\frac{1}{2}x^2 \Big/ \sum_{k=1}^{n} c_k^2\right).$$

Hint for (a). Let $f(z) := \exp(\theta z)$, $z \in [-c, c]$. Then, since f is convex,

$$f(y) \leq \frac{c-y}{2c} f(-c) + \frac{c+y}{2c} f(c).$$

Hint for (b). See the proof of (14.7,a).

Characteristic Functions

E16.1. Prove that

$$\lim_{T \uparrow \infty} \int_0^T x^{-1} \sin x \, dx = \pi/2$$

by integrating $\int z^{-1} e^{iz} dz$ around the contour formed by the 'upper' semi-circles of radii ε and T and the intervals $[-T, -\varepsilon]$ and $[\varepsilon, T]$.

E16.2. Prove that if Z has the U$[-1, 1]$ distribution, then

$$\varphi_Z(\theta) = (\sin \theta)/\theta,$$

and prove that there do not exist IID RVs X and Y such that

$$X - Y \sim \mathrm{U}[-1, 1].$$

E16.3. Suppose that X has the Cauchy distribution, and let $\theta > 0$. By integrating $e^{i\theta z}/(1+z^2)$ around the semicircle formed by $[-R, R]$ together with the 'upper' semicircle centre 0 of radius R, prove that $\varphi_X(\theta) = e^{-\theta}$. Show that $\varphi_X(\theta) = e^{-|\theta|}$ for all θ. Prove that if $X_1, X_2, \ldots X_n$ are IID RVs each with the standard Cauchy distribution, then $(X_1 + \cdots + X_n)/n$ also has the standard Cauchy distribution.

E16.4. Suppose that X has the standard normal N(0,1) distribution. Let $\theta > 0$. Consider $\int (2\pi)^{-\frac{1}{2}} \exp(-\frac{1}{2}z^2) dz$ around the rectangular contour

$$(-R - i\theta) \to (R - i\theta) \to R \to (-R) \to (-R - i\theta),$$

and prove that $\varphi_X(\theta) = \exp(-\frac{1}{2}\theta^2)$.

E16.5. Prove that if φ is the characteristic function of a RV X, then φ is *non-negative definite* in that for complex $c_1, c_2, \ldots, c_n$ and real $\theta_1, \theta_2, \ldots, \theta_n$,

$$\sum_j \sum_k c_j \bar{c}_k \varphi(\theta_j - \theta_k) \geq 0.$$

(*Hint.* Express LHS as the expectation of ... .) **Bochner's Theorem** says that φ is a characteristic function *if and only if* $\varphi(0) = 1$, φ is continuous, and φ is non-negative definite! (It is of course understood that here $\varphi : \mathbf{R} \to \mathbf{C}$.) E18.6 gives a simpler result in the same spirit.

E16.6. (a) Let $(\Omega, \mathcal{F}, \mathbf{P}) = ([0,1], \mathcal{B}[0,1], \mathrm{Leb})$. What is the distribution of the RV Z, where $Z(\omega) := 2\omega - 1$? Let $\omega = \sum 2^{-n} R_n(\omega)$ be the binary expansion of ω. Let

$$U(\omega) = \sum_{\text{odd } n} 2^{-n} Q_n(\omega), \quad \text{where} \quad Q_n(\omega) = 2R_n(\omega) - 1.$$

Find a random variable V independent of U such that U and V are identically distributed and $U + \frac{1}{2}V$ is uniformly distributed on $[-1, 1]$.

(b) Now suppose that (on some probability triple) X and Y are IID RVs such that

$$X + \tfrac{1}{2}Y \quad \text{is uniformly distributed on} \quad [-1, 1].$$

Let φ be the CF of X. Calculate $\varphi(\theta)/\varphi(\frac{1}{4}\theta)$. Show that the distribution of X must be the same as that of U in part (a), and deduce that there exists a set $F \in \mathcal{B}[-1,1]$ such that $\text{Leb}(F) = 0$ and $\mathbf{P}(X \in F) = 1$.

E18.1. (a) Suppose that $\lambda > 0$ and that (for $n > \lambda$) F_n is the DF associated with the Binomial distribution $B(n, \lambda/n)$. Prove (using CFs) that F_n converges weakly to F where F is the DF of the Poisson distribution with parameter λ.

(b) Suppose that $X_1, X_2, \ldots$ are IID RVs each with the density function $(1 - \cos x)/\pi x^2$ on $\mathbf{R}$. Prove that for $x \in \mathbf{R}$,

$$\lim_{n\to\infty} \mathbf{P}\left(\frac{X_1 + X_2 + \cdots + X_n}{n} \leq x\right) = \tfrac{1}{2} + \pi^{-1} \arctan x,$$

where $\arctan \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

E18.2. Prove the Weak Law of Large Numbers in the following form. Suppose that $X_1, X_2, \ldots$ are IID RVs, each with the same distribution as X. Suppose that $X \in \mathcal{L}^1$ and that $\mathbf{E}(X) = \mu$. Prove by the use of CFs that the distribution of

$$A_n := n^{-1}(X_1 + \cdots + X_n)$$

converges weakly to the unit mass at μ. Deduce that

$$A_n \to \mu \quad \text{in probability.}$$

Of course, SLLN implies this Weak Law.

Weak Convergence for Prob[0, 1]

E18.3. Let X and Y be RVs taking values in [0,1]. Suppose that

$$\mathbf{E}(X^k) = \mathbf{E}(Y^k), \qquad k = 0, 1, 2, \ldots .$$

Prove that

(i) $\mathbf{E}p(X) = \mathbf{E}p(Y)$ for every polynomial p,

(ii) $\mathbf{E}f(X) = \mathbf{E}f(Y)$ for every continuous function f on [0,1],

(iii) $\mathbf{P}(X \leq x) = \mathbf{P}(Y \leq y)$ for every x in [0,1].

Hint for (ii). Use the Weierstrass Theorem 7.4.

E18.4. Suppose that (F_n) is a sequence of DFs with

$$F_n(x) = 0 \text{ for } x < 0, \quad F_n(1) = 1, \qquad \text{for every } n.$$

Suppose that

$$(*) \qquad m_k := \lim_n \int_{[0,1]} x^k dF_n \text{ exists for } k = 0, 1, 2, \ldots .$$

Use the Helly-Bray Lemma and E18.3 to show that $F_n \xrightarrow{w} F$, where F is characterized by $\int_{[0,1]} x^k dF = m_k$, $\forall k$.

E18.5. Improving on E18.3: A Moment Inversion Formula

Let F be a distribution with $F(0-) = 0$ and $F(1) = 1$. Let μ be the associated law, and define

$$m_k := \int_{[0,1]} x^k dF(x).$$

Define

$$\Omega = [0,1] \times [0,1]^{\mathbf{N}}, \quad \mathcal{F} = \mathcal{B} \times \mathcal{B}^{\mathbf{N}}, \quad \mathbf{P} = \mu \times \text{Leb}^{\mathbf{N}},$$
$$\Theta(\omega) = \omega_0, \quad H_k(\omega) = I_{[0,\omega_0]}(\omega_k).$$

This models the situation in which Θ is chosen with law μ, a coin with probability Θ of heads is then minted, and tossed at times $1, 2, \ldots$. See E10.8. The RV H_k is 1 if the k^{th} toss produces heads, 0 otherwise. Define

$$S_n := H_1 + H_2 + \cdots + H_n.$$

By the Strong Law and Fubini's Theorem,

$$S_n/n \to \Theta, \quad \text{a.s.}$$

Define a map D on the space of real sequences $(a_n : n \in \mathbf{Z}^+)$ by setting

$$Da = (a_n - a_{n+1} : n \in \mathbf{Z}^+).$$

Prove that

$$(*) \qquad F_n(x) := \sum_{i \le nx} \binom{n}{i} (D^{n-i} m)_i \to F(x)$$

at every point x of continuity of F.

E18.6* Moment Problem

Prove that if $(m_k : k \in \mathbf{Z}^+)$ is a sequence of numbers in [0,1], then there exists a RV X with values in [0,1] such that $\mathbf{E}(X^k) = m_k$ if and only if $m_0 = 1$ and

$$(D^r m)_s \ge 0 \qquad (r, s, \in \mathbf{Z}^+).$$

Hint. Define F_n via E18.5(*), and then verify that E18.4(*) holds. You can show that the moments $m_{k,n}$ of F_n satisfy

$$m_{n,0} = 1, \quad m_{n,1} = m_1, \quad m_{n,2} = m_2 + n^{-1}(m_1 - m_2), \quad \text{etc.}$$

You discover the algebra!

Weak Convergence for Prob$[0,\infty)$

E18.7. Using Laplace transforms instead of CFs

Suppose that F and G are DFs on $\mathbf{R}$ such that $F(0-) = G(0-) = 0$, and

$$\int_{[0,\infty)} e^{-\lambda x} dF(x) = \int_{[0,\infty)} e^{-\lambda x} dG(x), \qquad \forall \lambda \geq 0.$$

Note that the integral on LHS has a contribution $F(0)$ from $\{0\}$. Prove that $F = G$. [*Hint.* One could derive this from the idea in E7.1. However, it is easier to use E18.3, because we know that if X has DF F and Y has DF G, then

$$\mathbf{E}[(e^{-X})^n] = \mathbf{E}[(e^{-Y})^n], \qquad n = 0, 1, 2, \ldots .]$$

Suppose that (F_n) is a sequence of distribution functions on $\mathbf{R}$ each with $F_n(0-) = 0$ and such that

$$L(\lambda) := \lim_n \int e^{-\lambda x} dF_n(x)$$

exists for $\lambda \geq 0$ and that L is continuous at 0. Prove that F_n is tight and that

$$F_n \xrightarrow{w} F \text{ where } \int e^{-\lambda x} dF(x) = L(\lambda), \qquad \forall \lambda \geq 0.$$

Modes of convergence

EA13.1. (a) Prove that $(X_n \to X, \text{ a.s.}) \Rightarrow (X_n \to X \text{ in prob})$.

Hint. See Section 13.5.

(b) Prove that $(X_n \to X \text{ in prob}) \not\Rightarrow (X_n \to X, \text{ a.s.})$.

Hint. Let $X_n = \mathrm{I}_{E_n}$, where $E_1, E_2, \ldots$ are independent events.

(c) Prove that if $\sum \mathbf{P}(|X_n - X| > \varepsilon) < \infty$, $\forall \varepsilon > 0$, then $X_n \to X$, a.s.

Hint. Show that the set $\{\omega : X_n(\omega) \not\to X(\omega)\}$ may be written

$$\bigcup_{k \in \mathbf{N}} \{\omega : |X_n(\omega) - X(\omega)| > k^{-1} \text{ for infinitely many } n\}.$$

(d) Suppose that $X_n \to X$ in probability. Prove that there is a subsequence (X_{n_k}) of (X_n) such that $X_{n_k} \to X$, a.s.

Hint. Combine (c) with the 'diagonal principle'.

(e) Deduce from (a) and (d) that $X_n \to X$ in probability if and only if every subsequence of (X_n) contains a further subsequence which converges a.s. to X.

EA13.2. Recall that if ξ is a random variable with the standard normal N(0,1) distribution, then

$$\mathsf{E}e^{\lambda\xi} = \exp(\tfrac{1}{2}\lambda^2).$$

Suppose that $\xi_1, \xi_2, \ldots$ are IID RVs each with the N(0,1) distribution. Let $S_n = \sum_{k=1}^{n} \xi_k$, let $a, b \in \mathbb{R}$, and define

$$X_n = \exp(aS_n - bn).$$

Prove that

$$(X_n \to 0, \text{ a.s.}) \Leftrightarrow (b > 0)$$

but that for $r \geq 1$,

$$(X_n \to 0 \text{ in } \mathcal{L}^r) \Leftrightarrow (r < 2b/a^2).$$

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Index

(Recall that there is a Guide to Notation on pages xiv-xv.)

ABRACADABRA (4.9, E10.6).

adapted process (10.2): Doob decomposition (12.11).

σ-algebra (1.1).

algebra of sets (1.1).

almost everywhere = a.e. (1.5); almost surely = a.s. (2.4).

atoms: of σ-algebra (9.1, 14.13); of distribution function (16.5).

Azuma-Hoeffding inequality (E14.2).

Baire category theorem (A1.12).

Banach-Tarski paradox (1.0).

Bayes' formula (15.7–15.9).

Bellman Optimality Principle (E10.2, 15.3).

Black-Scholes option-pricing formula (15.2).

Blackwell's Markov chain (E4.8).

Bochner's Theorem (E16.5)

Borel-Cantelli Lemmas: First = BC1 (2.7); Second = BC2 (4.3); Lévy's extension of (12.15).

Bounded Convergence Theorem = BDD (6.2, 13.6).

branching process (Chapter 0, E12.1).

Burkholder-Davis-Gundy inequality (14.18).

Carathéodory's Lemma (A1.7).

Carathéodory's Theorem: statement (1.7); proof (A1.8).

Central Limit Theorem (18.4).

Cesàro's Lemma (12.6).

characteristic functions: definition (16.1); inversion formula (16.6); convergence theorem (18.1).

Chebyshev's inequality (7.3).

coin tossing (3.7).

conditional expectation (Chapter 9): properties (9.7).

conditional probability (9.9).

consistency of Likelihood-Ratio Test (14.17).

contraction property of conditional expectation (9.7,h).

convergence in probability (13.5, A13.2).

convergence theorems for integrals: MON (5.3); Fatou (5.4); DOM (5.9); BDD (6.2, 13.6); for UI RVs (13.7).

convergence theorems for martingales: Main (11.5); for UI case (14.1); Upward (14.2); Downward (14.4).

d-system (A1.2).

differentiation under integral sign (A16.1).

distribution function for RV (3.10–3.11).

Dominated Convergence Theorem = DOM (5.9); conditional (9.7,g).

J.L. DOOB's Convergence Theorem (11.5); Decomposition (12.11); $\mathcal{L}^p$ inequality (14.11); Optional Sampling Theorem (A14.3–14.4); Optional Stopping Theorem (10.10, A14.3); Submartingale Inequality (14.6); Upcrossing Lemma (11.2) – and much else!

Downward Theorem (14.4).

Dynkin's Lemma (A1.3).

events (Chapter 2): independent (4.1).

expectation (6.1): conditional (Chapter 9).

'extension of measures': uniqueness (1.6); existence (1.7).

extinction probability (0.4).

fair game (10.5): unfavourable (E4.7).

Fatou Lemmas: for sets (2.6,b), 2.7,c); for functions (5.4); conditional version (9.7,f).

filtered space, filtration (10.1).

filtering (15.6–15.9).

finite and σ-finite measures (1.5).

Forward Convergence Theorem for supermartingales (11.5).

Fubini's Theorem (8.2).

gambler's ruin (E10.7).

gambling strategy (10.6).

Hardy space $\mathcal{H}_0^1$ (14.18).

harnesses (15.10–15.12).

hedging strategy (15.2).

Helly-Bray Lemma (17.4).

hitting times (10.12).

Hoeffding's inequality (E14.2).

Hölder's inequality (6.13).

Hunt's Lemma (E14.1).

independence: definitions (4.1); π-system criterion (4.2); and conditioning (9.7,k, 9.10).

independence and product measure (8.4).

inequalities: Azuma-Hoeffding (E14.2); Burkholder-Davis-Gundy (14.18); Chebyshev (7.3); Doob's $\mathcal{L}^p$ (14.11); Hölder (6.13); Jensen (6.6), and in conditional form (9.7,h); Khinchine – see (14.8); Kolmogorov (14.6); Markov (6.4); Minkowski (6.14); Schwarz (6.8).

infinite products of probability measures (8.7, Chapter A9); Kakutani's Theorem on (14.12, 14.17).

integration (Chapter 5).

Jensen's inequality (6.6); conditional form (9.7,h).

Kakutani's Theorem on likelihood ratios (14.12, 14.17).

Kalman-Bucy filter (15.6–15.9).

A.N. KOLMOGOROV's Definition of Conditional Expectation (9.2); Inequality (14.6); Law of the Iterated Logarithm (A4.1, 14.7); Strong Law of Large Numbers (12.10, 14.5); Three-Series Theorem (12.5); Truncation Lemma (12.9); Zero-One (0-1) Law (4.11, 14.3).

Kronecker's Lemma (12.7).

Laplace transforms: inversion (E7.1); and weak convergence (E18.7).

law of random variable (3.9): joint laws (8.3).

least-squares-best predictor (9.4).

Lebesgue integral (Chapter 5).

Lebesgue measure = Leb (1.8, A1.9).

Lebesgue spaces $\mathcal{L}^p$, L^p (6.10).

P.LÉVY's Convergence Theorem for CFs (18.1); Downward Theorem for martingales (14.4); Extension of Borel-Cantelli Lemmas (12.15); Inversion formula for CFs (16.6); Upward Theorem for martingales (14.2).

Likelihood-Ratio Test, consistency of (14.17).

Mabinogion sheep (15.3–15.5).

Markov chain (4.8, 10.13).

Markov's inequality (6.4).

martingale (Chapters 10–15!): definition (10.3); Convergence Theorem (11.5); Optional-Stopping Theorem (10.9–10.10, A14.3); Optional-Sampling Theorem (Chapter A14).

martingale transform (10.6)

measurable function (3.1).

measurable space (1.1).

measure space (1.4).

Minkowski's inequality (6.14).

Moment Problem (E18.6).

monkey typing Shakespeare (4.9).

Monotone-Class Theorem (3.14, A1.3).

Monotone-Convergence Theorem: for sets (1.10); for functions (5.3, Chapter A5); conditional version (9.7,e).

narrow convergence – see weak convergence.

option pricing (15.2).

Optional-Sampling Theorem (Chapter A14).

Optional-Stopping Theorems (10.9–10.10, A14.3).

optional time – see stopping time (10.8).

orthogonal projection (6.11): and conditional expectation (9.4–9.5).

outer measures (A1.6).

π-system (1.6): Uniqueness Lemmas (1.6, 4.2).

Pólya's urn (E10.1, E10.8).

previsible (= predictable) process (10.6).

probability density function = pdf (6.12); joint (8.3).

probability measure (1.5).

probability triple (2.1).

product measures (Chapter 8).

Pythagoras's Theorem (6.9).

Radon-Nikodým theorem (5.14, 14.13–14.14).

random signs (12.3).

random walk: hitting times (10.12, E10.7); on free group (EG.3–EG.4).

Record Problem (E4.3, E12.2, 18.5).

regular conditional probability (9.9).

Riemann integral (5.3).

sample path (4.8)

sample point (2.1)

sample space (2.1)

Schwarz inequality (6.8).

Star Trek problems (E10.10, E10.11, E12.3).

stopped process (10.9).

stopping times (10.8); associated σ-algebras (A14.1).

Strassen's Law of the Iterated Logarithm (A4.2).

Strong Laws (7.2, 12.10, 12.14, 14.5).

submartingales and supermartingales: definitions (10.3); convergence theorem (11.5); optional stopping (10.9–10.10); optional sampling (A14.4).

superharmonic functions for Markov chains (10.13).

symmetrization technique (12.4).

d-system (A1.2); π-system (1.6).

tail σ-algebra (4.10–4.12, 14.3).

Tchebycheff: = Chebyshev (7.3).

Three-Series Theorem (12.5).

tightness (17.5).

Tower Property (9.7,i).

Truncation Lemma (12.9).

uniform integrability (Chapter 13).

Upcrossing Lemma (11.1–11.2).

Uniqueness Lemma (1.6).

weak convergence (Chapter 17): and characteristic functions (18.1); and moments (E18.3–18.4); and Laplace transforms (E18.7).

Weierstrass approximation theorem (7.4).

Zero-one law = 0-1 law (4.11, 14.3).